Physics-Based Musculoskeletal Simulation and Inverse Modeling for Neural Delay Estimation

Thesis

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Abstract

Neural delay refers to a delay between a neural command and the onset of muscle activation. It plays a crucial role in human movement as it influences the rate at which muscles react and generate movement. Accurately estimating the delay from human movement data remains challenging. Existing approaches mainly use direct measurements such as electromyography (EMG). Noise and errors in identifying when muscles begin to activate can alter these measurements. Under controlled conditions, musculoskeletal simulation can generate detailed muscle and movement signals. But, in these simulations, the delay of the nervous system is typically known from the signals generated by the model, not measured. Therefore, estimation of neural delay from muscle activation/movement signals is still largely unexplored. In this study, a new inverse modelling approach is developed to solve this problem. OpenSim 4.5 and 3DGaitModel2354 lower-limb model were used to conduct forward dynamics simulations. 45,445 labelled observations of muscle activation and joint kinematics were created using neural delay values of 0.015 to 0.060s. The neural delay from these signals was estimated using a Random Forest regression model. A model was developed that obtained R^2>0.98 with a mean absolute error below 0.0001s. The muscle activation contributed approximately 93% of the predictive information. The most important signal was rectus femoris. The model also predicted the delay values that were not used for training and were found to remain the same even after femur mass changed by 7%. The results indicate that the hidden neural timing parameter can be retrieved from the muscle and movement signals it generates. The proposed approach is a new inverse method to estimate the neural delay from biomechanical signals.

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List of Acronyms

Acronym	Full form
ACh	Acetylcholine
CNN	Convolutional Neural Network
CNS	Central Nervous System
EMD	Electromechanical Delay
EMG	Electromyography
HD-sEMG	High-Density Surface Electromyography
IMP	Intramuscular Pressure
IMU	Inertial Measurement Unit
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MVC	Maximum Voluntary Contraction
NMD	Neuromechanical Delay
NMJ	Neuromuscular Junction
ODE	Ordinary Differential Equation
RF	Random Forest
R^2	Coefficient of Determination
RMSE	Root Mean Square Error
sEMG	Surface Electromyography
SVR	Support Vector Regression

1. Introduction

Human movement depends on precise temporal coordination between neural commands and muscle contraction. Motor commands must be generated in advance because neural conduction, synaptic transmission, and excitation–contraction coupling introduce delays between neural excitation and muscle activation [1]. The cascade of temporal processes collectively known as neuromechanical delay, are important footnotes in motor physiology. This is an important factor of movement quality, inter-muscular coordination, postural stability and the time response of neuromuscular system for unexpected perturbators during level walking or running [2]. Even though the neural delay parameter is physiologically grounded, it continues to pose some of the greatest challenges for obtaining experimental or computational biomechanical data. We can classify the methodological traditions of studying human movement into two broad families: experimental measurement, and physics-based musculoskeletal simulation. Electromyography (EMG), the primary experimental tool, can record the electrical correlates of timing of muscle activation but is noisy, sensitive to electrode placement, and completely unsuited to simulation environments [3]. By contrast, musculoskeletal simulation platforms such as OpenSim generate extensive, internally consistent biomechanical datasets under fully controlled conditions; however, neural delay is not a readily recovered parameter and remains embedded as a latent variable in the model muscle activation dynamics [4]. This creates an inverse modeling problem, whether the neural delay can be recovered from the muscle activation and movement signals generated by the simulation. This study addresses this problem using an inverse modeling approach within a controlled musculoskeletal simulation framework. Generalization to experimental EMG recordings or in clinical settings is one of the most important and remaining directions for future work but is beyond the scope of this study.

This paper bridges that gap by developing an inverse neuromechanical modeling framework where (i) a physics-based musculoskeletal forward simulation is implemented in OpenSim to generate a labeled dataset spanning a physiologically valid and highly informative subspace of neural delay conditions, and (ii) a Random Forest regression model learns to recover the latent neural delay parameter from resulting trajectories of joint kinematics and muscle activations. The framework achieves mean absolute estimation errors on the order of 10^{-4} seconds for coefficient of determination R^2 approaching unity under theoretical baseline conditions (allowing intra-trajectory sampling via randomization). Although this near perfect accuracy reflects the deterministic nature of the simulation, it also indicates that the neural delay is fully identifiable from the biomechanical signals generated by the model. [5]. Section 2 offers a systematic review of previous studies across EMD measurement, musculoskeletal simulation, and machine learning in biomechanics; enabling the identification and substantiation of the specific research gap.

1.1 Neuromechanical Control of Human Movement

Human movement is carefully regulated by a sensorimotor control architecture of the neuromusculoskeletal unit that integrates the nervous, muscular, and skeletal systems [6]. Motor

commands from the central nervous system are relayed by the corticospinal tract to spinal motor neurons which in turn excite skeletal muscle fibers through the neuromuscular junction. These contractile forces translate into joint torques that switch limb segments on and off, allowing for intentional motion coordinated across joints. However, this hierarchical cascade is not an open-loop system; proprioceptive feedback from muscle spindles and Golgi tendon organs, vestibular inputs, and visual cues are constantly integrated at the spinal and supraspinal levels to update the motor command on-line, ensuring that stable behavior can be maintained or adapted in response to behavioral demands as they arise during dynamic tasks such as walking, running and balance recovery [7]. Conceptually, the neuromuscular system is better related to a high dimensional, nonlinear feedback control system with intrinsic time delays from a control-theoretic standpoint [8]. Such delays stem from the electrical conduction speed of action potentials within peripheral motor neurons, the time it takes for chemicals to diffuse across synaptic junctions, calcium transient kinetics during excitation–contraction coupling and mechanical compliance of series-elastic tendon which must be pre-stretched before tension is externally recorded [2]. Since movement corrections must be reactive, but cannot happen instantaneously, the sensorimotor loop stability relies heavily on accurate predictions of these delays through internal forward predictive models generated by the CNS [1].

This investigation of sensorimotor integration during locomotion has demonstrated that joint mechanics are highly sensitive to neural timing. In structured gait, the CNS must balance dozens of lower-limb muscles by timing activation during the stance and swing phases of each gait cycle [9]. Neuromusculoskeletal models have shown that modest changes in timing of neural control can yield measurable gait deviations at the knee and hip, inter-muscular activation patterns and a greater locomotive cost [10]. The sensitivity of motor control to these temporal variations motivates the need for methods to quantify neural timing from observable biomechanical data. This problem is magnified with aging and neurological disease. With aging, there is a sustained increase in neuromechanical delay, indicative of altered motor unit recruitment (e.g. decline in muscle fiber synchronization), slower peripheral conduction and processes such as neuromuscular junction degeneration [11]. Other neuromuscular disorders like peripheral neuropathy, spinal muscular atrophy and myopathy further interrupt the timing of neural excitation and mechanical response with delays in severely affected populations exceeding several hundred milliseconds [2]. Kinematic studies have also established that changes in sensorimotor control translate directly into measurable differences in joint movement speed, trajectory smoothness and endpoint accuracy [12]. Thus, a robust and non-invasive measure of the neural delay of biomechanical observations would provide clinicians with an actionable biomarker that can be used for neuromuscular health assessment, rehabilitation guidance, and therapeutic evaluation within these populations.

1.2 Electromechanical Delay: Physiology and Clinical Significance

Electromechanical delay (EMD) is formally defined as the time interval between the onset of electrical muscle activity, as assessed using EMG, and the onset of a measurable mechanical force at either the level of the muscle or joint [13]. This delay arises from a sequence of physiological events including propagation of the action potential along the sarcolemma, transverse-tubule depolarization which activates calcium release from the sarcoplasmic reticulum,

calcium binding to troponin-C leading to cycling in the contractile element and mechanical elongation of the series-elastic components of the muscle–tendon unit (mostly tendon and aponeurosis) prior to external force transmission being measured [14]. EMD typically ranges from approximately 10 ms in maximally activated fast-twitch muscles to more than 60 ms under conditions such as slow contraction, fatigue, or aging, and may reach several hundred milliseconds in pathological conditions [11].

Experimental work by Norman and Komi [13] and Cavanagh and Komi [15] has shown that the EMD is not a constant physiological value, but rather a variable that changes with the type of movement, its intensity, the level of fatigue, the temperature of the muscle tissues, and the stiffness of the musculotendinous unit. Schmid et al. [14]. later showed that, under voluntary contraction, the temporal pattern of motor unit recruitment, as controlled by the CNS, is the major source of EMD (more than 20 ms of delay), and that the contribution of action potential conduction velocity is relatively small. Importantly, Del Vecchio et al. [2] showed experimentally that across a wide range NMD is actively modulated by the CNS in order to control the rate of force development, which means that NMD is not just a passive biophysical artefact, but a parameter of motor output that the CNS actively controls (as illustrated in figure 1).

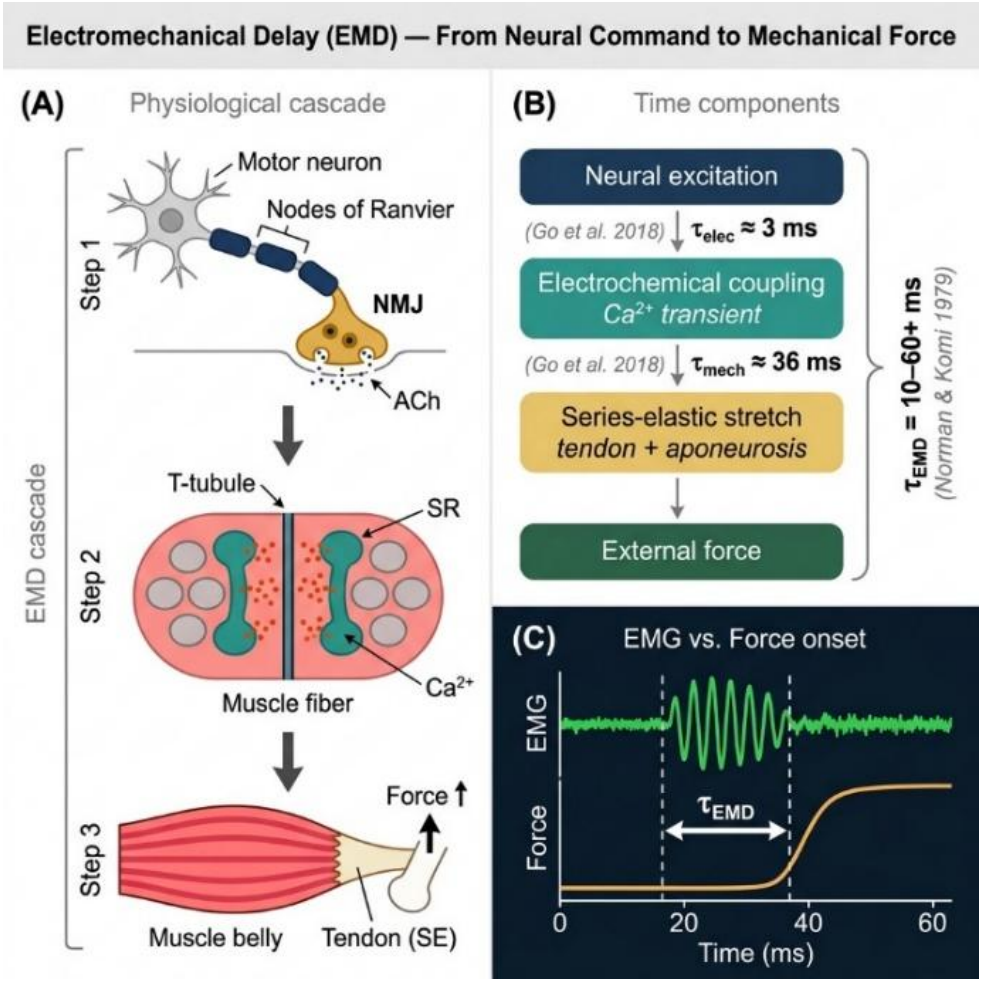


Figure 1: Physiological basis of electromechanical delay (EMD). (A) Sequential physiological cascade underpinning EMD: (Step 1) motor neuron depolarization propagating along the myelinated axon releases acetylcholine (ACh) at the neuromuscular junction (NMJ) onto sarcolemma; (Step 2) T-tubule depolarization triggers Ca^{2+} release from the SR which initiates actomyosin cross bridge cycling [14] and (Step 3) resulting force must first stretch series elastic

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The study by Go et al. [16] additionally quantified the electrochemical and mechanical aspects of EMD using intramuscular pressure as a surrogate for muscle tension and found that the electrochemical component (EMG to intramuscular pressure onset) was much shorter (about 3 ms) than the mechanical component (EMG to external force onset) which averaged 36 ms. This decomposition demonstrated the existence of a series-elastic tissue mechanics component to the externally observed delay, and a neural component to motor unit recruitment timing [14]. Most recently, Krätschmer et al. [11] showed the inverse relationship, finding that the mean NMD decreased by 31% from 175.8 ms at 10% MVc to 121.6 ms at 70% MVc, which they attributed to preferential recruitment of faster, higher threshold motor units at higher effort levels. These results make EMD and NMD biologically informative parameters that contain information regarding motor unit composition, tendon mechanics, CNS control strategy and the health status of the neuromuscular system. Neural delay quantification is clinically relevant across numerous populations and conditions. For musculoskeletal injury prevention, delayed activation of both knee flexors and extensors has been associated with increased risk of ligamentous injury and joint instability due to impaired protective reflex stiffening that occurs during impact loading [6]. Longitudinal EMD monitoring has significant clinical implications in rehabilitation medicine, as it offers an objective metric for assessing neuromuscular regenerative potential following injury, surgical intervention or neurological insult. EMD is used in sports science for executive function (training adaptations & effect of warm up protocols on fast-twitch muscle recruitment [13]. A non-invasive, simulation-compatible method for estimating the neural delay in each of these contexts would greatly enhance the scale and accessibility of neuromechanical assessment beyond specialized electrophysiology laboratories.

1.3 Limitations of Current Approaches to Neural Delay Estimation

In both experimental and clinical applications, the estimation of muscle activation timing is most commonly performed by electromyography. The electrical activity of recruited motor units is summed together and detected by the electrodes placed on the target muscle belly, allowing the estimation of the onset of muscle activation in comparison to a mechanical reference signal [3]. But surface EMG signals are also subject to contamination from neighboring muscle (crosstalk) to motion artifacts during dynamic activities and to variability due to precision of the placement of the electrodes, skin-electrode impedance, and thickness of the subcutaneous tissue. Such sources of variability cause timing detection errors of 10-30ms (on the order of the physiological EMD values being estimated) which make measurements unreliable in realistic conditions. Fine-wire intramuscular EMG offers greater specificity, but is also invasive, painful and not feasible for routine clinical monitoring or for large-cohort, long-term studies. Another fundamental drawback of EMG-based methods is their total inability to be used in a musculoskeletal simulation environment. A Physics-based simulation platform like OpenSim provides a detailed

biomechanical output, including muscle force, joint kinematics and joint contact forces, as they are produced via Hill-type actuators excited by neural signals [4]. The platforms have been crucial tools for biomechanical research as they allow for systematic and controlled investigation of parameters of the neuromuscular system that cannot be directly measured in experiments. But for these simulation models, the neural delay is not explicitly modeled and observable as an external parameter. Rather, it is indirectly embedded in the activation time constant of the muscle model (latent variable) that dictates the temporal development of the biomechanical outputs without being directly read or recovered from the outputs by any known method.

This establishes a basic methodological gap between experimental and computational biomechanics and neuromechanics. In the experimental sides, EMG can measure the timing of activation, but can access simulation environments and is corrupted by irreducible measurement noise. On the simulation side, forward dynamics models can yield an infinite number of biomechanical data sets when applied to any specific delay condition, but do not have a means to infer the delay parameter from the biomechanical data. Overall, the recovery of the neural delay from the macroscopic biomechanical signals is still a challenging problem, since high frequency temporal information is lost along the skeletal system. The literature has not yet included the estimation of the delay between the neural activity and the muscle response by a computational cascade of intermediate "virtual sensors" obtained by computation, which may be modular (Figure 2). A comprehensive systematic review of the relevant prior work is provided in detail in Section 2.

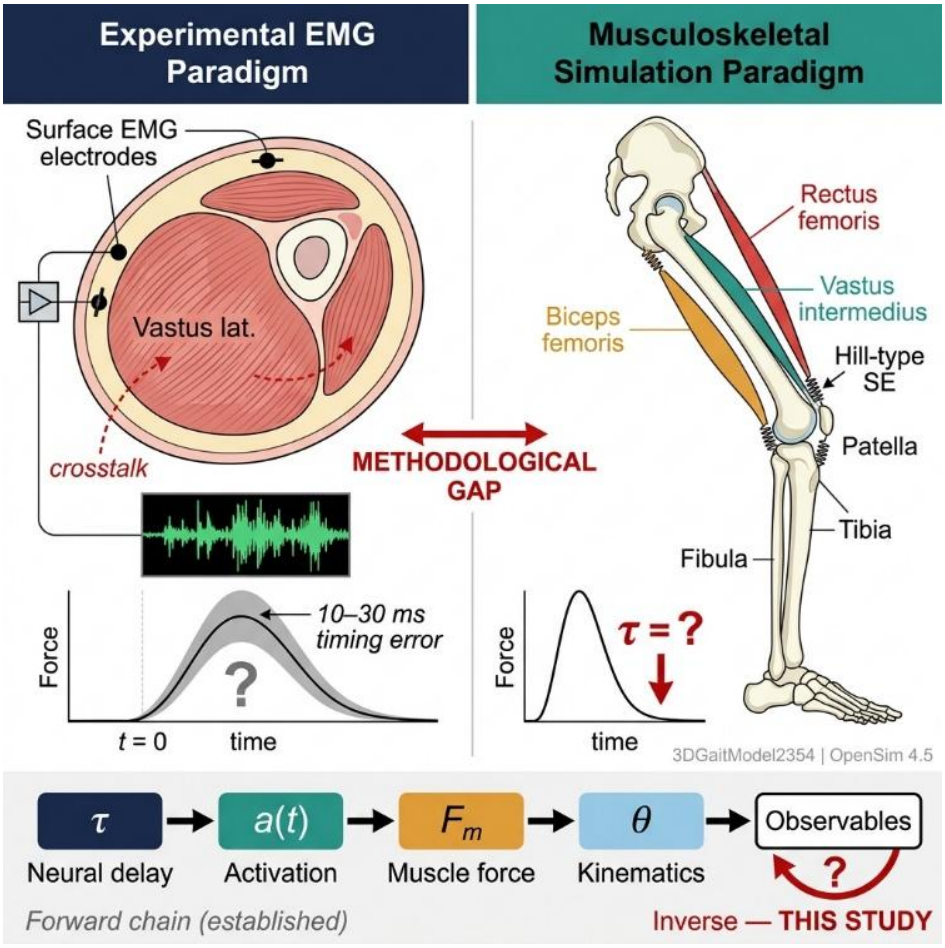


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crosstalk and signal noise introduce onset detection errors of 10–30 ms [3]—comparable to the physiological EMD rendering τ_EMD onset unidentifiable. (Right) Musculoskeletal simulation paradigm (3DGaitModel2354, OpenSim 4.5 [4][9]: rectus femoris, vastus intermedius, and biceps femoris are modelled as Hill-type actuators with series-elastic (SE) tendon elements; simulation outputs are deterministic, but τ cannot be recovered from them by any existing method [16]. (Bottom) The established forward neuromechanical causal chain (τ → activation → muscle force → joint kinematics → observables), the inverse problem.

1.4 Machine Learning as an Inverse Modeling Tool in Biomechanics

In simple terms, the inverse problem of estimating unknown physiological parameters from observable biomechanical signals can be formulated as a learning problem, where high-dimensional input signals are used to predict a specific target parameter [18]. This makes the problem suitable for supervised machine learning regression, in which a model learns the relationship between input signals and target values from labeled training data and then uses this relationship to make predictions on new data.

A key advantage of machine learning in this context is its ability to model complex, nonlinear relationships between biomechanical variables without requiring a predefined mathematical relationship between the inputs and outputs. In the present problem, this relationship is governed by coupled nonlinear processes involving muscle activation dynamics, Hill-type muscle mechanics, and skeletal rigid-body dynamics [19]. Musculoskeletal simulation provides a controlled way to generate labeled data for this supervised learning problem. By running forward dynamics simulations while systematically varying the neural delay parameter, the simulation environment provides the true delay value (τ) for each observation in the dataset. This reduces the label uncertainty associated with supervised learning based entirely on experimental data where ground-truth values may be affected by measurement noise and estimation errors [18].

In addition to ensuring through explorations of the parameter space, simulation-based data generation allows us to study extreme physiologic perturbations that would be unethical, impractical or intractable in human subjects [5] (Figure 3)

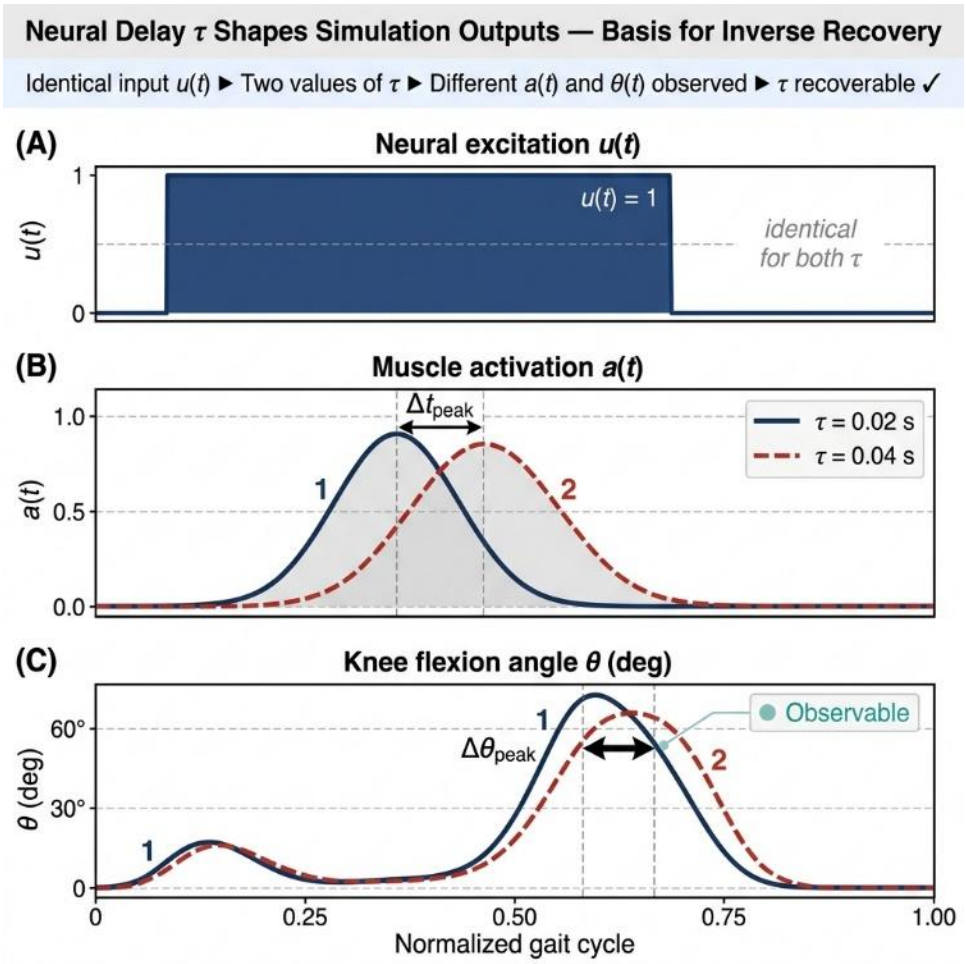


Figure 3: Encoding of neural delay τ in musculoskeletal simulation outputs as the basis for supervised inverse estimation. (A) Identical neural excitation u(t) applied to both simulated conditions, confirming that output differences arise solely from τ. (B) Muscle activation a(t) for two representative conditions (τ = 0.02 s, navy; τ = 0.04 s, crimson dashed); governed by activation (τ_act ≈ 15 ms) and deactivation (τ_deact ≈ 50 ms) time constants [33], larger τ produces a measurably delayed activation peak (Δt_peak). (C) Corresponding knee flexion angle trajectories θ(t) showing the τ-driven kinematic phase shift (Δθ_peak) at the swing-phase peak; this observable discriminability confirms that τ is recoverable from simulation outputs via supervised machine learning.

Ensemble tree-based regression methods appreciably perform both strong predictions and interpretable feature importance scores, with the latter providing results based on the specific contribution of every biomechanical signal to the overall prediction, especially on structured feature spaces created from biomechanical simulation [20]. This interpretability is critical in a biomedical inverse modeling context: for the learned mapping to be scientifically credible, the produced predictions have to both be highly accurate and also point at features whose functional relevance is consistent with what is known about them from the underlying causal structure of the neuromechanical system. If the most important variables turn out to be muscle activation trajectories which is perfectly aligned with the causal chain from neural delay (a value related to how quickly they respond) to activation dynamics, this provides mechanistic justification for the inverse framework above and beyond simple statistics about predictive performance.

More than interpretability, the robustness of the model response to biomechanical perturbation must be evaluated on any inverse neuromechanical framework. While inter-individual variability in musculoskeletal geometry, inertial properties and muscle architecture precludes training a model on simulation conditions and expecting the system to be able to generalize to other biomechanical systems outside of the training environment. The flux of the system induces a direct test of generalization that is similarly representable experimentally through cross-subject validation [21]; evaluating the stability of inverse mapping under systematic perturbation of musculoskeletal model parameters provides this principled test. In summary, the simulation-derived labeled data, ensemble regression with feature interpretability and perturbation-based robustness evaluation all together constitute the methodological basis of the inverse neuromechanical framework proposed herein.

1.5 Research Gap, Objectives, and Contribution of the Present Study

The systematic review in Section 2 demonstrates that no existing method EMG-based, simulation-based or data-driven-reported on a principled mechanism to recover the neural activation delay parameter from observable biomechanical signals under musculoskeletal forward dynamics simulations. The inability of EMG to operate within simulation environments, and the inability for simulation platforms to be able to perform inverse neural parameter estimation defined the key research gap that this work aims to address.

The main goal of this work is to provide a simulation-based proof-of-concept that the latent neural delay parameter τ can be recovered from the biomechanical signals it generates in a realistic, musculoskeletal forward dynamics simulation environment. More specifically, this study will: i) create a simulation dataset with realistic neural delay values (15-60 ms), within ranges observed in experimental literature on healthy neuromuscular function [11], [14], that parametrically span the activation time constant tau denoted above via the Thelen muscle model; ii) use data of joint kinematic and muscle activations produced by these same simulations to train and validate a supervised machine learning regression model to estimate tau; iii) assess which biomechanical signals carry discriminative information for identifying τ in the simulated context; iv) evaluate robustness of the learned mapping against systematic parameter perturbation in musculoskeletal models.

The main scientific contributions of this study are the following. The work begins with the development of an inverse neuromechanical estimation framework that integrates OpenSim forward dynamics simulation along with supervised Random Forest regression to generate a 45,445-sample labeled dataset of biomechanical observations spanning five discrete neural delay conditions. Second, with the use of simulation in a controlled setting, the framework can achieve sub-millisecond recovery precision (MAE ~ 10^{-4} s; R2>0.997). This degree of accuracy matches what would be theoretically predicted from inverting a noiseless set of five discrete τ conditions associated with the finite school first-order ODE system used to warp walking simulations into simulated time series (confirming that the activation dynamics generated by the Thelen muscle model contain enough information about tau to uniquely specify it from all solutions generated by our noiseless inverter.) This is a necessary proof-of-concept to extend towards real-world noisy

data, where the accuracy would be significantly lower [20]. Third, we performed systematic feature importance analysis found muscle activation trajectories and particularly those of rectus femoris to be the dominant carriers of neural delay information in agreement with known theoretical causal structure intrinsic to the Markovian formulation of muscle activation dynamics which yields mechanistic validation for the learned mapping [22]. Fourth, a robustness experiment with 7% perturbation of femur mass indicates that the mapping learned is bounded and structurally resilient to inter-individual biomechanical variability, moving towards potential application of the framework with actual experimental motion data. Overall, these contributions form the first scientific basis for inverse neuromechanical parameter estimation as a valid neural timing analysis. This framework, by positioning the model as the second phase of a modular clinical pipeline wherein static optimization (SO) is computationally used to infer muscle activations prior to ML ingestion has huge potential for subject-specific neuromechanical calibration. The rest of this paper is structured as follows. In Section 2, a systematic literature review of EMD measurement methods, musculoskeletal simulation and machine learning applications in biomechanics is presented wherefrom the research gap is systematically derived. In Section 3 the methodology is described including simulation framework and dataset construction, as well as machine learning model. Section 4 shows and analyzes the experimental results. Section 5 contains more general implications and limitations; Section 6 concludes the paper.

2 Literature Review

Current techniques for the characterization and estimation of neural activation delay in the human neuromuscular system are segregated into three analytically separate domains: (i) experimental electromyography-based methods to directly record electrical muscle activity to infer the time course of neural excitation governing mechanical force production; (ii) physics-based musculoskeletal simulation frameworks that can provide controlled, labeled biomechanical datasets using mathematical representations of whole-body motion systems subjected to external perturbation loads; and iii) data-driven machine learning methods that learn statistical mappings from observable biomechanical signals to physiological quantities of interest. Each domain represents a unique scientific advancement, and an equally unique, irreconcilable limitation. The remaining sections critically review each area in-depth including what each method accomplishes, but just as importantly where they fall short synthesizing their common shortcomings into a single, formally-defined research gap by the end of this review.

2.1 Electromyography-Based Methods for Electromechanical Delay Measurement

Electromechanical delay (EMD) defined as the time interval from electrical muscle recruitment detected by electromyography to actual mechanical force generation, is arguably the most important measurable metric of neural timing literature over the last five decades [23]. Its physiological basis is in a series of electrochemical and mechanical events: sarcolemma action potential propagation, transverse-tubule depolarization, sarcoplasmic reticulum calcium release, cross-bridge attachment and cycling within the contractile element and finally tensioning of the series-elastic components of the muscle–tendon unit before detectable external force occurs [24]. This cascade imposes an irreducible minimum delay between neural command and mechanical output and its duration is variable: it reflects the composite state of the neuromuscular system at any given moment making EMD a dynamically informative physiological parameter rather than a fixed tissue constant [25].

The experimental literature on EMD laid out the basic phenomenology across contraction conditions. Norman and Komi [26] showed that EMD during movement reflects a combination of electrochemical and mechanical processes, with estimates for the lower limb ranging from 25–65 ms under isometric conditions. Cavanagh and Komi [27] compared EMD conditions systematically under both concentric and eccentric contractions, demonstrating that contraction mode modulates delay considerably by changing the muscle mechanical state of the series-elastic tendon. This approach was extended by Zhou et al. [28] who found that, similar to our findings on coactivation and force variability, EMD evoked through voluntary contractions was higher (30–60 ms) than electrically stimulated (10–20 ms) because of central conduction latency contributions. Together, these classical findings established EMD as a multi-component parameter that encoded information about both peripheral tissue mechanics and central neural control strategy.

The mechanistic decomposition of EMD was progressively refined in the following decades. The most definitive computational account for any determinants of EMD to date was provided by Schmid et al. [29] which used a three-dimensional, finite-element muscle model and demonstrated that the motor unit recruitment pattern controlled by the CNS has more than 20 ms delay contribution (not attributable to peripheral tissue mechanics), thus greatly repositioning EMD from a tissue-mechanical artifact into a window on neural control architecture. This theoretical account was bolstered with empirical data by [16] who measured surface EMG, external force, and intramuscular pressure (surface EMG is a metric for electrically active motor units; we use NDI to quantify this concept in terms of muscle activation patterns) as a measure of contractile element tension simultaneously to demonstrate EMD can be decomposed into an electrochemical component (~3 ms from EMG onset to pressure onset) and a mechanical component (~36 Ms from pressure to external force), establishing the series-elastic tendon as the primary mechanically observable contributor to EMD while primarily manifesting a neural component that presents mostly through changes in recruitment timing of motor units.

Recent studies have shown EMD is not static but is dynamically modulated by the CNS as a goal-directed control mechanism. Use of high-density sEMG decomposition by Del Vecchio et al. [25] to revealed that the CNS varies NMD over a wide range (eg, 30 ms of voluntary modulation) to sculpt rate of force development about fine motor tasks — making NMD an active control variable and therefore also an erstwhile unrecognized regulatory variable as opposed to simply a passive outcome trait reflecting tissue mechanics. Krätschmer et al. [30] Under graded isometric conditions, these results were expanded to the quadriceps with a 31% reduction in mean NMD from 175.8 ms at 10% MVC to 121.6 ms at 70% MVC as a function of intensity-dependent motor unit recruitment dynamics. Together, these results place EMD and NMD as biologically meaningful, dynamically regulated signals providing information that can be mechanistically interpreted in the context of CNS control strategy, motor unit composition, musculotendinous stiffness and neuromuscular health status.

Despite these scientific advances, the fundamental limits of measuring EMD via surface EMG are irreducible. Surface electrodes capture the spatial summation of all underlying electrical activity in the muscle fibers that make up motor units activating simultaneously during a recording volume, thus making this signal very sensitive to placement precision and variability from inter-session repositioning as well as skin–electrode impedance effects due to changes such as subcutaneous tissue thickness [3]. Crosstalk from proximal muscles is a common and longstanding confound in multi-muscle measurements of the lower limb, where large muscles located in close proximity to one another generate significant electrical interference in adjacent electrode channels [31]. The application of onset detection algorithms on rectified, smoothed sEMG signals leads to timing errors based on diagnostic thresholds in a range of 10–30 ms—of a magnitude comparable to the neural timing differences under study—and they perform poorly during dynamic tasks due to movement artifacts and low fidelity as illustrated by locomotion [32]. While invasive fine-wire EMG reduces crosstalk and enhances signal specificity, this technique involves the insertion of a needle into muscle, preventing its application in non-invasive clinical monitoring, pediatric assessment or large-cohort longitudinal studies.

An additional and especially important limitation, not always emphasised in the EMD literature, is that surface EMG is inherently incompatible with a computational musculoskeletal

simulation environment. Musculoskeletal simulation platforms represent muscle force production using mathematical Hill-type actuators whose excitation is driven by abstract neural signals that have no functional means for ingesting, processing or interpreting recordings of electrode voltage. This has created a methodological gap: the most powerful controlled, labeled investigation of the parameter space of neural delay—physics-based simulation—cannot leverage EMG data and the most straightforward experimental route to delay measurement—EMG—is unable to enter simulation environments [33]. This exclusivity is the reason for the gap in research highlighted at the end of this review.

Table 1: EMG-based electromechanical delay studies (1979–2025). HD-sEMG = high-density surface EMG; IMP = intramuscular pressure; NMD = neuromechanical delay; MVC = maximum voluntary contraction.

Study	**Year**	**Method**	**Muscle/Task**	**Key Finding**	**Critical Limitation**
Norman & Komi [26]	1979	sEMG + force dynamometer	Lower limb, isometric	EMD 25–65 ms; establishes measurement protocol	No mechanistic decomposition; pre-HD-EMG era
Cavanagh & Komi [27]	1979	sEMG + force	Knee extensors/flexors	EMD varies 20–45 ms with contraction mode	Single muscle; no series-elastic analysis
Zhou et al. [28]	1995	sEMG + force, 3 stim modes	Soleus, isometric	Voluntary EMD > electrically evoked; central latency component identified	Peripheral neural drive only; no in-vivo imaging
Schmid et al. [24]	2019	3D biophysical FE model	Idealized muscle-tendon	CNS recruitment >20 ms of EMD; series-elastic mechanics quantified	Idealized geometry; not subject-specific
Go et al. [16]	2018	sEMG + IMP + force	Tibialis anterior, ramp	Electrochemical EMD ~3 ms; mechanical ~36 ms	Invasive IMP sensor; small n=8
Del Vecchio et al. [25]	2018	HD-sEMG + force	Finger flexors	CNS actively modulates NMD 20–50 ms range	Upper limb only; incompatible with simulation

Krätschmer et al. [30]	2025	HD-sEMG cross-correlation	Quadriceps 10–70% MVC	31% NMD reduction with intensity; motor unit basis	Isometric only; no locomotor generalization

2.2 Musculoskeletal Simulation Frameworks and the Latent Encoding of Neural Delay

Physics-based musculoskeletal simulation has become an essential technique for computational biomechanics, offering the controlled and analytical environments necessary to unravel the interplay of neural control, muscle mechanics and skeletal geometry in observable human movement. OpenSim is an open source platform first described by Delp et al. [9] and greatly extended by Seth et al. [33], is an open-source, modular framework for the optimal control of multi-body rigid-body dynamics with anatomically detailed Hill-type muscle–tendon actuators. It is built with support for inverse dynamics workflows (in which observed kinematics are used to calculate the joint moments necessary to produce them) as well as forward dynamics simulations (where neural excitation patterns through muscle activation dynamics produce contraction (mechanical contraction models), and the record of resulting joint kinematics). Forward simulation of this transformation therefore uses the activation dynamics formulation as a necessary interface, through which neural input is temporally mapped (an important part of the encoding in terms of timing) before modulating motor units to produce force [34].

The Hill-type models based formulations of muscle activation dynamics that was firstly developed by Thelen [34] and succeeded by Millard et al. [35], describes the transition from neural excitation signal u(t) to muscle activation a (t) as a first-order differential equation, with an activation time constant τ(act) ∼15 ms and a deactivation time constant τ(deact) ∼50 ms. These parameters define the rise and decay rates of the activation state due to excitatory neural impulses—essentially coding for a temporal lag in behavior observed between command from neurons and mechanical actuation that is present within the model. Importantly, systematic variation of these time constants over physiologically plausible ranges generates unique and quantifiably distinct temporal activation profiles that impact joint kinematics. This sensitivity has been demonstrated in model calibration studies Hicks et al. [36] specifically showed that improperly calibrated activation dynamics parameters yield systematic timing mismatches between simulated muscle coordination patterns and experimental EMG recordings, and other studies have highlighted the mechanistic sensitivity of simulation outputs to the activation time constants.

Human body musculoskeletal models (HSK) with high-fidelity properties offer simulation/reaction research researchers a wide range of bio fidelity virtual skeletons. 80-actuator Rajagopal et al [37] full-body model conducted a validation across walking and running tasks with joint moment errors lower than 3% RMSE, combined with the computationally efficient 3DGaitModel2354 embedded into the OpenSim environment, has supplied both anatomical infrastructure necessary for repetitive, controlled systematic neuromechanical perturbation studies. Buchanan et al. [31] in 2004, set forth the framework for the four-stage process of EMG-driven

neuromusculoskeletal modelling—activation dynamics, contraction dynamics, musculoskeletal geometry and equations of motion—that still define both simulation as well as hybrid techniques today. Zajac [38] established the theoretical basis for Hill-type muscle–tendon modeling providing both limb-length normalized dimensionless force–length and normalized dimensionless force–velocity relationships that serve as the rationale for all contemporary musculoskeletal simulation actuators. Millard et al. [35] tested several muscle model formulations against comparative data, with small perturbation (Thelen-type Hill models produce identical dynamical behavior for small perturbations within locomotor kinematics, and are biologically realistic over the ranges resulting during locomotion).

Hybrid methods linking calibrated EMG measurements with forward dynamics solvers have shown strong results for online neuromuscular estimation. Real-time emg-driven musculoskeletal modeling framework for generalizable subject-specific estimation of joint moments during novel human movements with data-driven calibration was developed by Durandau et al. [39] and validated on several lower-limb degrees of freedom. Sartori et al. [40] extended this paradigm to the context of personalized neurorehabilitation, where motor neuron firing patterns based on HD-sEMG decomposition directly drive musculoskeletal simulations, effectively closing the loop between physiological and biomechanical levels in the neuromechanical hierarchy. Nevertheless, they both rely on an automatic EMG recording as the input signal driving this first kind of neural processing. Neither framework can make the simulation run without electrode data. That is, the activation dynamics parameters that encode neural delay cannot be recovered from observation of simulation outputs alone: although the simulation can be driven forward from known excitation, it cannot adequately inverted to recover unknown excitation from observable outputs [41].

Another issue affecting the reliability of simulation-based analyses is that uncertainty in musculoskeletal model parameters. Gerus et al. [42] that uncertainty in muscle–tendon parameters can result in knee contact force predictions differing by as much as 2.1 body weights, highlighting the sensitivity of simulation outputs to parameter fidelity. Winter [43] defined the kinematic and kinetic reference standards for human gait on which validation of simulation outputs is mainly anchored in biomechanical literature. Gónçalves et al. [44] represents a rigorous cross-platform benchmark of muscle model validation in multibody dynamics environments. Among this body of validation work, however, there has been no study quantifying how uncertainty in activation dynamics time constants τ_{act} and τ_{deact} propagates into kinematic outputs and if such kinematic sensitivity is sufficiently strong to allow for inverse estimation of the neural delay parameter from those output metrics. To the best of our knowledge, the inverse simulation problem has not been treated at all in the literature.

Table 2:Musculoskeletal simulation studies and their treatment of neural delay. MSK = musculoskeletal; V&V = verification and validation; RMSE = root mean square error.

Study	**Year**	**Platform**	**Key Contribution**	**Activation Dynamics Role**	**Gap re. Neural Delay**
Zajac [38]	1989	Hill-type theory	Foundational muscle-tendon model & scaling laws	Implicit: f-l-v relationships	τ not identified as estimable parameter

Thelen [34]	2003	OpenSim Thelen2003	First-order activation dynamics, τ_{act}~15 ms	Explicit: τ governs u(t)→a(t) transform	τ is an input; no inverse estimation possible
Delp et al. [9]	2007	OpenSim	Open-source simulation platform	Supported; fixed per model	Forward-only; no τ recovery from outputs
Buchanan et al. [31]	2004	EMG-driven MSK	4-stage neuromusculoskeletal modeling framework	Stage 1: EMG→activation dynamics	EMG required; τ calibrated, not estimated from kinematics
Seth et al. [4]	2018	OpenSim extended	Neuromuscular control & animal movement	Embedded in Hill actuators	Outputs encode τ implicitly; no inversion framework
Rajagopal et al. [37]	2016	Full-body 80-actuator	Validated gait model <3% RMSE joints	Standard τ values applied	No investigation of τ sensitivity on kinematics
Durandau et al. [39]	2018	Real-time EMG-driven	Cross-task generalization of joint moments	EMG-driven activation	EMG required as input; τ cannot be recovered without it
Millard et al. [35]	2024	Muscle model benchmark	Comprehensive validation across perturbation sizes	Extended Thelen model tested	τ variation not studied systematically for inversion

2.3 Machine Learning Approaches to Biomechanical Signal Analysis

In particular, the application of machine learning to biomechanical signal analysis has witnessed a major methodological transformation over the last decade and a half, moving from traditional feature-engineering methods combined with shallow classifiers to deep end-to-end architectures capable of automatically discovering spatiotemporal representations directly from raw multi-channel time-series data [45]. This evolution has been stimulated by two converging forces: the increasing access to large traditional biomechanical datasets acquired via optical motion capture, wearable inertial sensors and force platforms; and the observation that nonlinear, high-

dimensional mappings from neural control signals to observable joint kinematics and kinetics cannot be appropriately described using linear regression or parametric physiological models alone. We can divide the literature into three technically-stable subdomains based on the architecture of the learning system and the nature of the learning target.

The earliest literature on EMG-based biomechanical estimation had rather classical ensemble and kernel methods. Support vector regression (SVR) was used very extensively between 2012 and the end of 2020 to estimate continuous angles and torques [46]. SVR was consistently able to outperform single decision trees and linear regressors in terms of cross-validation benchmarks for isometric torque prediction but its strict dependence on manually engineered features, sensitivity to kernel hyperparameters, and limited scalability with respect to dimensionality and processed input space complexity restrained abusive application under dynamic, multi-joint conditions. Random Forest regression [47], originally from the work of Breiman but with a modern ensemble flavor, gradually replaced SVR as the foremost classical choice for biomechanical feature-based regression tasks because it is robust to heterogeneity of the training set, insensitive to the scaling of features and able to provide vestibularized, out-of-bag measures of feature importance that would indicate which biomechanical variables are most likely causative determinants of consistency or variation in target quantity being predicted [48].

Since mid-2018 deep learning architectures have made significant improvements in the state-of-the art biomechanical time-series estimation from IMU data. In their study, long short-term memory (LSTM) networks and bidirectional LSTM architectures were used to continuous estimate hip, knee and ankle joint kinematic from IMU and sEMG time-series inputs—taking advantage of the recurrent unit's ability to learn temporal dependencies over an entire gait cycle that spans roughly 1 second at normal walking speed [49]. Previous studies using 3D printed mechanical devices have reported Pearson correlation coefficients above 0.90 between LSTM-predicted and optically measured joint angles, indicating that the temporal structure of the kinematic trajectory is learnable purely on the basis of wearable sensor data in laboratory conditions. Even gait phase detection and event identification from IMU data have been implemented with convolutional neural networks (CNNs), achieving over 97% classification accuracies in controlled environments with sliding-window multi-channel approaches [50]. Hybrid CNN–LSTM frameworks, which provide a unified model of spatial feature extraction and temporal sequence modeling in one end-to-end architecture, have shown better generalization performance on pathological populations like Parkinsons disease and cerebral palsy [51].

The physics-informed machine learning paradigm has recently emerged as a promising and both evidence-based and data-efficient fusion of computational biomechanics/statistics with the modern direction of statistical learning. Using forward musculoskeletal simulations as a synthetic data generation engine, researchers can build training datasets where every biomechanical observation is labeled with the specific, noise-free physiological parameter that produced it - or ground-truth labeling — otherwise structurally inaccessible to experimental data collection in which target parameters must be estimated from imperfect measurements [52]. It allows systematic, controlled mapping of physiologically relevant parameter spaces including rare pathological conditions, pediatric body proportions and surgically modified anatomies unavailable from large experimental cohorts. Durandau et al. [39] as an initial example of simulation-informed learning in neuromusculoskeletal modeling, demonstrated that arrival at a Hill-type model

calibrated on subject-specific EMG enabled generalization of moment predictions across untrained dynamic tasks. No existing study has, however, adopted this paradigm within the framework of physics-informed estimates specifically to inverse estimation of the neural delay parameter: the forward simulation serves as a labeling engine for learning the inverse mapping from kinetic outputs to latent neural timing parameter.

Feature importance analysis, a native outcome of ensemble tree-based methods, has been used in biomechanical regression studies to estimate each individual input signals relative predictive contributions to the target variable estimations. Phinyomark et al. [53] used a systematic approach to identify the informativeness of EMG time-domain and frequency-domain features for neuromuscular classification tasks which showed that activation-onset-related temporal features are disproportionately informative in tasks with temporally meaningful estimates such as those requiring estimation of duration parameters. If muscle activation trajectories, not just kinematic signals, carry the greatest amount of neural timing information then this may have mechanistic implications for solving the neural delay estimation problem since it is consistent with a causal structure that follows from the activation dynamics formulation where τ represents a rate parameter dominating the transfer function via which excitation transfers to activation. However, this hypothesis has never been tested specifically in the context of our simulation-derived neural delay estimation method and there is also no existing study that characterizes its feature importance landscape.

Table 3:Machine learning methods for biomechanical signal estimation. sEMG = surface EMG; IMU = inertial measurement unit.

Study	**Year**	**Architecture**	**Input Signal**	**Target Variable**	**Performance**	**Key Limitation**
Breiman [47]	2001	Random Forest	Generic features	Classification / regression	Foundational: low variance, interpretable	Requires labeled data; static features
SVR biomechanics [46]	2012 – 2020	SVR	sEMG time/freq features	Joint torque/angle	Competitive on small datasets	Kernel tuning; poor scalability; EMG mandatory
Durandau et al. [39]	2018	EMG-driven + calibration	sEMG + kinematics	Joint moments (real-time)	Cross-task generalization demonstrated	EMG required; τ not estimated
LSTM/BiLSTM [49]	2019 – 2022	Recurrent LSTM	IMU/sEMG time-series	Joint angle trajectories	$r > 0.90$ in lab conditions	Large corpus needed; poor interpretability

CNN + IMU [50]	2020 – 2023	1D/2D CNN	IMU sliding windows	Gait phase (classification)	>97% accuracy in lab	Cross-subject drop; activity-specific
CNN–LSTM hybrid [51]	2021 – 2023	CNN + LSTM end-to-end	Multimodal sensor	Joint kinematics / pathology	Improved clinical population accuracy	High data requirement; black-box
Phinyomark et al. [53]	2012	Feature analysis	sEMG features	Classification	Temporal features most informative	No forward-to-inverse application

2.4 Cross-Domain Analysis: Complementary Capabilities and Shared Deficiencies

As the review of methodological domains above reveals, their technical capabilities do not overlap, displaying structured complementarity: equally novel and structured constraints are established through complementary limitations that create a definable methodological vacuum around inverse estimation of neural delay. This cross-domain analysis across eight capability dimensions of direct relevance to the neural delay estimation problem is formalized in Table 4.

Table 4:Cross-domain capability matrix for neural delay estimation. RF = Random Forest

Capability Dimension	**EMG-Based Methods**	**Musculoskeletal Simulation**	**Machine Learning Methods**
Direct access to neural excitation signal	Yes (via electrode recordings)	No (u(t) is a model input)	No (requires external labeling)
Neural delay τ as explicit estimable target	Partially (indirect, noisy)	No (τ is input, not output)	Not attempted in literature
Compatible with simulation environment	No (hardware incompatible)	Yes (native)	Yes (if simulation-derived data used)
Requires electrode hardware	Yes (mandatory)	No	No (when simulation-driven)
Provides exact ground-truth labels for τ	No (τ estimated, not known)	Yes (τ is the specified input)	No (requires external labeling source)
Systematic parameter space coverage	No (experimental cost)	Yes (unlimited computation)	Yes (once training data provided)
Interpretable signal-to-delay mapping	Partially (biophysical models)	Not investigated	Yes (feature importance in RF)

Demonstrated cross-population generalization	Partially (aging, fatigue)	N/A (no inverse established)	Partially (limited to kinematic targets)

As shown in Table 4, it reveals a precise methodological alignment problem. Methods based only on EMG gain direct access to neural excitation, but cannot be used to label simulation data, and do not function without electrode hardware. Thus, even though simulation provides us an exact, noise-free ground-truth labels for τ, they are the only means of providing such data and cannot be used to solve the inverse problem that would use those labels for training. The only logical class of method that could learn to invert the mapping between observable signals and τ are machine learning methods, but these require labeled training data which can really be provided only by simulation. The answer is obvious: these three domains need to couple, where simulation gives labels and machine learning learns the inverse mapping so that the emerging modular model converts our computational estimates of internal states (such as SO-derived activations) into latent neural parameters without relying solely on invasive sensing of localized tissue. However, none of the previous work are built upon this integration for neural time delay estimation. We have a good formulation of the forward chain ($\tau \rightarrow$simulation outputs) but we can say nothing about the inverse chain (simulation outputs$\rightarrow \tau$).

2.5 Historical Trajectory of the Field and the Outstanding Research Gap

Neural delay research follows an identifiable intellectual trajectory that maps onto four phases of mythological maturity with each successive expansion of the scientific understanding of the problem leaving its ultimate inverse formulation untouched. The first phase (1979–1999) consisted of preliminary experimental studies characterizing the phenomenology of EMD itself in terms of its dependence on contraction mode, intensity and muscle identity using single-channel surface EMG and isometric force dynamometers [13], [54], [55]. Although these studies showed that EMD is a realistic, measurable and physiologically variable parameter, no mechanistic explanation of its internal structure or mathematical relationship with muscle model parameters was provided. During a second phase from 2000 to 2012, the original Hill-type musculoskeletal simulation platforms were developed that incorporated activation dynamics formulations as an explicit computational representation of the neural delay, and EMG-driven modeling frameworks provided typically higher anatomical fidelity connections between excitation generated by neurons and biomechanical output [19], [31], [56].

During the third phase (2012-2020), machine learning applications in biomechanics accelerated eightfold, yielding high-accuracy joint kinematic and kinetic estimations from both wearable sensor data and sEMG. Quantifying this output, Random Forest, SVR, LSTM and hybrid deep architectures showed that observable biomechanical signals are rich and learnable sources of information regarding the underlying neuromuscular state [46], [47], [49], [50]. At the same time, real-time EMG-driven musculoskeletal models had become sufficiently fast enough for clinical use [39], [40]. Yet, throughout this large machine learning literature on biomechanical signals, investigators systematically selected observable kinematic or kinetic quantities as learning targets—the latent neural control parameters whose variation was responsible for producing those

observables were never chosen. The fourth and ongoing phase (2020–present) has been a contemporary emergence of Physics-Informed machine learning explicitly as new paradigm, in which we treat simulation data as labeled training supervision for models that learn to generalize to experimental conditions [52]. This paradigm provides the computational framework which is in fact needed for casting and solving the inverse neural delay estimation problem. The biophysical motivation is well established now [29], [2], [11]; the simulation infrastructure is mature and validated already [14, 15, 19]; the machine learning methods are capable [47], [49]; but the integration of all these approaches targeting neural delay has not be done yet.

Table 5:Historical evolution of the neural delay research field across four methodological phases. Gap: the inverse neural delay estimation problem remains unaddressed in all four phases.

Phase	Period	Key Advances	What Remained Unresolved
Phase 1: EMD Phenomenology	~1979–1999	EMD measured via sEMG+force; contraction-mode dependence established [26], [15], [28]	No decomposition of neural vs. mechanical EMD; no simulation representation
Phase 2: Simulation Frameworks	~2000–2012	Hill-type activation dynamics formulation; EMG-driven modeling; OpenSim release [9], [19], [31], [56]	Forward-only; no inverse estimation of τ from biomechanical outputs
Phase 3: Machine Learning Expansion	~2012–2020	RF, SVR, LSTM, CNN for joint kinematics; real-time EMG-driven models [56], [46], [47], [49], [50]	ML targets kinematic outputs; latent neural parameters never used as learning target
Phase 4: Physics-Informed ML	~2020–present	Simulation-supervised learning paradigm; HD-sEMG NMD quantification [2], [11], [18]	Integration for neural delay inverse estimation: τ as ML target from simulation data—not yet executed

Among all the dimensions to consider within this gap, perhaps the most salient scientific implication is that the forward neuromechanical causal chain $\tau \rightarrow$ activation dynamics $\rightarrow$ muscle force $\rightarrow$ joint kinematics has never been inverted. The question of whether the biomechanical outputs that are observable from musculoskeletal forward simulation contain enough discriminative information about initiating neural delay parameter to permit accurate recovery is ultimately a fundamental, and open scientific question. If that is indeed true, which the theoretical sensitivity of forward simulation outputs to activations time constant variation would suggest [19],

[36], then we are on our way to a physics-grounded computational method of neural timing analysis that has both two-stage and population scale capabilities, virtual sensing that bridges macroscopic kinematics and internal neural delay. If the answer is no, then limits on observable neural timing parameters from biomechanical signals would be established with important consequences for future designs of neuromechanical monitoring systems. Either outcome would be considered a major contribution to science and we do not know which is the case now.

3 Methodology

This study investigates whether the neural activation delay parameter τ which acts as a latent control variable within the Thelen muscle activation model in musculoskeletal simulation can be recovered from the biomechanical signals. It produces in a controlled, noiseless simulation environment, using a supervised machine learning inverse modeling approach. This study does not use experimental EMG data; all biomechanical observations are produced by the OpenSim forward dynamics pipeline below. To test this hypothesis, a simulation dataset was created from a detailed musculoskeletal model where neural delay was simulated by altering muscle activation dynamics. This study then processed the output to create a complete dataset of joint kinematics and muscle activation during known neural delay conditions. These biomechanical signals were used to train machine learning models in order to predict the underlying neural delay parameter. Besides baselines inverse estimation experiments, additional robustness analyses were made with biomechanical perturbations and noise injection in order to evaluate the stability of the proposed framework. Overall approach of musculoskeletal simulation, data-driven modeling and systematic validation experiments for inverse neuromechanical parameter estimation.

3.1 Musculoskeletal Simulation Framework

The biomechanical dataset used in this study was generated through physics-based musculoskeletal simulations. The simulation framework integrates anatomical musculoskeletal modeling with physiologically realistic muscle activation dynamics in order to reproduce the mechanical behavior of the human lower limb during locomotion. By controlling specific parameters governing muscle activation dynamics, the simulations enable systematic investigation of how neural timing influences observable biomechanical signals. The simulated signals are the building blocks for creating the dataset to use in the inverse neural delay estimation framework.

3.1.1 Musculoskeletal Simulation Platform

The biomechanical simulations in this study were implemented within OpenSim (version 4.5), a free open-source software for the modeling and simulation of musculoskeletal systems. OpenSim offers an all-inclusive framework for integration of multi-body skeletal dynamics with detailed, physiologically-accurate muscle actuation models and is thus an excellent tool to investigate neuromechanical coupling in the context of human movement [57]. By integrating anatomical structure with physics-based equations of motion, the platform allows human locomotion simulations under precise experimental control [58].

The present work employed OpenSim to simulate a controlled dataset connecting unseen neural features to measurable movement-based outputs. The simulations are based on a formulation of forward dynamics, where muscle forces and external loads act to produce motion of the skeletal segments. Thus, joint trajectories are calculated directly from the underlying force equations of motion, which guarantees physical consistency with Newtonian kinematics [21]. This

property is important for creating high-quality ground-truth datasets to screen inverse modeling tasks.

The simulation environment was concerned with the right lower limb and this used 3DGaitModel2354 musculoskeletal model to simulate human gait dynamics. Key Features: The model contains well defined representations of the lower extremity segments and musculature commonly used in gait analyses. This study links muscle activation patterns with the resulting joint kinematics (hip and knee flexion dynamics) by simulating motion within this biomechanical framework. This allows to modulate the effect of differences in the timing of neural activation on the mechanical behavior of the musculoskeletal system during locomotion [22]. As shown in Fig. 4, the musculoskeletal model implemented in OpenSim was used to simulate gait and generate the dataset used for inverse neural delay estimation.

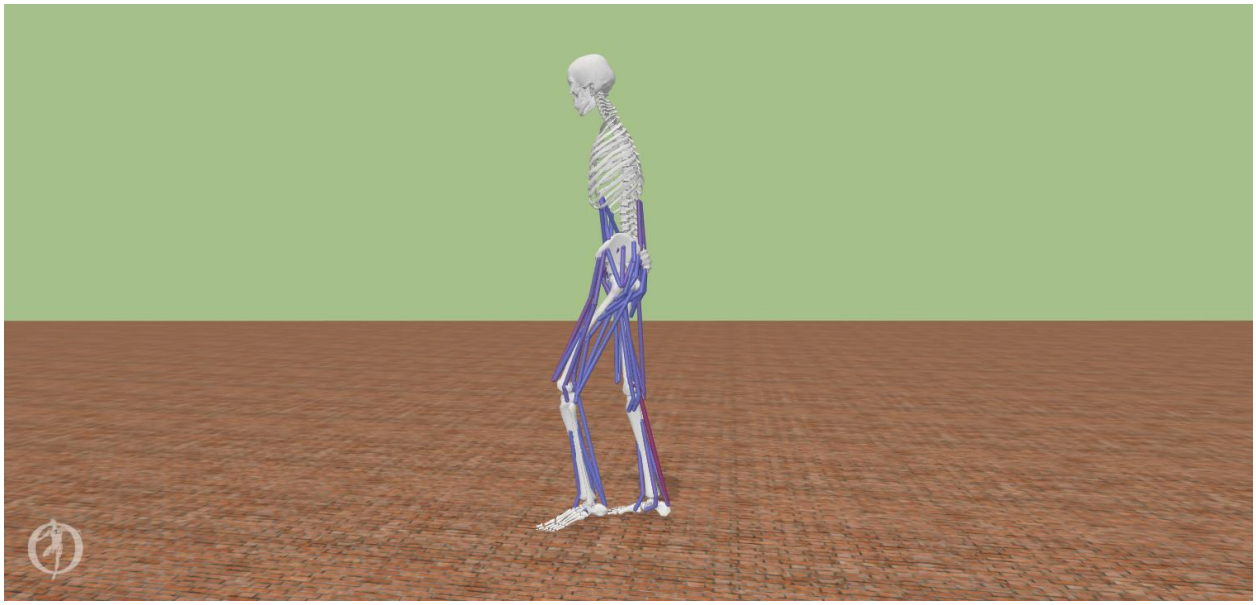


Figure 4:The musculoskeletal simulation environment implemented in OpenSim, illustrating the lower-limb musculoskeletal model and muscle paths employed in the gait simulations used to generate the dataset for inverse neural delay estimation.

3.1.2 Musculoskeletal Model Description

The simulations performed in this study using the OpenSim framework to serve as a digital counterpart of the 3DGaitModel2354 musculoskeletal model. This model has extensively been implemented in gaitanalyses and neuromuscular control, enabling a comprehensive representation of lower-limb biomechanics. We created a model with several rigid body segments (pelvis, femur, tibia, and foot), connected by kinematic Joints corresponding to the degrees of freedom at the human lower limb joints. These anatomically-grounded models allow for the simulation of realistic motion of skeletal elements due to musculo-skeletal forces [57].

This model includes 3DGaitModel2354 and combines anatomically realistic muscle paths with physiological parameters that govern muscle–tendon behavior. These properties enable the model to represent the dynamic interaction between muscle contractions and skeletal movement

during locomotion. Since the musculoskeletal geometry and inertial properties are explicitly encoded, joint kinematics remain physically consistent with the equations of motion at a given movement. In this study, the analysis focuses on the right lower limb, specifically targeting three muscles selected for their critical roles in modulating knee and hip joint trajectories:

- **Rectus femoris**, a bi-articular muscle contributing to both hip flexion and knee extension.
- **Vastus intermedius**, a mono-articular muscle acting as a primary knee extensor and stabilizer during the stance phase.
- **Biceps femoris long head**, a key antagonist muscle that facilitates knee flexion and hip extension, contributing to the control of the swing phase and deceleration of the lower limb.

The resulting joint kinematics of these joints during simulation gait are determined by the coordinated activation of these muscles. The timing of neural control signals has a strong influence on the activation patterns of these muscles, since they span both hip and knee joints. Parsing these muscle activation signals along with the associated joint angles and angular velocities, the simulations form a physiologically realistic database that acts as primary inputs to the inverse model framework described in this study.

3.1.3 Muscle Activation Dynamics

In the simulations, muscle force generation is defined using the Thelen2003 Muscle Model in a Hill-type muscle–tendon formulation implemented with OpenSim as an optimal simulation approach. It offers a reduced-order, physiologically justified approximation of the connection between neural excitation and muscle activation, as well as the mechanical force generated by the muscle–tendon unit. With activation dynamics in conjunction with force–length and force–velocity relationships, the model can simulate realistic muscle behavior during dynamic movements like walking.

The Thelen muscle formulation consists of three main components: a contractile element that mimics the behavior of actin and myosin filaments within the sarcomere, a parallel elastic element, and a series elastic tendon element that absorbs energy through deformation during stretch. In this formulation, the contractile element accounts for active force generation by muscle fibers, whereas elastic elements abstract passive mechanical properties originating from connective tissues and tendons. Collectively these are the components that determine how joint kinematics and muscle activation relate to the resulting torques at each joint that cause skeletal motion.

Modeling the dynamics of muscle activation — which describes how these neural excitation outputs are transformed into the performance of muscular contractions over time — is an important part of this formulation. There are activation and deactivation time constants that dictate how quickly muscle acts based on neural control. As muscle activation does not change instantaneously, these dynamics create a lag between neural excitation and the mechanical response of the muscle that is dependent on time.

Muscle activation dynamics are especially important under this context of simulation due to the fact that they provide a way to indirectly characterize the timing of neural control in musculoskeletal simulations. Differences in the activation dynamics are important for controlling when and how much force is exerted by these muscles, ultimately resulting in differences in the lower limb joint kinematics. Investigating these biomechanical responses, we could explore how neural timing parameters are reflected in observable musculoskeletal motion.

3.2 Representation of Neural Delay

While musculoskeletal simulations generate tension and movement of skeletal motion as well as from muscle, they lack direct representations of neural conduction processes. In any neuromuscular systems neural commands produced by the central nervous system take a finite amount of time to travel through motor neurons to produce mechanical responses in muscles. This phenomenon is called Electromechanical Delay (EMD) and leads to a delay between muscle activation (neural excitation) and tension development in skeletal muscle. This delay has a significant impact on the timing at which muscle force is produced, and, as a result, affects the biomechanical movement of the musculoskeletal system.

Neural delay was indirectly represented via the modification of the muscle activation dynamics within a physics-based musculoskeletal simulation framework to examine how neural timing can affect advantages and disadvantages. In particular, τ, the time constant to respond was used in the activation dynamics of converting neural excitation into muscle activation. This method produces a physiological delay between neural input signals and mechanical muscle response while still being compatible with the Hill-type muscle models already implemented in this simulation environment.

3.2.1 Mathematical Formulation of Activation Dynamics

In this study, neural delay is represented through the activation dynamics formulation of the Thelen muscle model [41]. The relationship between the neural excitation signal $u(t)$ and the resulting muscle activation $a(t)$ is governed by a first-order differential equation describing the rate of change of activation:

$$\frac{da(t)}{dt} = \frac{u(t) - a(t)}{\tau(u, a)} \tag{3.1}$$

Where:

- $u(t)$ represents the neural excitation signal,
- $a(t)$ represents the muscle activation state,
- $\tau(u, a)$ represents the activation time constant governing the rate of transition between neural excitation and mechanical activation.

In this formulation, the neural delay parameter tau is a latent temporal control variable controlling the transition between neural excitations and mechanical muscle activations. As a result of this, changes in tau lead to unique differences in the signals from the muscles over time. Since the simulation is noise free and that our ODE is first-order, an important note to highlight here about the signal of activation a(t) for fixed excitation u(t): it is deterministic with respect to tau. This means that the inverse estimation problem considered in this work can be mathematically phrased as learning the inverse of a mapping associated with this ODE from data, which is well defined and non-ambiguously so within the noiseless simulation setting. This would be fundamentally different from performance characteristics and realizability of this inversion on real EMG noise induced by biological motion, which constitutes a further research problem. The time constant $\tau(u, a)$ depends on whether the muscle is activating or deactivating and is defined as:

$$\tau(u,a) = \begin{cases} \tau_{\text{act}} \cdot (0.5 + 1.5a) & \text{if } u > a \\ \dfrac{\tau_{\text{deact}}}{(0.5 + 1.5a)} & \text{if } u \leq a \end{cases} \tag{3.2}$$

Here, τ_{act} and τ_{deact} represent the activation and deactivation time constants respectively. These parameters control how quickly muscle fibers transition from a resting state to an active force-producing state.

Variations in these parameters introduce a temporal delay in the development of muscle activation. This delay propagates through the Hill-type muscle-tendon model, influencing the timing of muscle force generation. The resulting muscle force F_{mt} can be expressed as:

$$F_{mt} = a(t) \cdot F_{m0} \cdot \left[f_l(l_m) f_v(v_m) + a_p \right] \tag{3.3}$$

where:

- F_{m0} represents the maximal isometric force,
- $f_l(l_m)$ represents the force-length relationship,
- $f_v(v_m)$ represents the force-velocity relationship,
- a_p represents passive muscle force components.

Changes in the neural delay parameter thus influence the temporal variation of excitation over time, causing a modification of the muscle activation dynamics and therefore a different (pair or couples) force output profile and joint torques during motion.

3.2.2 Neuromechanical Pathway and Simulation Framework

In this study, the conceptual neuromechanical pathway used is illustrated in Figure 5. Neural excitation signals $u(t)$ are transformed into muscle activation through the activation dynamics described above. The muscle activation generated is then used as input into a Hill-type muscle model to produce force, which affects the skeletal segments of the musculoskeletal model.

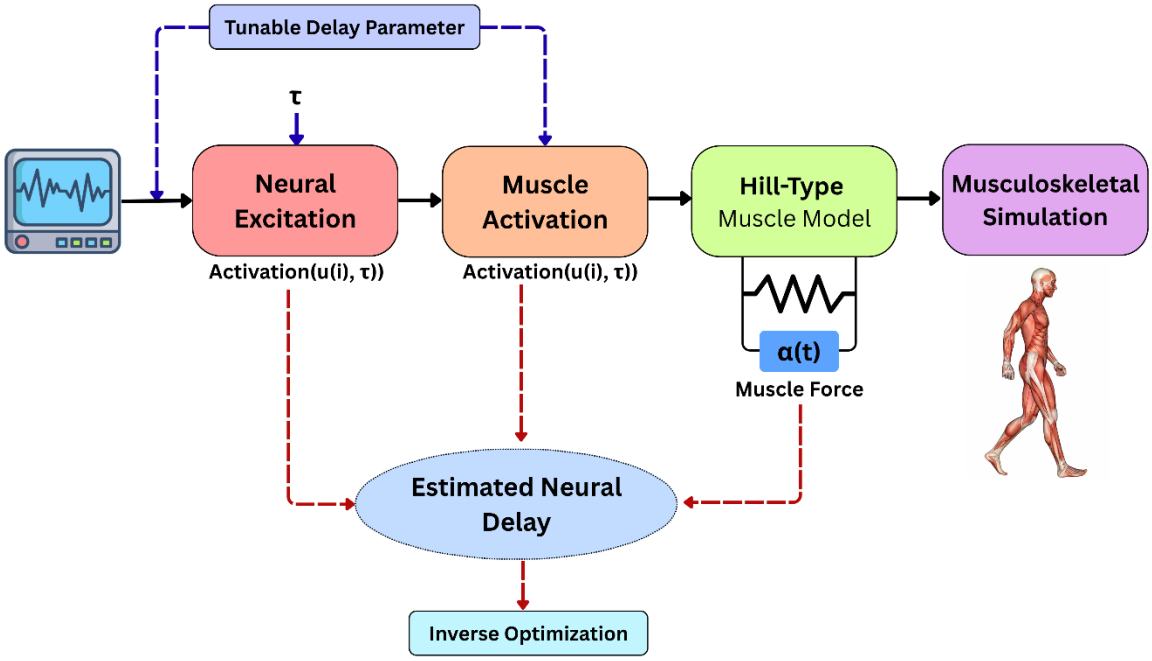


Figure 5:Neuromechanical modeling framework.

Features of the framework are shown where it couples causal chain from neural excitation to muscle activation, force generation, and kinematics at joints in a musculoskeletal simulation. One purpose of the inverse modeling framework developed in this study is to provide a way to estimate the neural delay parameter τ from the underlying observable biomechanical signals arising as a consequence of this process. Using this framework with τ as a varying parameter was used to simulate multiple neural delay conditions. From these simulations, we generated datasets of kinematics and muscle excitation signals associated with known values of neural delay which is showed in table 6.

Table 6: Neural delay parameter values used in the simulation matrix and their physiological interpretation.

Neural Delay Parameter (τ)	**Clinical Description**	**Physiological Context**
0.015 s	Healthy baseline	Consistent with high-performance neuromuscular response [59][16]
0.020 s	Standard delay	Typical value used in musculoskeletal simulations [16]
0.030 s	Moderate lag	Observed during early fatigue states [55]
0.040 s	Increased delay	Associated with neuromuscular aging [59]

0.060 s	Pathological delay	Observed in sarcopenia and neuropathy [29]

3.2.3 Validation of the Simulation Matrix

The selected delay range between 15–60 ms was chosen based on physiological, computational, and machine learning considerations.

First, due to differential muscle condition (e.g. archimedes more than uni-joint), contraction type (eccentric more than concentric) and neuromuscular health, physiological studies indicate that electromechanical delay differs significantly. In healthy young adults, the EMD is generally around 10–30 ms while delays longer than 50–60 ms have been found with older populations [59], in a state of muscle fatigue or neuromuscular disorders. The chosen range, then, covers the spectrum from sound neuromuscular competence to delayed pathologic states.

Second, looking at the equation for activation dynamics numerically, we can see that one of its parameters—the delay—functions as a temporal filter: it dictates the speed with which muscle activates. Choosing delay values across the physiological range guarantees that the simulation generates a clear biomechanical response which does not lead to numerical instability in the integration scheme used within the musculoskeletal simulation framework [60].

Finally, as proposed by recent developments of physics-informed modeling from the machine learning perspective, one can learn through a set of discrete meaningful values of system parameter that regression models capture the underlying continuous relationship between system parameters and observable outputs [29]. Trained on representative delay values across the observably reasonable range of the system, the learning model can interpolate previously unseen intermediate neural delay values and recover latent neural parameter estimates from biomechanical signals. Because the neural delay parameter enters the activation dynamics equation as a continuously time-scaling coefficient, the resulting biomechanical responses vary smoothly with respect to changes in τ. This in turn means that by sampling a limited number of evenly-spaced parameter values over the physiological range, enough information is determined for the learning regime to map an approximate continuous relationship from biomechanical features and observations to its mapping on a latent neural delay parameter.

Therefore, the chosen delay values represent a reasonable trade-off between being physiologically feasible and computationally tractable while also ensuring that the model has good generalization ability to simultaneously develop an inverse modeling framework capable of estimating neural delay via musculoskeletal motion data.

3.3 Forward Dynamics Simulation

This section explains the forward dynamics simulation protocol for generating the biomechanical data used in the inverse neural delay estimation framework. These simulations were completed in the OpenSim musculoskeletal modeling environment discussed previously. Forward simulations were performed systematically adjusting the neural delay parameter within the muscle activation dynamics formulation to produce biomechanical signals associated with specific neuromechanical states.

Forward dynamics simulation provides a physics-based method to predict skeletal motion, driven by muscle forces. Within this framework, a neural excitation signal is processed through the activation dynamics model to produce muscle activation and then muscle forces are applied to skeletal segments via joint constraints acting on these skeleton segments according to rigid-body dynamics equations. Simulated joint trajectories are biologically compliant to the anatomical and biomechanical model since they are resolved numerically in this simulation environment.

The simulation was set up in the OpenSim graphical interface, specifying muscle parameters and other simulation specifications just before running a forward dynamics solver. Specifically, the indirect additive time delay of rectus femoris (the key muscle from Thelen model that controls temporal behavior of muscle response) was used to implement neural delay. The OpenSim simulation environment used to set up the musculoskeletal model configuration and gait simulations performed in this work is illustrated in Figure 6.

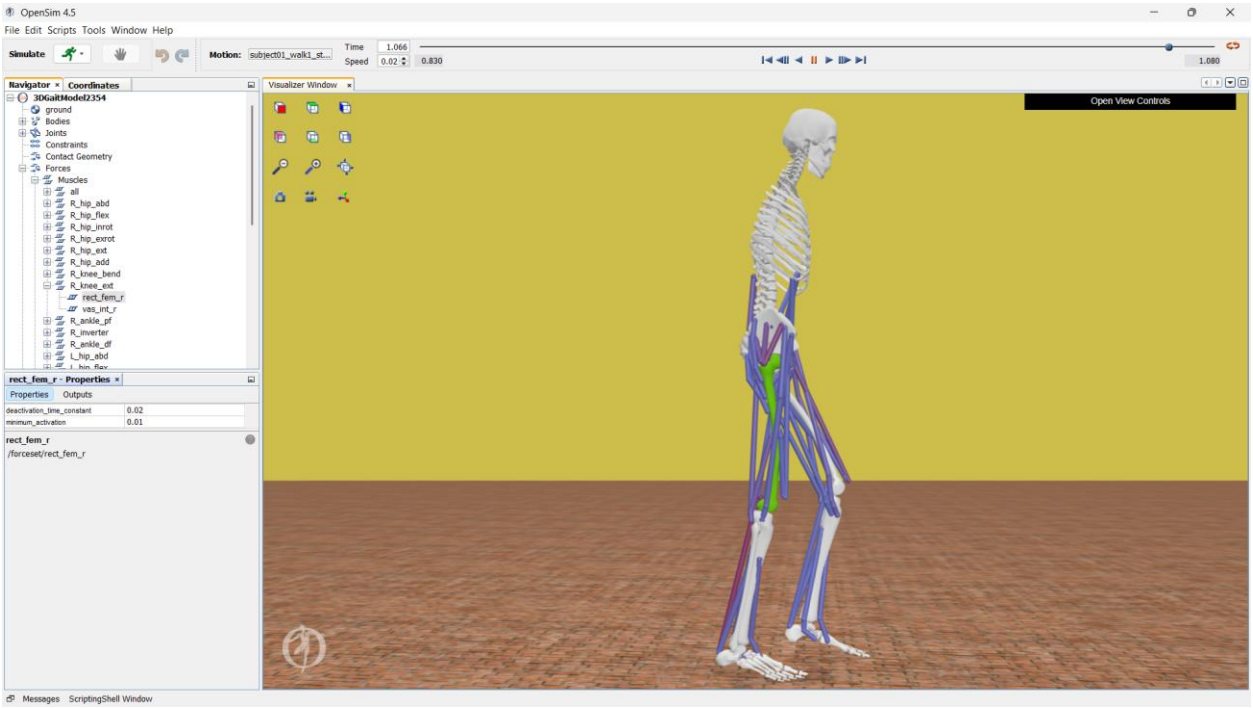


Figure 6: OpenSim simulation environment used for forward dynamics analysis. The 3DGaitModel2354 musculoskeletal model is configured within the OpenSim graphical interface to simulate lower-limb biomechanics during gait. The femur segment, shown in color green, is the anatomical region of interest where the rectus femoris muscle a) originates and b) where modification of the activation dynamics is applied. This visualization helps to visualize where the paired neural delay parameter has been altered in the musculoskeletal model.

For each selected neural delay value τ, the parameters of the OpenSim model were set and then a forward dynamics simulation was performed. At each simulation, muscle-generated forces propelled the skeletal segments, yielding time-resolved joint kinematics and muscle excitation trajectories over the simulated gait cycle. The OpenSim solver logged the dynamic state of musculoskeletal system at every integration time step to create a detailed record of biomechanical variables that captured the changes in motion in each time step.

The raw outputs from the simulations were exported as state trajectory files consisting of approximately 150 biomechanical variables, representing joint angles, angular velocities, muscle activation signals, muscle-tendon forces and internal state variables of the musculoskeletal system. One can think of those as a complete description of the simulation state, but not all variables can be immediately relevant to neural delay estimation.

Thus, a dataset construction stage was conducted to keep only the most relevant biomechanical signals in context of lower-limb neuromechanics for machine learning analysis in the next section. The selected variables are joint kinematics of the hip and knee together with activation trajectories of the rectus femoris, vastus intermedius, and biceps femoris muscles. Using these physiologically meaningful signals, the constructed dataset encompasses biomechanical impacts of neural timing variations and reduces redundancy in the simulation output space.

Figure 7 provides a summary of each step in the dataset generation workflow, you start with a neural delay parameter modification, then you forward simulate and create your dataset construction.

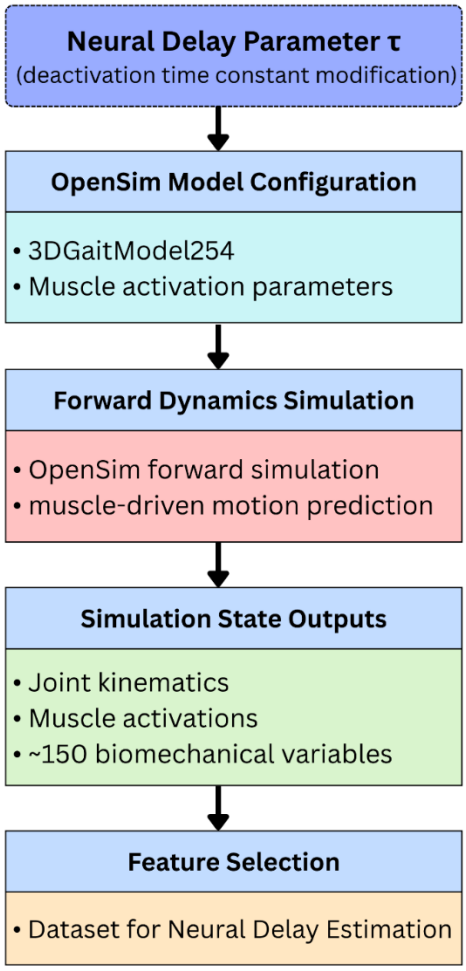


Figure 7:Biomechanical dataset generation pipeline using forward dynamics simulation. The existing activation dynamics of the rectus femoris is modified to include the neural delay parameter τ. The muscle motion of the musculoskeletal system then calculated by means of OpenSim forward dynamics solver. The simulation state outputs include around 150 biomechanical variables and are

exported to be processed by feature selection in order to create the data set for neural delay estimation.

3.4 Dataset Construction and Feature Extraction

In this sub-section, we processed the raw simulation outputs into a structured dataset for machine learning–based neural delay estimation based on the forward dynamics simulations described in Section 2. The time-resolved state trajectories that described the dynamic behavior of the musculoskeletal system during each gaits simulated by OpenSim were obtained from all configurations. Within these trajectories are joint kinematics, muscle activation signals, muscle-tendon forces, and other internal solver variables that describe the dynamic state of the musculoskeletal model.

The raw simulation outputs consisted of roughly 150 biomechanical variables per time step that captured the full dynamic state of the musculoskeletal system simulated with the OpenSim forward dynamics solver. Although all of these variables give a complete description of the simulated motion, not all of them are useful for estimating delays in neural activation directly. Automating joint parameters are often intermediate states of the numerical integration process (corresponding to muscle-tendon slack length) or redundant biomechanical quantities that do not reflect norms of the targeted neuromuscular control dynamics.

Hence, a stage of constructing the datasets was applied where only those biomechanical signals with the highest relevance for estimating neural delay were preserved. This feature selection step centers around variables that describe the timing of neural activation, dynamics of muscle activation and excitation signals in relation to joint motion of the lower limb. The features chosen for selection are joint kinematics of the hip and knee joints in conjunction with activation trajectories of three key muscles participating during lower limb locomotion (the rectus femoris, vastus intermedius and biceps femoris long head). Joint kinematics are familiar, observable phenomena that describe skeletal motion whereas muscle activation signals contribute to the underlying neuromuscular processes responsible for generating this motion. Combined, these variables provide physiologically interpretable signals from which we can infer the latent neural delay parameter.

The entire dataset construction process is shown in Fig. 8, which provides an overview of how OpenSim simulation outputs could be transformed to produce the machine learning dataset utilized for neural delay estimation.

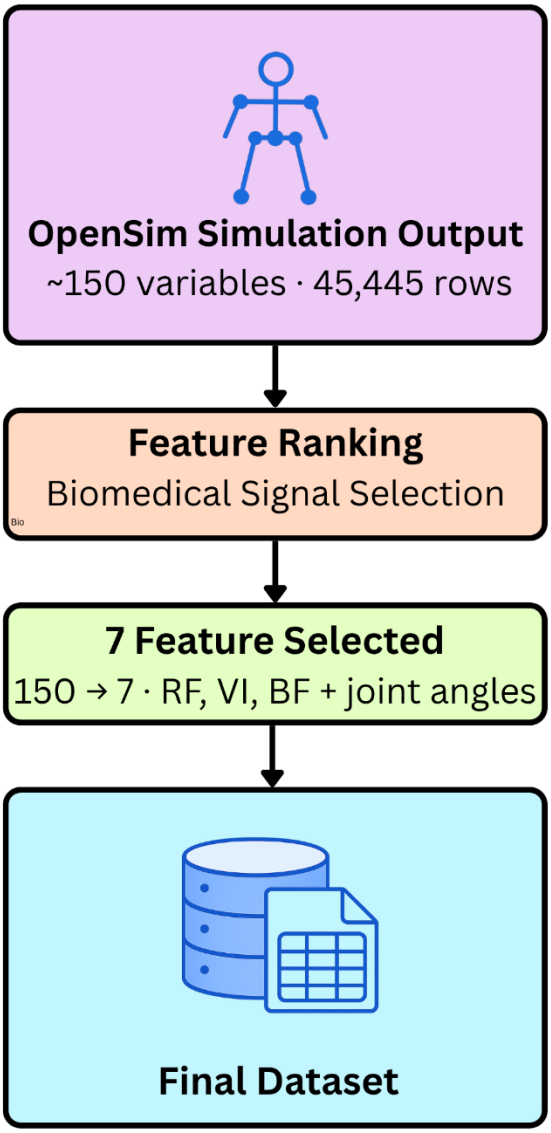


Figure 8:Dataset construction pipeline used in this study. Raw state trajectories generated by the OpenSim forward dynamics simulations contain approximately 150 biomechanical variables describing the dynamic state of the musculoskeletal system. Relevant biomechanical signals, including joint kinematics and muscle activation trajectories, are extracted through feature selection and organized into a structured dataset used for neural delay estimation.

3.4.1 Feature Extraction

The purpose of the feature extraction stage is to extract the biomechanical signals that are involved in the dynamic with small neural activation time changes. Approximately 150 variables that the OpenSim forward dynamics simulations generate, a subset was chosen that carry physiological relevance to create features for the machine learning input. The chosen variables characterise not only the visible lower limb motion but also the neuromuscular activation dynamics which create this motion. Joint kinematics of the hip and knee were included to quantify the mechanical response of the musculoskeletal system, while muscle activation trajectories were included to characterize the internal neuromuscular control signals driving this motion. With this

rationale, seven biomechanical variables were chosen as inputs of the neural delay estimation framework which includes:

- knee joint flexion angle
- knee joint angular velocity
- hip flexion angle
- hip angular velocity
- rectus femoris activation
- vastus intermedius activation
- biceps femoris long head activation

These variables were chosen because they collectively describe the interaction between neural excitation, muscle activation, and skeletal motion during the simulated gait cycle. The rectus femoris and vastus intermedius muscles act as primary knee extensors, while the biceps femoris functions as a key antagonist muscle contributing to knee flexion and hip extension. Differences in timing with which these muscles are activated modulate activation dynamics, and in turn modify joint motion through the musculoskeletal system.

The selected feature set encodes both the causal (neuromuscular activation) and observable (joint movement) effects of timing differences in neural control by combining muscle activation signals with joint kinematics. Coupling Biomechanical and Neural DataLevel of ObservationThe biomechanical observations are defined by a relationship between those parameters and the machine learning model, which allows the latent neural delay parameter to be inferred from biomechanical observations. Together with these input features, the neural delay parameter τ chosen in each simulation was added to the dataset as a target variable for supervised learning. So, each sample in the dataset is represented by the biomechanical signals selected with their values and corresponding neural delay value while those simulated. The resulting dataset has 45445 samples of biomechanical measure: seven input variables and one target variable describing the neural delay parameter.

Table 7:Biomechanical variables selected as input features for neural delay estimation. The dataset combines joint kinematics and muscle activation signals to capture the relationship between neuromuscular control and observable musculoskeletal motion

Feature	**Description**
knee_angle_r	Right knee joint flexion angle
knee_speed_r	Right knee angular velocity
hip_flexion_r	Right hip flexion angle
hip_speed_r	Right hip angular velocity
rect_fem_activation	Right femoris muscle activation
vas_int_activation	Vastus intermedius activation
bifemlh_activation	Biceps femoris long head activation
τ	Neural delay parameter (target)

3.4.2 Biomechanically Perturbed Dataset for Robustness Evaluation

Even though the main dataset produced from forward dynamics simulations indicates that neural delay may be predicted from biomechanical signals, in this study, it needed to test if this was still true when the underlying musculoskeletal model was perturbed slightly away from its pre-trained pose. These simulation-based datasets are often based on idealized model parameters while real-world biomechanical systems will have variability due to anatomical differences, measurement noise, and modeling uncertainties. Consequently, this study devised a robustness validation experiment to determine whether the inverse neural delay estimation framework is robust to biomechanical perturbations.

To perform this evaluation, a second dataset was created by perturbing the musculoskeletal model used in the simulations with a controlled perturbation. More specifically, a 7% increase in the mass of the right femur segment compared to the model. This change affects the inertial properties of the lower limb, which in turn has an impact on how muscle activation interacts with joint torques and subsequent resulting motion of the limb over a gait cycle simulated.

This perturbation along the structural system was expressed as knee flexion angle trajectories throughout a gait cycle for both the nominal and perturbed (W1, W2) models as a quantification of how perturbations in structure respond by altering biomechanical signals in musculoskeletal simulation. Simulations were performed at different neural delays (τ = 0.015 – 0.06 s), and the knee kinematic trajectories were extracted from the OpenSim state files as data processing output. Since the forward simulations produced trajectories with slightly different temporal lengths, each knee angle signal was resampled into a normalized gait cycle representation between 0–100%. This temporal normalization technique guarantees that biomechanical events will always be aligned across simulations, so it can directly compare joint kinematics due to different neural delays.

For every model configuration, knee flexion profiles of the neural delay associated trajectories were averaged and a standard deviation envelope was obtained. The average trajectory is thus the mean expected knee kinematic response which results from changing the neural delay parameter, while its standard deviation envelope reflects the range of motion of that joint, as a result of varying antagonistic nervous control (relative to synergies). This leads to a comparison between the nominal and perturbed models given in Figure 9.

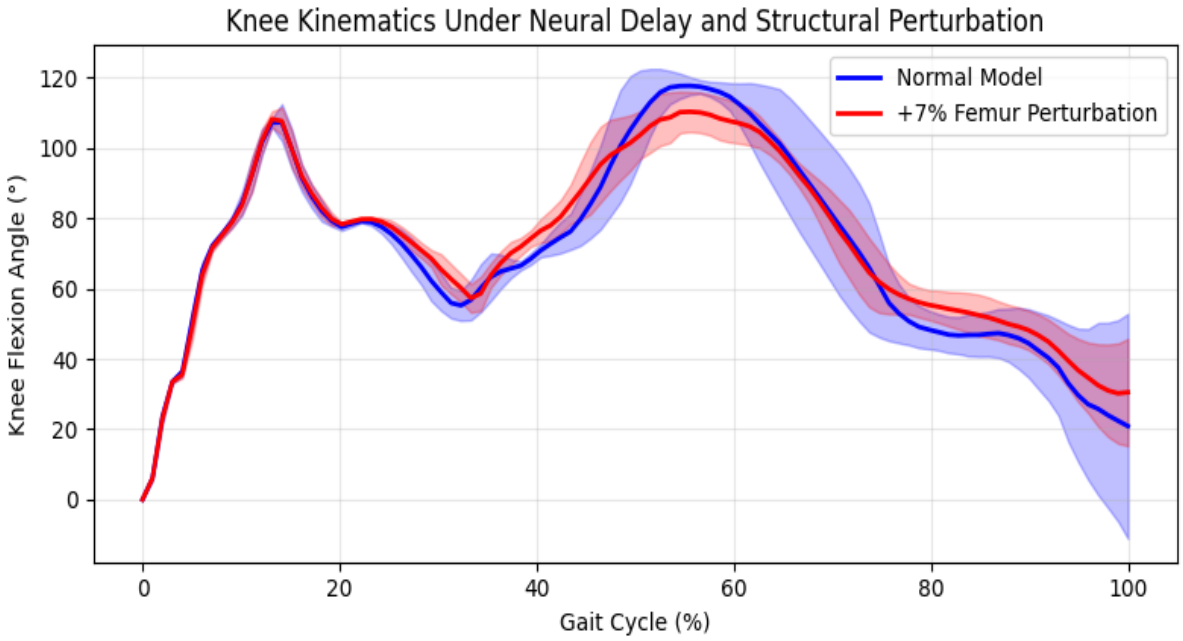


Figure 9: Knee kinematics under neural delay variations and structural perturbation of the musculoskeletal model. The blue curve represents the mean knee flexion trajectory obtained from the nominal model, while the red curve corresponds to the perturbed model with a 7% increase in femur mass. The shaded regions represent ±1 standard deviation across simulations performed under different neural delay conditions (τ = 0.015–0.06 s). The comparison illustrates how neural delay variations and structural perturbations interact to influence knee joint kinematics during simulated gait**.**

As shown in Fig. 9, over the pattern of a gait cycle, the nominal and perturbed models share similar patterns; yet they differ in both amplitude and variability of the knee flexion trajectories. Its gait is characterized by marginally different kinematics in multiple phases of the gait cycle, indicating that limb inertia produces a systematic effect on neuromuscular dynamics and suggesting that limb function could also be approached using feedback control principles. Since femur mass directly influences how force or perturbation get translated into inertial properties of the lower limb, the muscle forces cause different joint responses during simulation. Critically, the integrity of the gait cycle is still preserved even with the introduction of a perturbation (suggesting that gross motor function and underlying neuromechanical dynamics are stable). In fact, this observation suggests that in the case of small variations in musculoskeletal parameters, neural delay leads to consistent biomechanical signatures. As a result, the perturbed dataset serves as an influential robustness check for the inverse neural delay estimation framework introduced hereafter in this study.

Importantly, all other aspects of the simulation framework were kept identical to those used for the original dataset. The same OpenSim simulation environment, musculoskeletal model configuration, neural delay parameter values and forward dynamics simulation procedures were exactly the same used. The neural delay values remained

$$\tau \in \{0.015, 0.020, 0.030, 0.040, 0.060\} \text{ seconds}$$

And the final structure of the dataset was generated using the same feature extraction pipeline discussed in Section 3.4.1. It should be noted that in the perturbed dataset, all biomechanical variables and dimensions of the dataset remain unaltered; only the mass distribution of the femur segment is changed.

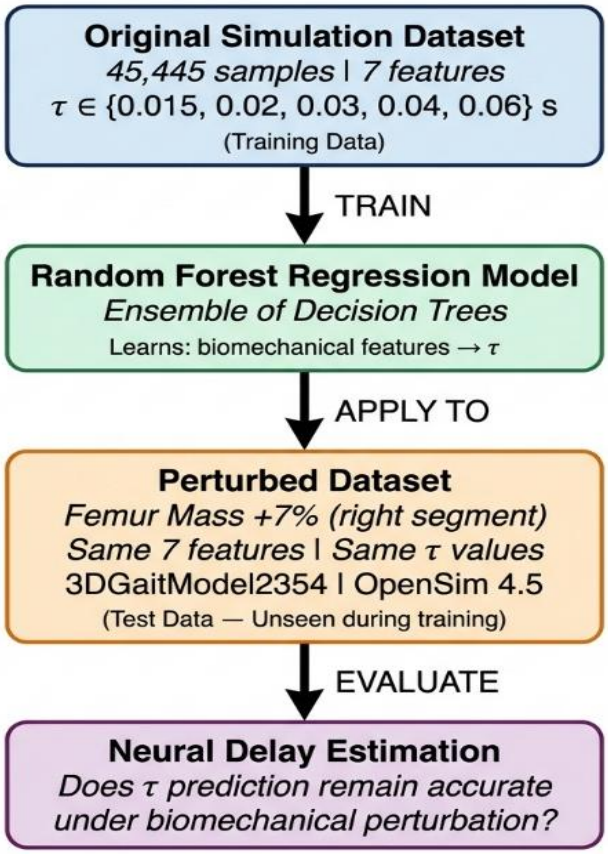


Figure 10: Robustness validation experiment using a biomechanically perturbed dataset. The machine learning model is trained using the original dataset generated from the OpenSim forward dynamics simulations. A second dataset is created by introducing a 7% increase in the femur mass of the musculoskeletal model. The trained model is then evaluated on the perturbed dataset to determine whether neural delay estimation remains accurate under variations in musculoskeletal parameters.

Similarly, the forward dynamics simulations were repeated under each neural delay condition after changing the femur mass parameter. The simulated outputs were processed using the same feature extraction and dataset construction procedures outlined above and as a result produced a second dataset comprised of the exact same feature columns but with altered biomechanical dynamics because of the different limb inertia in this additional simulation. This perturbed dataset is meant to test whether the machine learning model can still correctly recover the neural delay parameter when the biomechanical system differs from that used during training. In this validation experiment, the learning model was trained with the undisturbed dataset and tested in turn on manipulated dataset. If the predictor for neural delay can still predict even after this biomechanical perturbation than that means the learning of a mapping from biomechanical signals to neural delay does not depend on functioning with a true model simulation.

Instead, the observed performance on the perturbed dataset suggests that neural delay may produce consistent and detectable signatures in joint kinematics and muscle activation trajectories across a wide range of underlying musculoskeletal parameters. Importantly, this result corroborates the interpretation of neural delay as a physiologically meaningful latent variable because it is manifesting in biomechanical motion patterns and not merely an artifact arising due to a specific type of simulation configuration.

From a methodological perspective, the introduction of the perturbed dataset strengthens the proposed inverse neuromechanical modeling framework in multiple ways. First of all, it demonstrates that the relation of neural activation timing to musculoskeletal motion is apparent even in circumstances with substantial biomechanical variability. Second, it provides a more rigorous test case in which the learning method is required to generalize beyond the precise simulation conditions employed to create the training set. In conclusion, this robustness evaluation serves to increase the appeal of the proposed framework for analyzing experimental gait data, where musculo-skeletal parameters will inevitably differ between individuals.

3.5 Machine Learning Framework

The objective of the proposed framework is to estimate the latent neural delay parameter τ from observable biomechanical signals generated by the musculoskeletal simulations. While the forward dynamics simulations described in the preceding sections explicitly define neural delay as an input parameter, the present stage addresses the inverse neuromechanical problem of recovering this parameter from the resulting biomechanical motion and muscle activation patterns. This inverse estimation problem can be formulated as a supervised regression task in which a machine learning model learns the relationship between biomechanical observations and the neural delay parameter used during simulation. The biomechanical features extracted from the dataset described in Section 3.4 are presented as an input to the learning algorithm. Such features are joint kinematics (of the hip and knee) and activation trajectories of tonic key lower-limb muscles. The output of the model is the estimate of the neural delay parameter τ for each biomechanical measurement. The inherently nonlinear relationship between neural activation timing and resulting musculoskeletal motion makes this a problem particularly well suited to machine learning methods. Different neural delays change how muscles are activated and that affects muscle–tendon force generation and joint torque on the skeleton. You are exposed to ground reaction forces through these interactions, which transfer and amplify through the musculoskeletal model leading to substantial differences in joint kinematics and muscle activation trajectories. Consequently, neural timing information is indirectly represented in the observable biomechanical signals produced during simulated movement.

A Random Forest regression model was used to estimate neural delay from the chosen biomechanical features in this study. Random forest models are an ensemble learning method that creates multiple decision trees during training time, then poll all individual trees to obtain a final prediction estimate (for regression). This is especially ideal for biomechanical datasets as it can discover non-linear relations between variables while also being insensitive to noise, interaction among features and small shifts in data distribution under the hood. Ensemble tree-based methods also offer better stability when compared to single decision tree models and high flexibility with

strong capabilities for modeling complex mappings between input features and a target variable, while not making too strong assumptions on the functional form of the relationship. Thus, the learning framework works by mapping biomechanical observations to the corresponding neural delay parameter that produced them in simulation. The model is trained on finding joint kinematics as well as muscle activation patterns related to different neural delay conditions. After training, the model estimates neural delay from new biomechanical observations leading to the general formulation of a neural activation timing estimate for musculoskeletal motion data.

The overall workflow of the machine learning framework is illustrated in Figure 11, which summarizes how biomechanical signals extracted from the simulation dataset are used as inputs to the learning model in order to estimate the neural delay parameter.

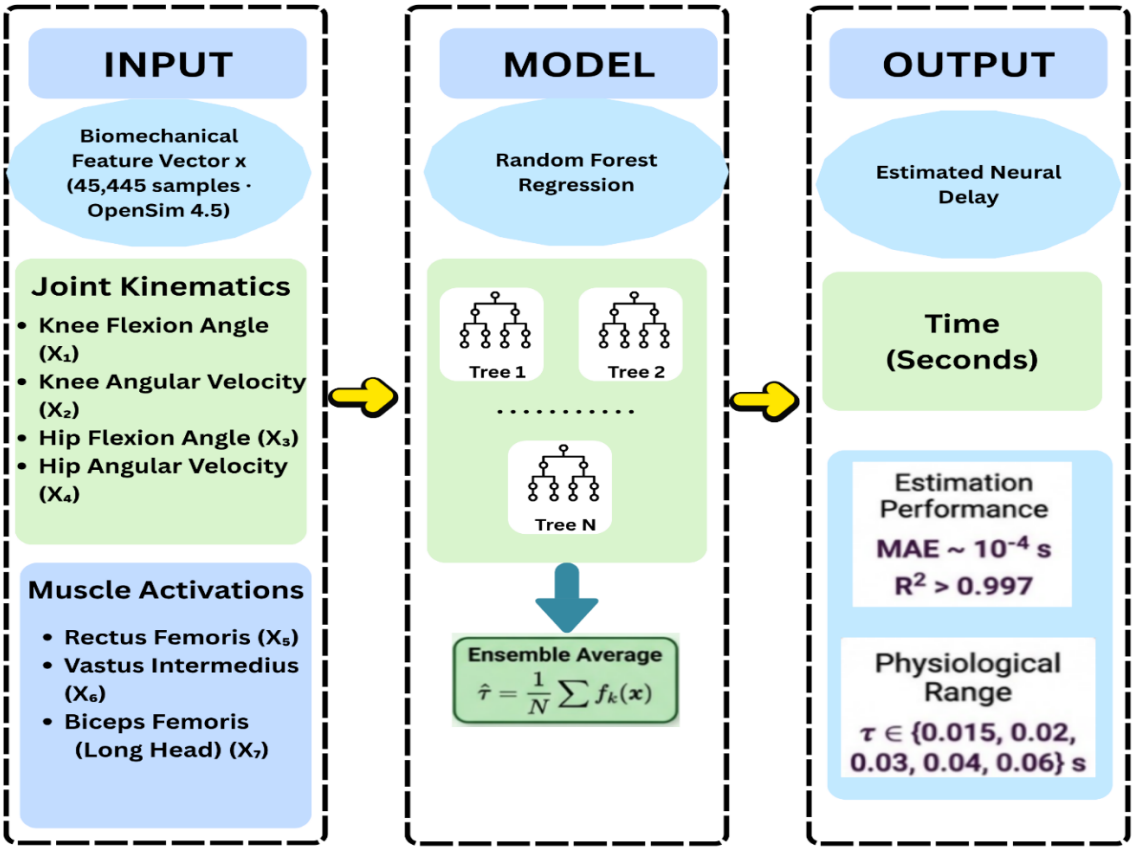


Figure 11: Machine learning framework for inverse neural delay estimation.

Biomechanical features extracted from the simulation dataset, including joint kinematics and muscle activation trajectories, are used as inputs to a Random Forest regression model. The model learns the nonlinear mapping between these biomechanical signals and the neural delay parameter τ, enabling the estimation of neural activation timing directly from musculoskeletal motion data. As shown Figure 11, the learning model receives biomechanical signals that represent the dynamic behavior of the musculoskeletal system and outputs an estimate of the neural delay parameter given those signals. This framework inverting the forward neuromechanical pathway outlined in Section 3.2, trying to estimate when neural control occurs based on observable biomechanical motion properties. The following subsections describe in detail the mathematical formulation of the inverse regression problem, the architecture of learning model, strategy for

optimizing model parameters via training and validation as well as metrics for prediction accuracy evaluation.

3.5.1 Problem Formulation of Neural Delay Estimation

The objective of the proposed framework is to estimate the neural delay parameter τ from raw biomechanical signals available from musculoskeletal simulations. The forward dynamics simulations in previous sections generated biomechanical trajectories given some latent neural delay parameter, the present stage considers the inverse problem of estimating this latent neural value from observed joint kinematics and activations. Let the dataset generated from the OpenSim simulations be represented as:

$$\mathcal{D} = \{(\boldsymbol{x}_i, \tau_i)\}_{i=1}^{N} \tag{3.4}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the biomechanical feature vector corresponding to the i-th observation and τ_i represents the neural delay parameter used in the simulation that produced those biomechanical signals. The dataset therefore consists of N pairs of biomechanical observations and their associated neural delay values.

In this study, each feature vector contains seven biomechanical variables that describe the dynamic state of the lower-limb musculoskeletal system:

$$\mathbf{x}_i = \left[\theta_i^{\text{knee}}, \dot{\theta}_i^{\text{knee}}, \theta_i^{\text{hip}}, \dot{\theta}_i^{\text{hip}}, a_i^{RF}, a_i^{VI}, a_i^{BF}\right] \tag{3.5}$$

Where θ^{knee} and θ^{hip} denote knee and hip joint angles, $\dot{\theta}^{\text{knee}}$ and $\dot{\theta}^{\text{hip}}$ represent their corresponding angular velocities and a^{RF}, a^{VI}, a^{BF} correspond to activation levels of the rectus femoris, vastus intermedius, and biceps femoris long head muscles, respectively. In combination, these variables account for the observable joint motion and the underlying muscle activation dynamics that arise during simulated gait cycle [61].

The inverse neuromechanical relationship explored in this study can thus be expressed as a functional mapping from biomechanical variables to the neural delay parameter:

$$\tau = f\left(\theta_{\text{knee}}, \dot{\theta}_{\text{knee}}, \theta_{\text{hip}}, \dot{\theta}_{\text{hip}}, a_{RF}, a_{VI}, a_{BF}\right) \tag{3.6}$$

This Equation describes the inverse neuromechanical relationship investigated in this study, where the neural delay parameter is estimated as a function of observable biomechanical variables including joint kinematics and muscle activation signals [62]. Thus, the neural delay estimation problem can be seen as a regression task in which a function $f(\cdot)$ is learned to map biomechanical observations to the corresponding neural delay value:

$$\hat{\tau}_i = f(\mathbf{x}_i) \tag{3.7}$$

where $\hat{\tau}_i$ denotes the predicted neural delay parameter produced by the regression model. The learning objective is to find the function $f(\cdot)$ of the dataset that minimizes the prediction error between the estimated neural delay and the true delay values. This objective can be expressed

using the mean squared error loss function:

$$\mathcal{L}(f) = \frac{1}{N}\sum_{i=1}^{N} \left(\tau_i - f(\mathbf{x}_i)\right)^2 \tag{3.8}$$

where $\mathcal{L}(f)$ denotes the regression loss function, N is the total number of samples in the dataset, τ_i is the neural delay value used in the simulation, and $f(\mathbf{x}_i)$ denotes the neural delay predicted by the regression model for the corresponding biomechanical feature vector $\mathbf{x}_i$ [63].

The minimization of this loss function is designed to have the regression model identify patterns in biomechanical signals that corresponded to different neural delay conditions. On a neuromechanical level this formulation is the inverse of the forward musculoskeletal dynamics process described previously. Neural delay in the forward simulations is evident at the level of the muscle activation dynamics [64]; i.e., neural signals governing muscle contractions indirectly govern the force output and joint motion. Such interactions relay through the musculoskeletal system, generating quantifiable biomechanical signals, e.g., joint kinematics and muscle activation trajectories. Consequently, the differences in neural delay produce distinguishable signatures in the biomechanical signals extracted from the simulation outputs. By learning the mapping between such biomechanically defined patterns and a neural delay parameter allows one form of regression model to be treated as a computational device for estimating when neurons are activated purely from musculoskeletal motion data.

3.5.2 Random Forest Regression Model

A Random Forest regression model is used in this study to approximate the inverse neuromechanical mapping as defined in Section 3.5.1. Random Forest is an ensemble learning method that creates a multitude of decision trees and provides their outputs, to provide a strong regression estimate. The model reduces variance and increases generalization performance compared to a single decision tree model by averaging outputs of the many trees that were trained on multiple bootstrap samples of the data.

In the current study, our Random Forest model is learning to estimate the non linear mapping of biomechanical features derived from musculoskeletal simulations and their corresponding neural delay parameter. Let $\mathbf{x} \in \mathbb{R}^d$ denote the biomechanical feature vector defined above. Every decision tree in the forest makes a prediction for the given neural delay parameter, built from recursive binary partitioning of the feature space. Averaging the predictions of all trees in an ensemble gives us final prediction of Random Forest Model:

$$\hat{\tau}(\boldsymbol{x}) = \frac{1}{T}\sum_{t=1}^{T} h_t(\boldsymbol{x}) \tag{3.9}$$

where T represents the number of decision trees in the forest and $h_t(\mathbf{x})$ is the prediction produced by the t -th tree. This ensemble averaging bidirectionally mitigates prediction variance, while allowing the model to account for complex nonlinear interaction among biomechanical

variables. Each decision tree uses a bootstrap sample taken from the data set used for training. A random set of input features is taken into account on each split when the algorithm decides optimal patterns to partition feature space during tree construction. With training data conditioned to data noun in this random feature selection mechanism, the trees of the ensemble become de-correlated and enhance the model predicting capabilities [65].

For several reasons, the present neuromechanical estimation problem is particularly well suited for the Random Forest algorithm. The coupling between muscle activation dynamics, muscle–tendon mechanics and skeletal motion renders the relationship between neural activation timing and resultant musculoskeletal motion inherently nonlinear for a number of reasons: On the other hand, models based on tree ensembles can learn such non-linear dependencies without making explicit assumptions about the functional form of the underlying relationship. Second, when creating musculoskeletal simulations, biomechanical features usually have correlations. However, Random Forest models are resilient to such correlated inputs since they will select only one of them as the most informative when making predictions. The biomechanical dataset described in Section 3.4 is used to train the Random Forest regression model for this study. During training, the input feature vector consisted of seven biomechanical signals each representing joint kinematics and muscle activation trajectories and the target variable was designated as the neural delay parameter associated with that sample of simulation. As part of the training phase, it learns characteristic patterns in the biomechanical signals corresponding to different levels of neural delay. After training, the model is able to regress the neural delay parameter from biomechanical observations not previously observed during training and generated by the same simulation pipeline specifically, observing a new estimated τ value within… reference or determined range of values included in the training dataset (0.015 – 0.06s). This in-simulation generalization ability is the proof-of-concept of what the proposed framework achieves. However, evaluations on the model's generalization to experimental data, musculoskeletal models with different characteristics (i.e. unloaded locomotion versus loaded locomotion) or other subjects with different neuromuscular properties must be done via retraining under those conditions and are not evaluated in this study.

3.5.3 Model Training and Evaluation Procedure

In order to test how well the suggested framework can estimate neural delay through biomechanical signals, the constructed dataset was used to train and validate a Random Forest regression model. All observations come from a physics-based forward dynamics simulation of an isolated limb undergoing a known neural delay parameter value used during simulation, which is one such biomechanical state.

The dataset was structured as supervised learning pairs of biomechanical feature vectors and neural delay labels. The rows of the feature vectors consist of seven biomechanical variables that describe the dynamic state of the lower-limb musculoskeletal system. These variables are joint kinematic quantities combined with muscle activation signals for important muscles driving knee and hip motion. Together, these variables are a reflection of the observable mechanical behavior of the limb and the unobservable neuromuscular activation dynamics that determine when muscles

generate force. The dataset was split into training and testing subsets for the model training and performance assessment respectively. The Random Forest regression model was then built to learn the mapping from biomechanical observations in the training subset to corresponding neural delay values. The testing subset was used only for evaluating performance and excluded from training. This separation ensures that the predictive capability of the model is evaluated on previously unseen biomechanical observations, thereby providing an unbiased estimate of generalization performance.

In the training phase, several decision trees are constructed based on bootstrap samples drawn from the training dataset. At the node of a random tree, the optimal split of the feature space is considered with only a randomly selected subset of biomechanical features. This randomized training strategy reduces correlations among individual trees and improves the stability and predictive accuracy of the ensemble model. The final prediction of the Random Forest model is obtained by averaging the outputs of all trees within the ensemble. The predictive performance of the trained model was evaluated using standard regression metrics commonly employed in machine learning studies. In particular, the Mean Absolute Error (MAE) was used to quantify the average magnitude of prediction errors between estimated and true neural delay values. In addition, the coefficient of determination (R^2) was calculated to measure how well the predicted neural delay values explain the variance in the ground-truth delay parameters. Additional validation experiments were performed for the inverse neuromechanical modeling framework under perturbed biomechanical conditions to better assess its robustness. Biomechanical signals obtained from simulations with small modifications of musculoskeletal parameters were used to evaluate the model that has been trained with respect to these experiments. This assessment tests if the learned mapping between features of biomechanical data and neural delay is robust when the biomechanical system (or its configuration) varies from that which was used for training. This study rigorously explores if neural delay can consistently be predicted using observable biomechanical signals arising from musculoskeletal dynamics through this training and evaluation process.

4 Results and Discussion

This section presents the experimental results obtained from the proposed inverse neuromechanical modeling framework. The objective of the experiments is to determine whether the neural delay parameter τ, embedded as a latent variable in musculoskeletal forward dynamics simulations, can be recovered from the simulation-generated biomechanical signals it produces, using a supervised Random Forest regression model trained and tested entirely within the simulation framework. Experimental results in this section are based on simulated data from controlled, noiseless scenarios using the OpenSim 3DGaitModel2354 framework described in Section 3. The analysis aims to identify how variability in the timing of neural activation is expressed as differences in joint kinematics and muscle activation patterns generated by musculoskeletal simulations. The experiments seek to establish if the latent timing parameter of neural delay produces detectable signatures in the observable dynamics that arise within the biomechanical structure. To test this hypothesis, machine learning models were trained to obtain neural delay from biomechanical features extracted from the simulation dataset. The experimental results are analyzed from three complementary perspectives. First, the learning model predictive capability is evaluated as to whether it can recover neural delay from biomechanical observations. Second, the neural delay-biomechanical variable relationship is assessed as it relates to how the timing of neural signals dictates musculoskeletal motion. At last, robustness experiments are performed to ensure the stability of the learned neuromechanical mapping against perturbations in biomechanics and noisy signals.

Collectively, these analyses demonstrate empirical support for the feasibility of inverse neuromechanical parameter estimation, a method where latent neural control parameters are inferred from measurable biomechanical motion features.

4.1 Baseline Neural Delay Estimation Performance

The first experiment evaluates if the neural delay parameter can be accurately estimated from data generated by musculoskeletal forward dynamics simulations. The objective of this experiment is to determine whether the observable biomechanical variables extracted from the simulated gait dynamics contain sufficient information to reconstruct the neural activation timing parameter that governed the simulations. Neuromuscular delays and feedback influence the neuromechanical dynamics of the musculoskeletal system via excitation contraction coupling of muscle models (in Section 3). In other words, the delay parameter controls how fast neural drive signals are transformed to muscle activation states. Presuming that muscle activation is the primary determinant of force generation in contractile elements of a muscle–tendon unit, differences in neural delay are correlated with peripheral effects on timing of muscle force production. The Joint torques generated by these forces then initiate skeletal motion. Therefore, the neural delay parameter affects the entire neuromechanical pathway

neural excitation → muscle activation → muscle force → joint torque → joint kinematics.

Neural delay cannot be directly observed from biomechanical measurements, but its temporal effects propagate through the causal chain and become represented in the biomechanical outcomes produced. Analysis of the joint kinematic trajectories emerges from this musculoskeletal forward simulation can provide a direct way to investigate the influence of neural delay on observable biomechanical motion. This is because the knee joint, one of the key joints in lower-limb biomechanics, exhibits a relatively informative signal as it is driven by activation dynamics from both rectus femoris and vastus muscle groups - two groups that play integral roles in controlling gait mechanics. Changes in the timing of neural activations adjust how quickly muscle force builds and decays, which then modifies these torques at the joints that act on the skeletal segments. Consequently, the resulting knee flexion trajectory is a direct biomechanical correlate of neural timing in simulation of locomotion.

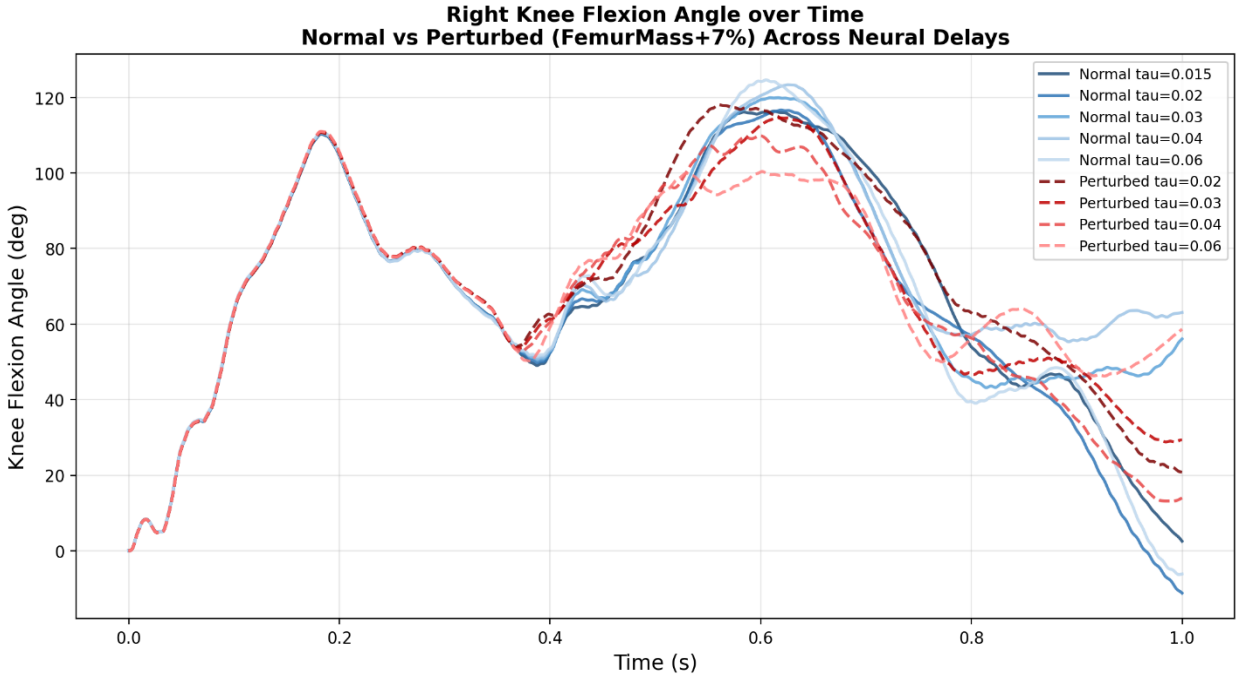


Figure 12:Right knee flexion angle trajectories obtained from the OpenSim forward dynamics simulations across multiple neural delay conditions. Solid curves correspond to simulations performed with the nominal musculoskeletal model, while dashed curves represent simulations with a biomechanical perturbation consisting of a 7% increase in femur mass. Each trajectory corresponds to a different neural delay parameter value (τ = 0.015–0.06 s). The divergence of the trajectories during the gait cycle illustrates how neural activation timing influences the resulting joint kinematics.

It was found that the knee joint is sensitive to changes in the neural activation delay in the figure above. In the initial part of the gait cycle, the trajectories for various values of delay are fairly close together, reflecting the fact that the mechanical state of the system at the beginning of the gait cycle is the most important one for determining the dynamics of the gait. The paths start to diverge, however, during later portions of the gait cycle, especially during the swing phase when knee flexing is at its greatest. The larger the neural delay value, the more delayed and amplified is the flexion peak in the simulation, and this corresponds to the slower activation dynamics of the

muscles for knee extension. These differences illustrate the transmission of temporal changes in neuromuscular control from the muscle and neural system to the joints as observable changes in joint kinematics. The behavior can be understood by the dynamics of the activation of the muscle model. The neural delay parameter τ is a measure of how fast the neural excitation dies down before becoming muscle activation, as described in Section 3.2. A higher delay parameter would make the activation slower, which would lead to a temporal delay in muscle force development. The temporal lag is cascaded through the biomechanical system based on the causal relationship, because muscle force produces joint torque based on the Hill-type muscle–tendon formulation.

neural delay → muscle activation → muscle force → joint torque → joint motion

As a result, there is measurable variability in the time course of knee joint kinematics due to neural delayHence, the central idea of the proposed inverse neuromechanical modeling framework is that the biomechanical variables measured in the observed motion include enough information to be used to reconstruct the underlying neural delay parameter.

From a mathematical perspective, this inverse estimation problem can be formulated as the existence of a functional mapping between the biomechanical feature vector X and the neural delay parameter τ

$$\tau = f(X) \tag{4.1}$$

where X is the biomechanical feature vector which is defined in Section 3.5.1 where joint kinematics and muscle activation signals are stored. The machine learning model attempts to approximate this unknown mapping using a regression function $\hat{f}$

$$\hat{\tau} = \hat{f}(X) \tag{4.2}$$

where $\hat{\tau}$ represents the neural delay value predicted by the regression model.

To prediction capability of the inverse neuromechanical framework, the predicted delay values are compared with the ground-truth delay parameters used during the forward simulations. The prediction error for each observation can therefore, be written as

$$e_i = \tau_i - \hat{\tau}_i \tag{4.3}$$

where τ_i denotes the true neural delay used during simulation and $\hat{\tau}_i$ denotes the delay estimated by the regression model for the corresponding biomechanical observation.

The baseline evaluation was based on a random split of the time-series data of 45,445 biomechanical samples for the pooled time-series data set. This method shares the same continuous simulation paths as testing points with training data, which enables intra-trajectory interpolation. As a result, the regression model can attain theoretical mean absolute error of the order of 10^{-4} seconds and the R^2 can be close to unity. This baseline experiment was intentionally constructed without testing clinical generalization and was aimed at setting the theoretical maximum of identifiability – demonstrating that the continuous manifold of the data mathematically encodes the latent parameter τ.

For comparison, the size of the estimation error can be compared to the scale of the neural delay values which were used in the simulations.

For this data set, the delay is found to be within the range of neural delay

$$\tau \in [0.015, 0.06]$$

which corresponds to a total delay span of

$$\Delta\,\tau = 0.06 - 0.015 = 0.045\ \text{s}$$

The mean absolute prediction error is on the order of 10^{-4} s. Therefore, the relative error scale can be approximated as:

$$\frac{\text{MAE}}{\Delta\tau} \approx \frac{10^{-4}}{4.5 \times 10^{-2}} \approx 2.2 \times 10^{-3}$$

The prediction error is more than two orders of magnitude smaller than the range of neural delay values, implying that the regression model can recover the neural delay with extremely high precision compared to the physiological variation in neural delay that is simulated.

Another way of looking at the degree of accuracy of the estimation can be through the separation between the discrete delay states employed in the simulation matrix. The simulated neural delay values were defined as $\tau \in \{0.015, 0.02, 0.03, 0.04, 0.06\}$ s. The smallest difference between adjacent delay levels is thus equal to

$$\Delta\tau_{min} = 0.02 - 0.015 = 0.005\text{s}$$

The accuracy of the absolute estimation error (in terms of mean) is around 10^{-4} s and the prediction error is more than one order of magnitude less than the smallest delay separation in data. As a result, the regression model can be used to resolve neural delay states that are separated by only a few milliseconds. This outcome shows that the model's inverse neuromechanical mapping has the sensitivity needed to capture the small temporal differences in neural activation that are contained within the biomechanical signals.

In order to show the relationship between predicted and true delay parameters, a scatter plot is shown in Figure 13, which compares the predicted neural delay values to the ground-truth delay parameters for the forward simulations.

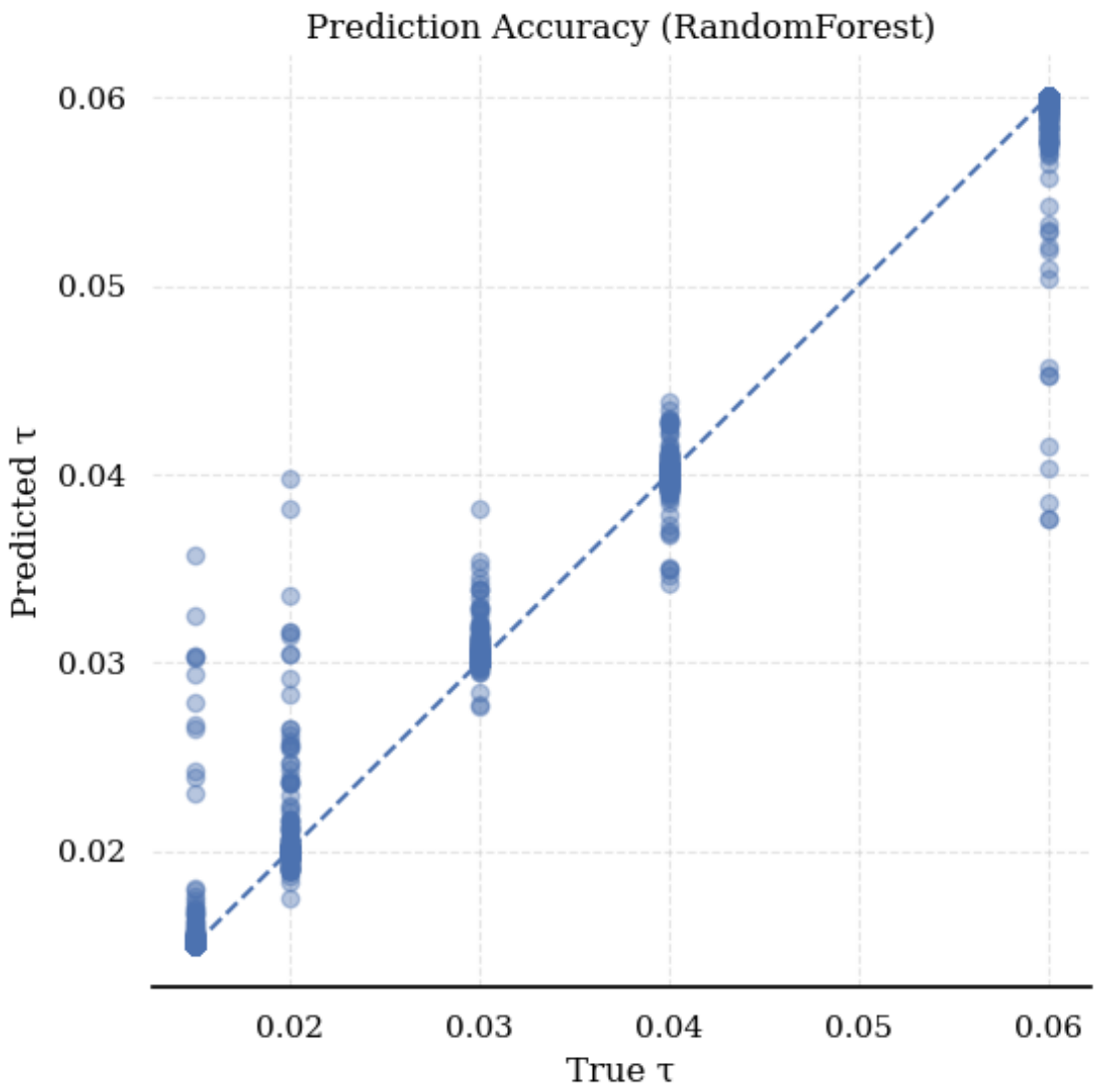


Figure 13:Comparison between predicted neural delay values and ground-truth delay parameters obtained from the OpenSim forward dynamics simulations. Each point represents a biomechanical observation from the dataset, while the diagonal line represents perfect prediction.

It can be seen in Fig. 13 that the delay values predicted are extremely close to the identity line which represents perfect agreement between predicted and true delay parameters. It is important to note, however, that in the figure, there are groups of predicted delay values that correspond to the ground-truth delay parameters used in the simulation matrix. Neural delay was not sampled continuously in the forward simulations but rather given the five discrete values of $\tau \in \{0.015, 0.02, 0.03, 0.04, 0.06\}$ seconds. The real delay distribution is therefore a finite set of delay states, and not a continuous spectrum.

This observation is an insight into the behavior of the inverse estimation model. If the underlying neuromechanical relationships were not replicated by the regression model, the delay values estimated by the regression model would be expected to be distributed widely throughout the range of delay. In contrast, the predictions are concentrated in narrow bands of discrete delays, corresponding to those used in the biomechanical simulations.

In mathematical terms, such behavior means that the regression model is modelling the inverse mapping.

$$\tau = f(X) \tag{4.4}$$

In a way that maintains the separable structure of the underlying delay states. Let τ_k denote the k-th delay value used in the simulations and $\hat{\tau}_i$ the predicted delay for observation i. The clustering visible in Fig. 13 implies

$$\hat{\tau}_i \approx \tau_k \text{ for observations generated under delay state } \tau_k$$

for the majority of samples. In other words, the learned mapping accurately divides the biometric space into areas corresponding to the discrete neural delays.

This finding is another confirmation of the presence of identifiable neural delay signature in the biomechanical signals produced by the forward simulations. The machine learning model is therefore not just modeling the numbers but also the underlying delay regimes that are captured in the musculoskeletal dynamics.

The recovery of neural delay can be understood if one looks at the effect of the delay parameter on muscle activation dynamics. The activation dynamics formulation presented in Section 3 shows that the neural delay parameter is a temporal scaling that influences the time evolution of muscle activation. The longer the delay parameter the longer the time from the start of neural activation to the onset of muscle activation, which is a time lag in muscle force production. To illustrate this relationship numerically, the relationship between delay and accuracy is shown numerically in two typical cases for the simulation matrix. With neuromuscular delay of τ = 0.02 s, muscle activation will react relatively fast to the signals of excitation and generate a muscle force in a fast manner, resulting in synchronized joint motion. In contrast, a longer delay of τ = 0.04 s, the activation dynamics delay the transition between the states of excitation and activation. This results in measurable temporal delay in the generation of muscle force, which in turn creates a change in the joint torque timing and the change in the joint velocity profile. These temporal differences are written into the dataset, which is used to train the regression model since the biomechanical feature vector contains both joint kinematics and muscle activation trajectories. Thus the machine learning algorithm learns patterns in the biomechanical signals corresponding to different neural delay conditions.

While Fig. 13. demonstrate the accuracy of predicted delay values with respect to the ground truth is shown in Fig. 13, a second complementary performance analysis can be made by studying the distribution of errors on neural delay estimation. Fig. 14 shows the distribution of prediction errors in the test dataset.

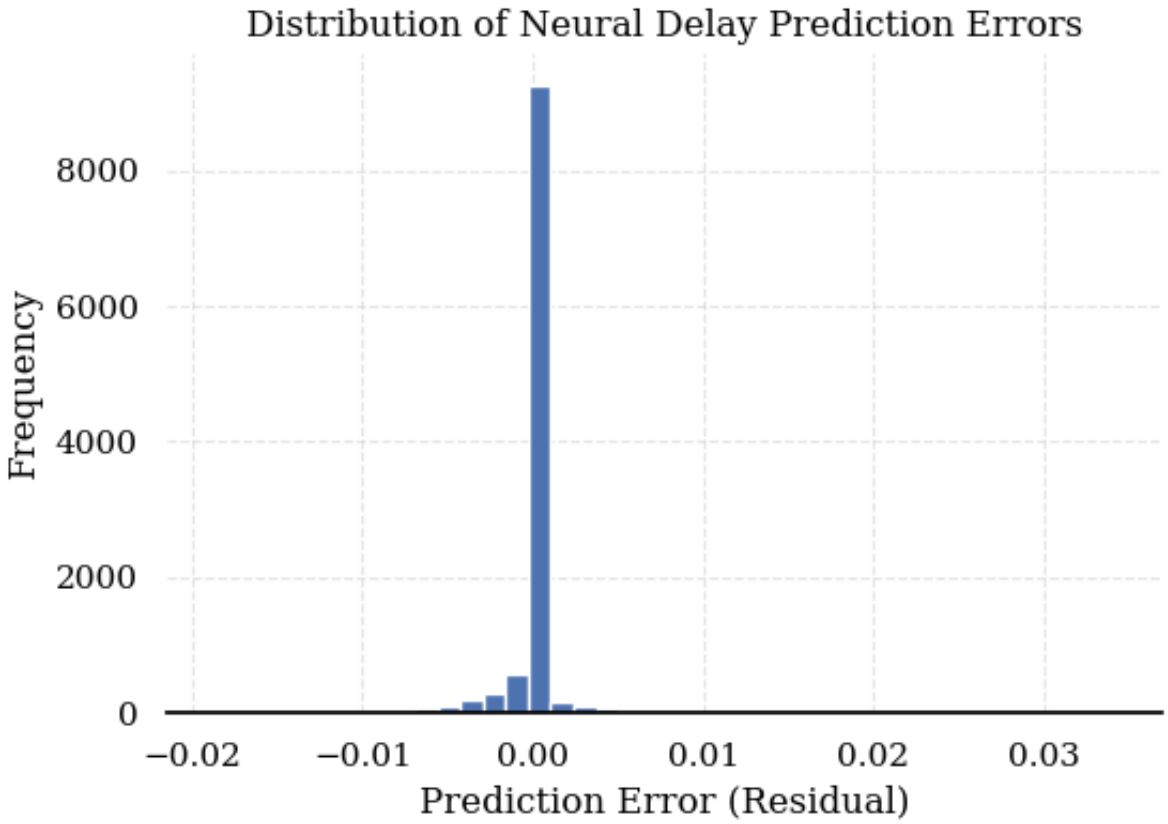


Figure 14:Distribution of neural delay estimation errors for the baseline experiment. The narrow distribution centered around zero indicates that the predicted delay values closely match the ground-truth neural delay parameters.

The error distribution can be seen in the Fig. 14 is strongly centered around zero, meaning most predictions are very close to the true delay values. An additional property evident from Fig. 14 is the approximate symmetry of the residual distribution around zero. If the regression model systematically overestimated or underestimated the neural delay parameter, the residual distribution would exhibit a noticeable shift away from zero. Instead, the residuals look scattered evenly on both sides of the zero-error line, meaning that we can expect a prediction error satisfying

$$E[e_i] \approx 0$$

This observation indicates that the estimator is unbiased, which means that there are no systematic biases favoring larger or smaller delay values in the regression model or dataset across entire records. In particular, it is an important condition for reliable inverse parameter estimation. This result from a neuro-mechanical perspective further provides strong evidence for one of the main hypotheses of this study. While it does not directly affect biomechanical measurements (i.e., neural delay), the temporal effects of activation dynamics cascade through the musculoskeletal system and manifest as signatures in joint motion and muscle activation trajectories. The regression model leverages the learned mapping from these biomechanical signatures to infer the underlying neural delay parameter, recovering latent neural timing information embedded within the simulated movement data.Thus, the findings of this baseline study show that inverse

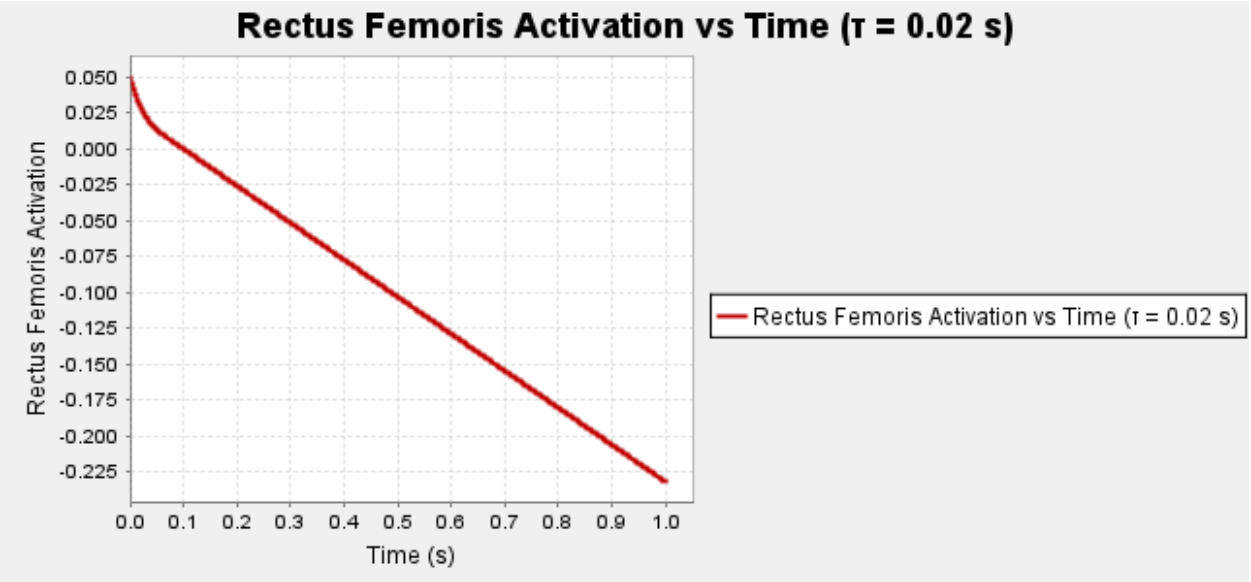
neuromechanical parameter estimation is feasible. The fact that the regression model can also recover the neural delay parameter from biomechanical signals means that timing of neural activation leaves detectable marks in musculoskeletal motions generated by the simulation framework. The results indicate that latent neuromechanical seasonals controlling muscle activation dynamics can be the effective subject of observed biomechanical input signals through data-driven reverse modeling.

4.2 Biomechanical Signature of Neural Delay

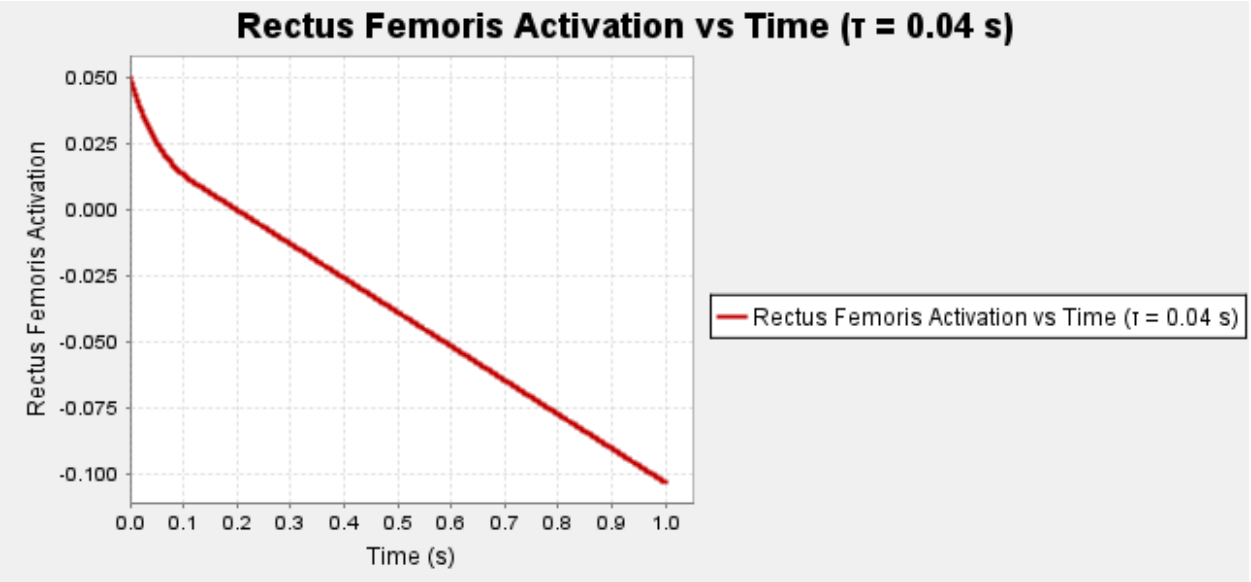
Although the evidence presented in Section 4.1 clearly shows that the neural delay parameter can be well identified from biomechanical signals, it is also important to study how neural delay works by influencing the underlying musculoskeletal dynamics. This relationship is crucial for an interpretation of the inverse estimation results in terms of physiology. If neural delay has a real effect on biomechanical motion then quantifiable features will be detected in the biomechanical variables employed as inputs in the regression model.

In the neuromechanical framework employed in this study, a neural delay affects muscle–tendon units activation dynamics. The delay parameter determines the time interval between when a neuron fires and when it translates into a muscle contraction, which affects the timing of muscle force generation. Since muscle forces create joint torques that propel skeletal movement, timing changes in neural activation pass through the biomechanical system and are indirectly measurable from muscle activation signals and joint kinematics. Therefore, changes in neural delay will be expected to manifest as distinguishable biomechanical signatures within the simulated trajectories.

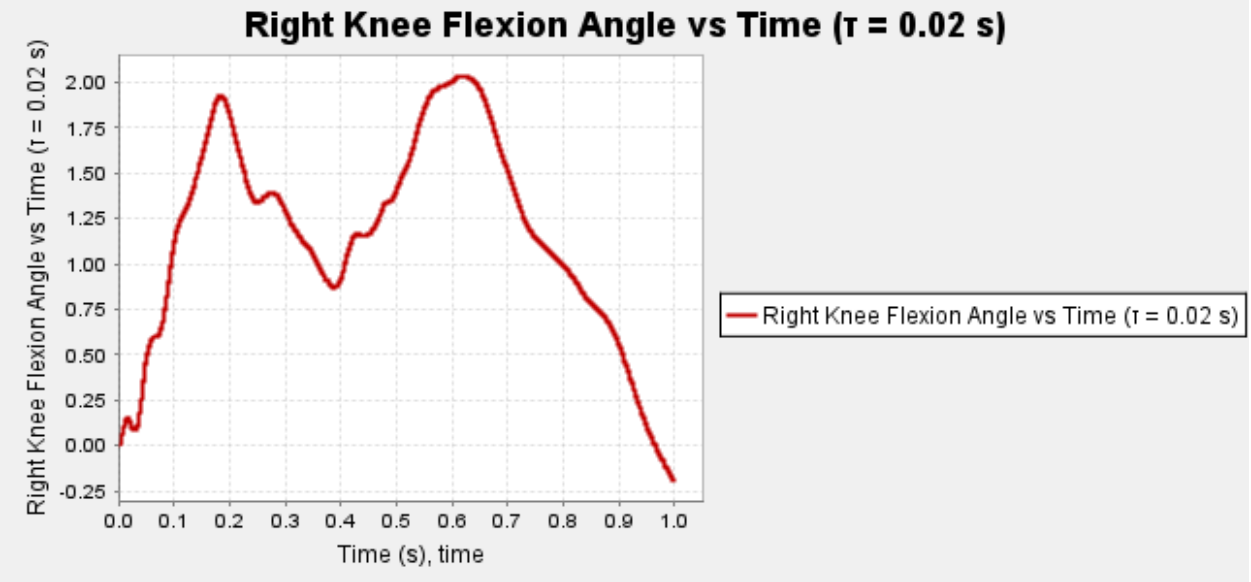
In this study, representative time-series trajectories were extracted from the OpenSim forward dynamics outputs to illustrate how variations in the neural delay parameter influence the biomechanical signals generated by the musculoskeletal simulations. The analysis included two neural delay conditions were considered in this analysis ($\tau = 0.02$ s and $\tau = 0.04$ s). For each delay condition, both muscle activation dynamics and joint kinematic signals were visualized. Muscle activation is a signal of how the motor control interacts with the biological system to drive musculoskeletal motion and joint kinematics represents that kinetic interaction by estimating mechanical behavior due to muscle activation in limb segments. This enables researchers to see how changes in timing of neural activation are distributed throughout the neuromuscular system and how they affect the resulting patterns of biomechanical motion, when comparing these signals across conditions involving different amounts of neural delay. Fig. 15.



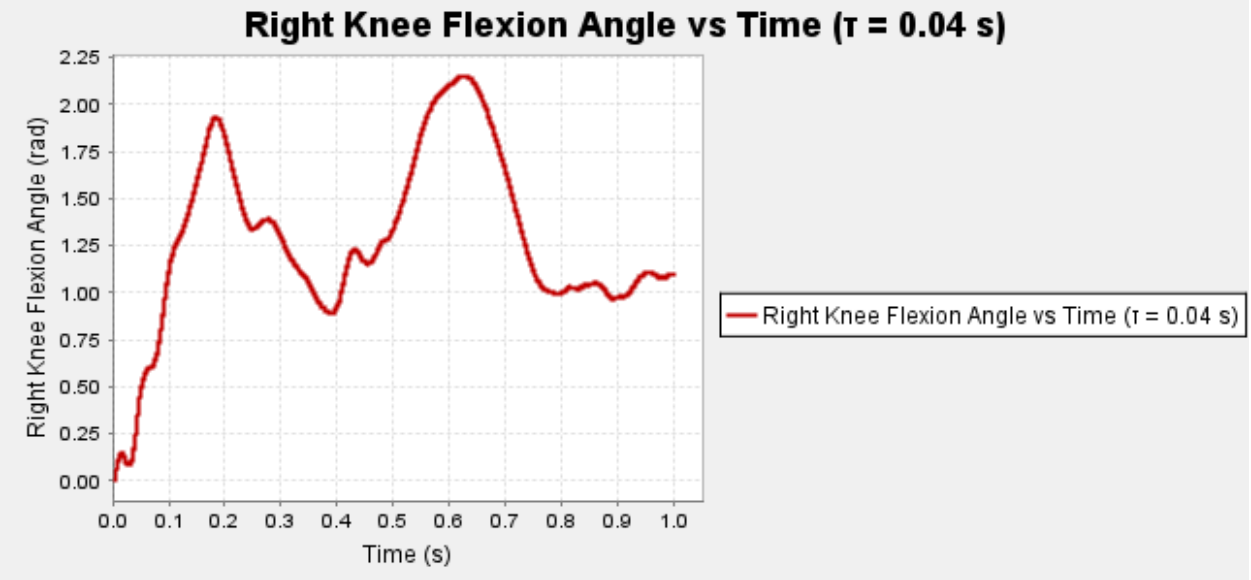
15(a) Rectus femoris activation τ=0.02



15(b) Rectus femoris activation τ=0.04



15(c) Knee flexion angle τ=0.02



15(d) Knee flexion angle τ=0.04

Figure 15: Influence of neural delay on biomechanical signals generated by the OpenSim forward dynamics simulations. Panels (a–b) show rectus femoris activation trajectories for two neural delay conditions ($\tau = 0.02$ s and $\tau = 0.04$ s). Panels (c–d) illustrate the corresponding right knee flexion angle trajectories for the same simulations. Increasing the neural delay parameter alters the temporal characteristics of muscle activation dynamics and produces observable differences in the resulting joint kinematics.

The biomechanical trajectories shown in Fig. 15 illustrate how variations in the neural delay parameter affect both muscle activation dynamics and joint kinematic behavior within the musculoskeletal simulation. In the activation plots, the rectus femoris muscle exhibits clear differences in its temporal response when the neural delay increases from $\tau = 0.02$ s to $\tau = 0.04$ s. Specifically, the activation trajectory becomes temporally shifted and slightly broadened,

reflecting slower activation and deactivation dynamics within the muscle model. The temporal differences that exist at the muscle cooperation level propagate to the mechanical level of the musculoskeletal system, since muscle activation directly controls the timing and magnitude of muscle force generation in the Hill-type muscle formulation. As a result, the knee joint trajectories illustrated in panels (c) and (d) demonstrate analogous shifts in time. While these differences in joint motion are smaller than the one observed for the activation curves, they indicate the downstream mechanical effects of changes in neuromuscular activation timing. These observations also suggest that different neural delays produce quantitatively distinct muscle activation patterns and joint kinematics in the simulation. The trajectories shown in Fig. 15 also shows how the neural timing propagates within the neuromechanical system. The musculoskeletal simulation framework has a parameter called neural delay, which can describe the dynamics of activation that determine how quickly neural excitation results in muscle activation. Thus, variability in neural delay alters the time course of muscle activation. Since the activation of a muscle determines the generation of muscle force according to the Hill-type muscle model, adjustments in time-to-activity lead to an adjustment in the temporal profiles of muscle forces applied on each segment. These forces create torques at the joints that ultimately cause motion in the skeleton and manifest through the joint angles measured and consumed by an external observer. Hence, neural delay variation sequentially propagates through the neuromechanical pathway from neural delay to muscle activation, from muscle activation to muscle force generation, and further on from muscle force to joint torque and finally joint motion. This propagation process is reflected in the difference between the trajectories at τ =0.02 s and at τ =0.04 s. This biomechanical signature provides the physical basis for estimating neural delay based on observable musculoskeletal motion patterns using the inverse modeling framework. Mathematically, the forward neuromechanical relationship governing this process can be written as:

$$\tau \rightarrow a(t) \rightarrow F_m(t) \rightarrow M_j(t) \rightarrow q(t) \tag{4.5}$$

where $a(t)$ represents muscle activation, $F_m(t)$ is muscle force, $M_j(t)$ represents joint torque, and $q(t)$ denotes joint kinematics. According to this relationship, Neural delay directly impacts muscle activation dynamics; downstream biomechanical consequences of these changes in neural control are evident in joint kinematics as predicted. Thus, if neural delay is contained in the biomechanical signals enabling the regression model, its most pronounced signatures should be observed within muscle activation trajectories.

In order to test this hypothesis, a series of complementary analyses was performed to detect which biomechanical variables contain information related to the neural delay parameter. Analyses are performed using random Forest regression to estimate feature importance, mutual information to perform an information-theoretic dependency analysis between biomechanical variables and prediction accuracy of a letter-matching task, and feature ablation experiments that quantify the causal contribution of each biomechanical variable for predicting accuracy. Collectively, these experiments yield several independent lines of evidence that neural delay imparts a measurable biomechanical signature to the simulated musculoskeletal dynamics.

4.2.1 Dominance of Muscle Activation Signals

As the Random Forest regression model produces features importance, to identify which biomechanical variables encode information about neural delay, a feature importance analysis was performed. Feature importance scores provide quantitative values that measure how much each input variable contributes to making accurate predictions by measuring immutable of the features combined influence over prediction error reduction in the ensemble of decision trees. These importance scores quantify the amount of information available in various biomechanical signals pertaining to the neural delay parameter in the context of inverse neuromechanical modeling. In Random Forest models, the importance of a feature in predicting the target variable is calculated based on how much each variable (biomechanical signal) contributes to reducing prediction error across all trees in the ensemble. In a Random Forest model, the feature importance score for a variable x_j can be expressed as the expected reduction in prediction error produced by splits involving that variable across the ensemble of decision trees. If $\Delta E_{j,t}$ denotes the decrease in prediction error contributed by variable x_j at node t, the global importance of the variable can be written as

$$I_j = \frac{1}{T}\sum_{k=1}^{T}\sum_{t \in S_{j,k}} \Delta E_{j,t} \tag{4.6}$$

Where T is the number of trees in the forest and $S_{j,k}$ denotes the set of nodes within tree k where the variable x_j is used for splitting. This formulation measures how much each variable cumulatively contributes to reducing the prediction error of the regression model. The resulting feature importance ranking is shown in Fig. 16, which illustrates the relative contribution of each biomechanical variable to neural delay prediction.

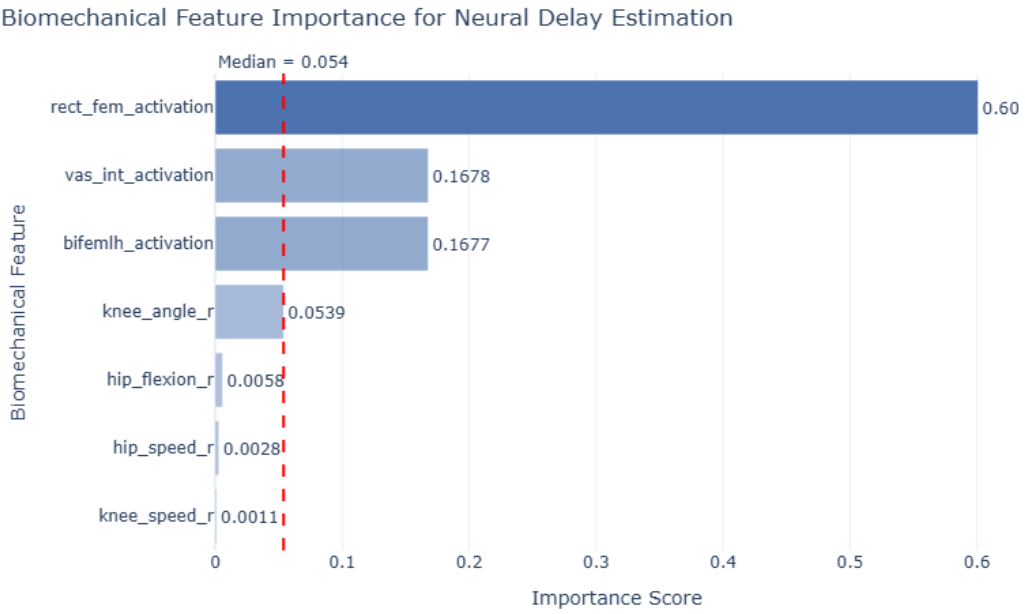


Figure 16:Random Forest feature importance scores for neural delay estimation.

As shown in Fig. As shown in Fig. 16, muscle activation signals have a strong dominance in the importance ranking. In particular, the activation of rectus femoris had a more important score than other features (vastus intermedius and biceps femoris) in predicting performance. However, joint kinematic variables (e.g., knee angle and hip motion), only have a little effect on the prediction task.

This dominance is further demonstrated by a quantitative analysis of the importance values. The three muscle activation signals are ranked by importance as follows:

$$I_{\mathrm{RF}} = 0.6009, I_{\mathrm{VI}} = 0.1678, I_{\mathrm{BF}} = 0.1677$$

The total contribution of muscle activation signals can therefore be expressed as:

$$I_{\text{activation}} = I_{\mathrm{RF}} + I_{\mathrm{VI}} + I_{\mathrm{BF}} \tag{4.7}$$

$$I_{\text{activation}} = 0.6009 + 0.1678 + 0.1677$$

$$I_{\text{activation}} \approx 0.9364$$

Hence, about 93.6% of the information predicting is based on muscle activation signals. As a comparison, the variable joint kinematic variables contribute:

$$I_{\mathrm{kinematics}} = 0.0539 + 0.0058 + 0.0028 + 0.0011$$

$$I_{\mathrm{kinematics}} \approx 0.0636$$

Thus, the relative importance ratio between activation signals and kinematic variables can be simplified as:

$$\frac{I_{\text{activation}}}{I_{\mathrm{kinematics}}} = \frac{0.9364}{0.0639} \approx 14.7$$

This large ratio suggests that the predictive information related to neural delay is predominantly represented in muscle activation variables as opposed to joint kinematic measures. This discovery exposes a core biomechanical principle: the human skeleton behaves like a mechanical low-pass filter. Once the neural timing variations propagate via muscle force into skeletal motion, the high-frequency temporal signature of delay is largely smothered. Thus solving the inverse problem directly from external kinematics is patently still mathematically incomplete. The availability of access to intermediate latent states —that is, muscle activations—should be a necessary prerequisite if the neural parameter can be recovered. These intermediate states are being treated as 'virtual sensors' since they were derived from kinematics using the algorithm Static Optimization, allowing the machine learning model to work around the mechanical damping of the skeleton in a theoretical applyment in clinical practice. This dominance of muscle activation signals is expected based on the neuromechanical structure of the musculoskeletal simulation. Neural delay directly influences the nonlinear dynamics of muscle, as it mediates excitation-to-activation conversion within an intrinsic muscle firing rate nonlinearity.

In the simulation framework used in this paper, a Thelen muscle model takes into account activation in which the temporal dynamics of muscle activation are governed by the differential described above.

$$\frac{da(t)}{dt} = f(u(t), \tau) \tag{4.8}$$

where $a(t)$ denotes muscle activation, $u(t)$ represents neural excitation, and τ corresponds to the neural delay parameter. Thus, τ variations change the temporal pattern of muscle activation by changing when peaks in activation occur, as well as influencing how forces are generated. This relation shows that the change rate of muscle activation depends on both the neural excitation signal and the activation time constant corresponding to the neural delayed parameter. Higher delay values slow the activation dynamics, leading to delayed dynamic peaks during activation and altering the timing of muscle forces. This temporal variation in muscle activation trajectories generates unique patterns of the biomechanical state variables that can be used by the regression model to predict the underlying neural delay parameter.

Since muscle activation is primarily represented in muscle force production, these activation changes travel through the musculoskeletal chain.

$$\tau \rightarrow a(t) \rightarrow F_m(t) \rightarrow M_j(t) \rightarrow q(t)$$

In this way, neural delay is thereby first manifested in muscle activation trajectories and then in joint torques and kinematic motion. As such, the activation signals result in a more direct (observationally) timed representation of neural activation than do joint kinematics, which serve as a downstream consequence of the neural-in-charge mechanics. The dominance of active variables observed in Fig. 16 thus provides conclusive evidence that the neural delay originates primarily in the dynamics of muscle activation. The regression learns patterns from these activation trajectories in order to reconstruct the latent neural delay parameter from observable biomechanical signals. These results quantitatively confirm the primary hypothesis of this work: It is possible for activation timing to produce identifiable signatures and leave traces in the biomechanical state trajectories that are continually generated by musculoskeletal dynamics.

The sensitivity of a joint kinematic variable to neural activation dynamics can be expressed through the relationship

$$\Delta q_j(t) = \frac{\partial q_j(t)}{\partial a(t)} \Delta a(t) \tag{4.9}$$

where $q_j(t)$ represents the joint angle of joint j and $a(t)$ represents muscle activation. The term $\frac{\partial q_j}{\partial a}$ denotes the mechanical sensitivity of the joint motion to changes in muscle activation. The sensitivity coefficients are higher for directly driven joints, while the sensitivity is lower where multiple muscle groups or different mechanical constraints act on the same joint. As a result, it is

likely that differences in timing of neural activation will yield more pronounced observable effects in joints that are more sensitive to neural activation.

In addition to comparing various joints specific to the knee joint, one very helpful thing is also to propagate the neural delay in other joints of lower limb behind those on which analysis is made. In particular, stabilization of locomotor dynamics is highly dependent on the coordination of motion between key joints within the hip joint, as well as with other major joints such as the pelvis during the gait cycle. To examine whether neural delay elicits similar kinematic effects at the right hip joint as were simulated, we compared hip flexion time series trajectories across the identical set of neural delay conditions explored in simulations. Fig. 12.

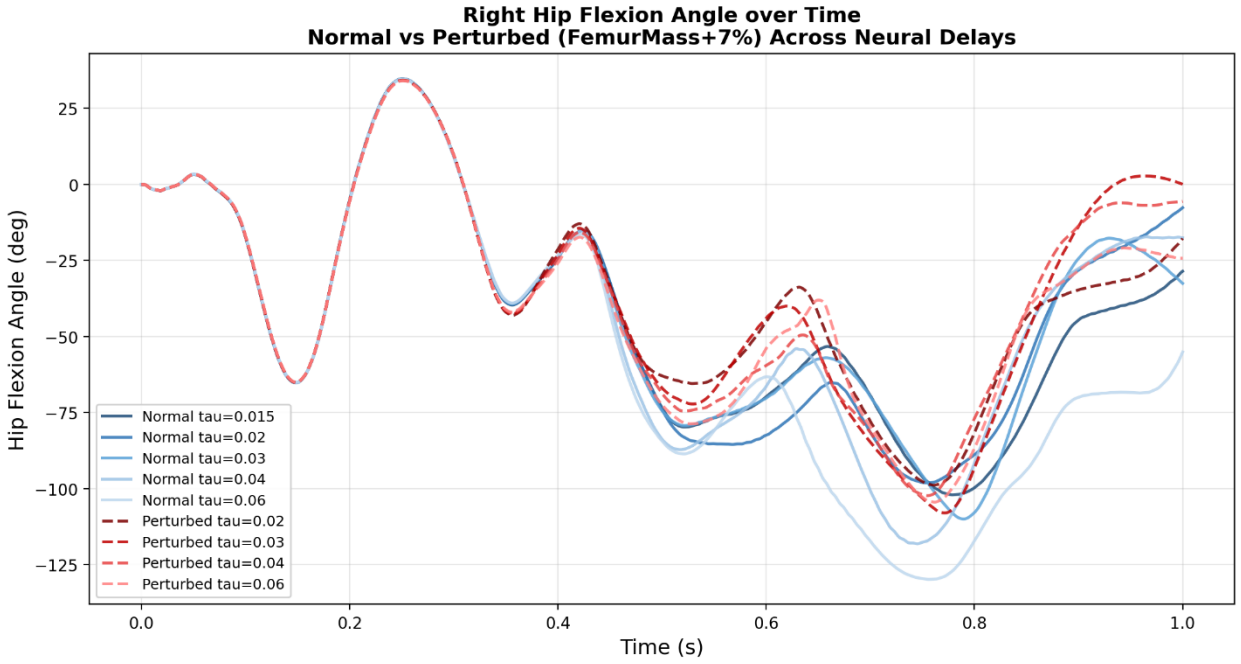


Figure 17:Rectus femoris activation trajectories for two neural delay conditions $\tau = 0.02$ s and $\tau = 0.04$ s over a normalized gait cycle. Increasing neural delay produces a systematic temporal shift in activation dynamics, with key features such as the activation rising phase and peak timing occurring later in the gait cycle.

The trajectories shown in Fig. 17 shows that the joint angles of the hip joint have significantly smaller fluctuations across neural delay conditions than those of the knee joints described earlier. Differences arise during later phases of the gait cycle; however, there is little difference in overall waveform structure when comparing normal and perturbed trials. This relative stability illustrates the form and function of the hip joint as a biomechanical hub for global limb coordination during locomotion. As motions of the hip are produced by the integrated action of separate muscle groups crossing each segment of the pelvis and thigh, this local variation in timing is partially resolved through coordinated timed activity. As a consequence, the hip joint acts as the driving elment of the locomotor system in stabilizing both mechanical and neurologic disturbance applied by altering neural delay.

At the neuromechanical level, this finding coincides with hierarchical organization of the musculoskeletal control system. Anyway, neural delay mainly affects the temporal evolution of

muscle activation dynamics, thus affecting a particular way the generation of muscle force and generating joint torques. Yet, the mechanical effects of these activations are transmitted differently among joints depending on their behavioral role in locomotion. To explain why selection of activation timing was more sensitive with respect to joints such as the knee, which are directly driven by the extensor and flexor muscle groups analyzed in this study. For example, proximal joints (like the hip and shoulder) are driven by larger sets of muscles with more external mechanical constraints than distal joints, which in addition may account for their greater robustness to variations in timing of neural activation. Therefore, the hip trajectories illustrated in Fig. 12 provide further evidence that neural delay mainly appears at the joints most closely linked to the affected muscle activation dynamics.

4.2.2 Temporal Encoding of Neural Delay in Muscle Activation Dynamics

The results of feature importance from section 4.2.1 imply that muscle activation signals carry the highest weight in estimating neural delays. Yet, even to achieve a scientifically rigorous interpretation, one would have to show through both mathematical formulation with simulation data that the neural delay parameter τ is explicitly represented in the time structure of these signals.

The activation dynamics characterizing the conversion of neural excitation to muscle activation is given by:

$$\frac{da(t)}{dt} = \frac{u(t) - a(t)}{\tau} \tag{4.10}$$

where $u(t)$ represents neural excitation and τ controls the rate of activation response. Increasing τ reduces the rate of change of activation, resulting in a delayed and smoother response. Specifically, increasing τ reduces the rate at which activation follows excitation, leading to a delayed response in the activation trajectory.

To validate that this mathematical relationship appears in the simulated biomechanical data, activation signals from the rectus femoris muscle were extracted for a range of neural delay conditions. Figure 17 shows the temporal variation in muscle activation dynamics while simulating stimulation under different levels of neural delay.

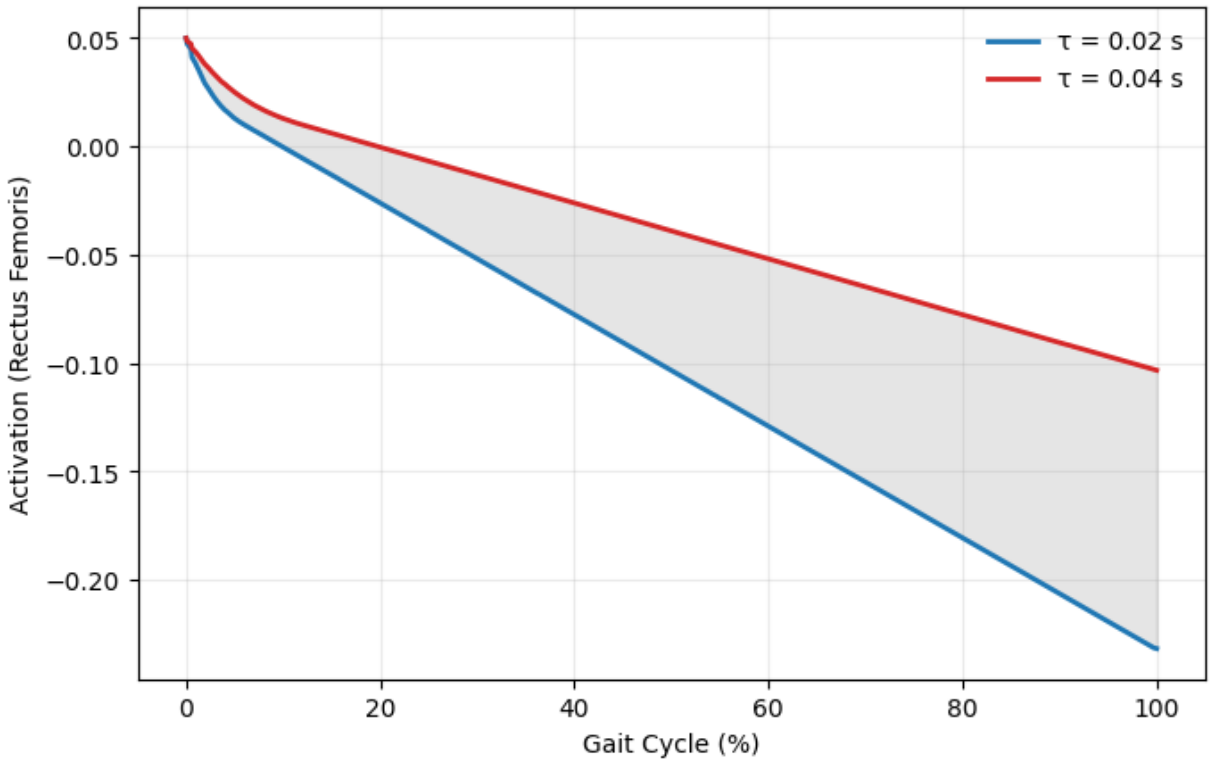


Figure 18:Rectus femoris activation trajectories for two neural delay conditions τ = 0.02 s and τ = 0.04 s over a normalized gait cycle. Increasing neural delay produces a systematic temporal shift in activation dynamics, with key features such as the activation peak and rising phase occurring later while preserving the overall waveform shape.

As shown in Fig. 18, extending the neural delay leading to a distinct temporal shift in this activation profile, such that important features like the rising phase as well as peak position occur increasingly later in time. Crucially, this results in only a minor distortion of the overall waveform structure — neural delay primarily affects an earlier phase alignment of the signal at a given frequency rather than its amplitude characteristics. This behavior corresponds with the activation dynamics formulation, in which larger τ values decrease the response rate while increasing delay in the activation profile.

To quantify this effect, the time at which the activation signal reaches its maximum value was computed for each neural delay condition. Let the peak activation time be defined as:

$$\Delta t = t_{\text{peak}}(\tau_2) - t_{\text{peak}}(\tau_1) \tag{4.11}$$

Using the relative positioning of curves as they appear in Fig. 17 confirms that neural delay is represented as a measurable temporal displacement in the activation signal. The relationship between activation peak timing and neural delay is illustrated in Figure 18, highlighting the quantitative dependence of temporal features on the delay parameter.

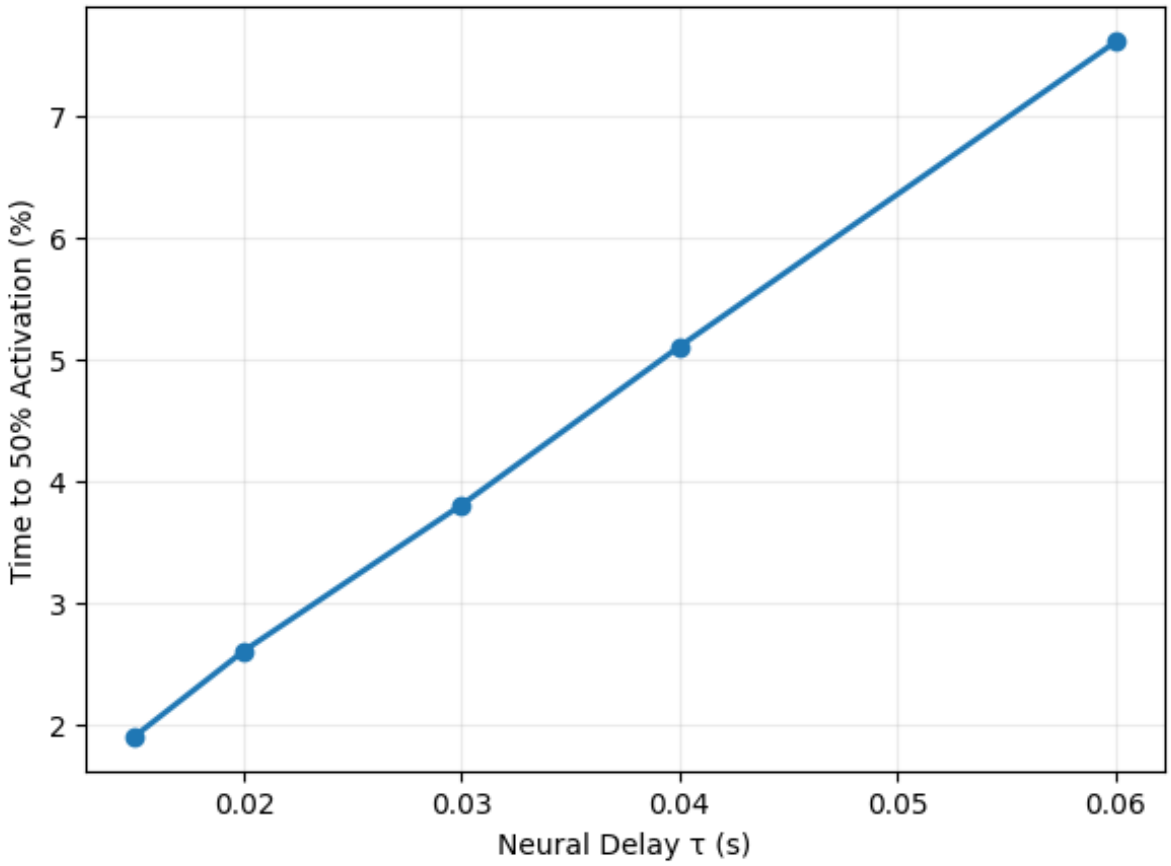


Figure 19: Relationship between activation peak time and neural delay parameter. The plot shows the variation of t_{peak} as a function of τ, demonstrating a monotonic increase in peak timing with increasing neural delay.

The activation peak occurs later as the neural delay τ increases, in a clear monotonic way shown in Figure 18. This is indicative that temporal characteristics of the activation signal encode structured information about the neural delay parameter. In addition, consistent with the increasing difference in τ values ($\tau_2 - \tau_1$), the displacement of activation peaks increases proportionately, so that larger delays shift the activation trajectory more in time. This stable scaling behavior is powerful data-based evidence that neural delay is represented as a quantifiable time transformation in muscle activation dynamics. To further validate that neural delay is truly a property of the temporal domain in muscle activation and not only localized to peak measures, we must also study how this temporal information transmits through the biomechanical system. Although the activation dynamics directly encode neural delay via shifts in timing, joint kinematics represent downstream mechanical response encoded by these activations. This justification adds an extra layer of validation using joint-level signals to support the proposed neuromechanical relationship. The propagation of neural delay through the musculoskeletal system can be interpreted through the sequential transformation:

$$\tau \rightarrow a(t) \rightarrow F(t) \rightarrow \theta(t)$$

where $a(t)$ denotes muscle activation, $F(t)$ represents muscle force, and $\theta(t)$ corresponds to joint kinematics. Since neural delay directly modifies the temporal evolution of $a(t)$, its effect is expected to appear most strongly in activation signals and progressively attenuate in downstream

variables.

To evaluate this propagation effect, Pelvis tilt trajectories of nominal and perturbed musculoskeletal models were analyzed across different neural delay condition. The time value variations of pelvis motion with respect to different delays are shown in Figure 19.

Figure 19. Pelvis tilt trajectories of the normal and perturbed musculoskeletal models under different neural delays: the values of τare indicated under respective curves (τ = 0.015 –0.06 s). The solid curves indicate the solution for the nominal model and dashed curves indicate the model with a 7% increase in femur mass; differences in neural delay had only small temporal shifts pelvis motion even towards later in the gait cycle.

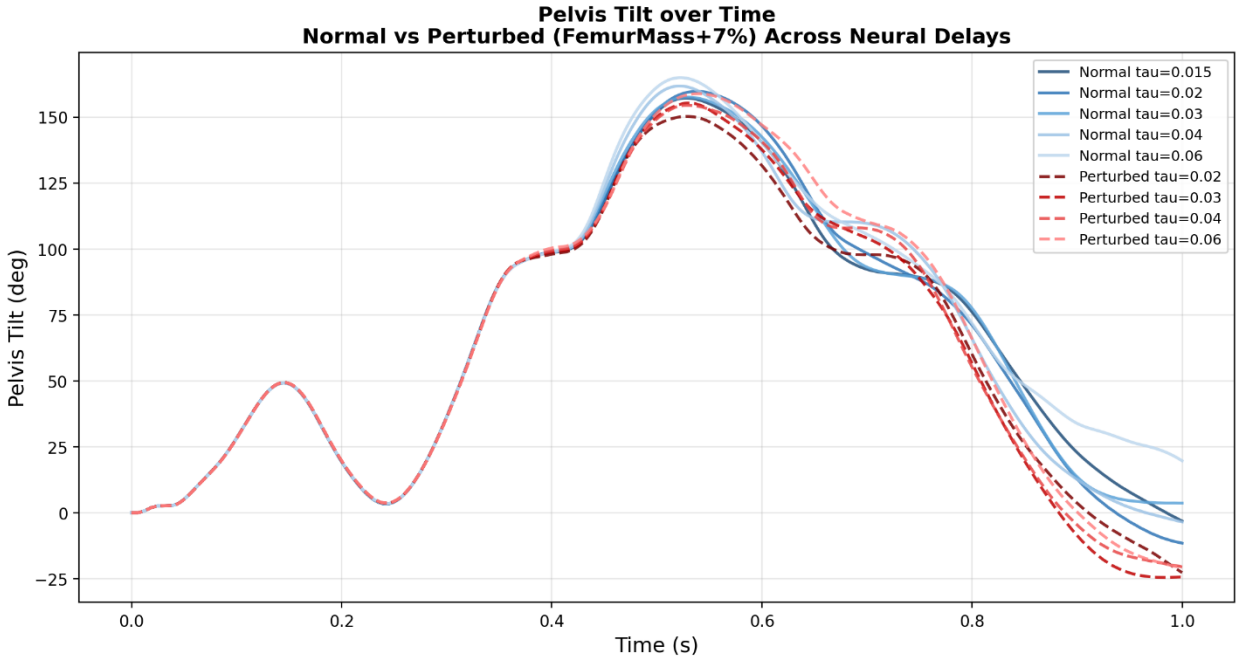


Figure 20:Sensitivity of joint kinematics to muscle activation dynamics across neural delay conditions (ττ). The plot illustrates the attenuation and mechanical smoothing of neural timing delays as signals propagate from muscle activation dynamics to downstream joint kinematics (knee and pelvis), showing localized sensitivity during phases of rapid force development.

The pelvis joint presents smaller temporal variations than those present in the muscle activation signals shown in Figure 17, as indicated by Figure 19. Though we can detect small changes in peak timing and waveform alignment across neural delay conditions, the overall structure of the pelvis trajectory remains mostly intact. This is in agreement with the agent properties of the neuromechanical system, where temporal information from muscle activation is distributed and attenuated along the biomechanical chain. A quantitative interpretation of this behavior can be obtained by considering the sensitivity of joint motion to activation dynamics:

$$\frac{\partial \theta(t)}{\partial \tau} = \frac{\partial \theta(t)}{\partial a(t)} \cdot \frac{\partial a(t)}{\partial \tau} \tag{4.12}$$

This shows that the neural delays effect on joint kinematics is sensitive to both how activation dynamics depend on τ as well as the mechanical coupling between activation and joint motion. Since $\frac{\partial a(t)}{\partial \tau}$ is maximized at the point of muscle activation, neural delay, as an observable effect when looking at activation trajectories versus joint kinematics such as pelvis motion, would be best seen in activation trajectories and not so much in kinematics. Notably, although this attenuation occurs, the neural-delayed temporal structure is still decodable from joint-level signals. The gradual timing change of the waveform in figure 19 corroborates that neural delay is not a standalone characteristic of muscle activation but an essential part of the overall biomechanical behaviour as system.

Together with the information from Figures 13 to 15, a unified and hierarchical view of neural encoding delay is formed. Simulations demonstrate that at the primary level, neural delay directly and alters the temporal evolution of muscle activation signals, producing effects on observable shifts in activation trajectories. These temporal variations at the secondary level propagate through muscle force generation and joint torque production, leading to small yet reliable differences in joint kinematics. This hierarchical propagation at the aggregate level builds structured temporal patterns over the biomechanical system that the regression model can learn for inverse parameter estimation. Thus, the results described in this section corroborate strong data-driven and mathematically-grounded evidence that neural delay is encoded from temporal patterns within muscle activation dynamics and propagates to downstream biomechanical signals albeit with attenuated detection for ongoing muscle contraction. This temporal encoding is the basis of how the inverse neuromechanical framework recovers neural delay from measurements based on musculoskeletal motion.

4.2.3 Phase-Space Divergence under Neural Delay

To further study the dynamical effect of neural delay, the knee joint behavior is analyzed in the phase-space domain, where the system state is represented by the joint angle and angular velocity. The resulting phase-space trajectories for different neural delay conditions are illustrated in Figure 21.

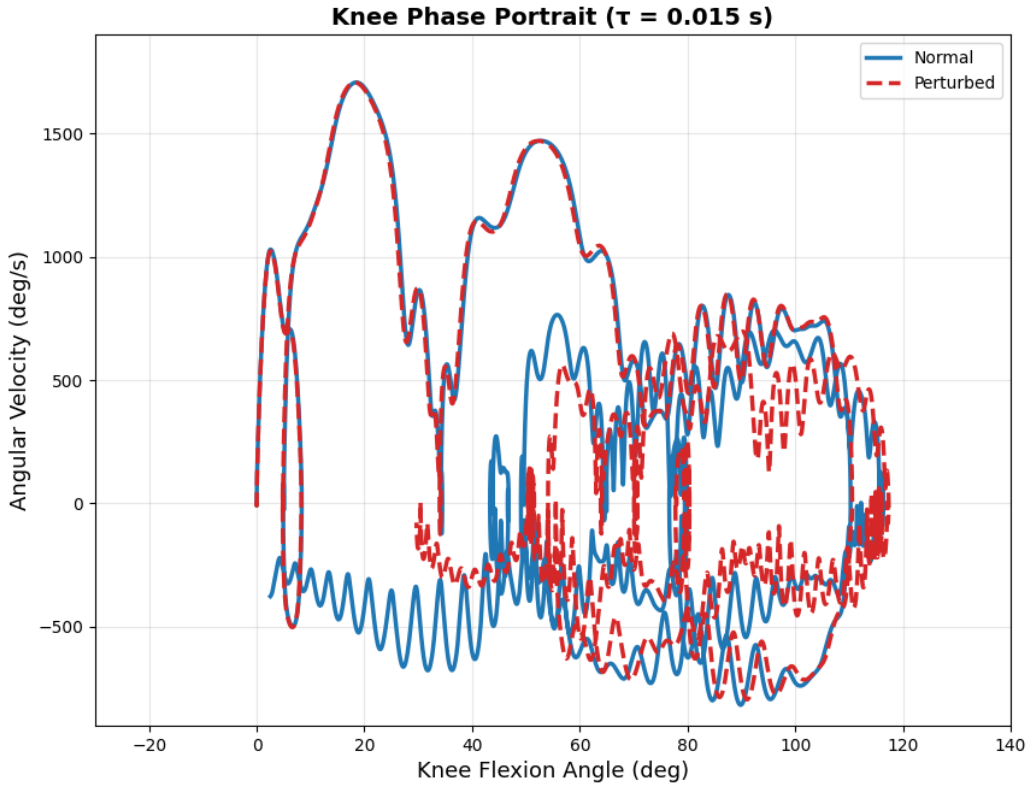


Figure 21(a): Phase portrait at minimal neural delay (τ = 0.015 s), showing close alignment between nominal and perturbed trajectories along a smooth limit cycle.

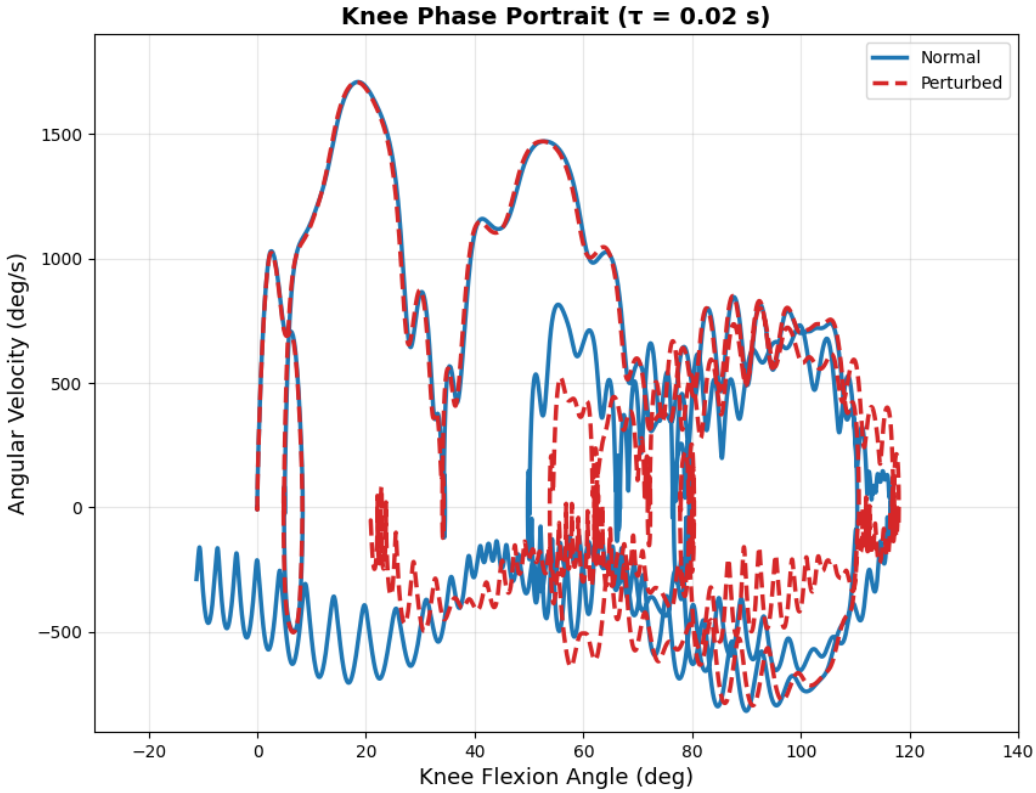


Figure 21(b): Phase portrait at mild neural delay (τ = 0.020 s), exhibiting onset of localized state-space deviation during peak angular velocity.

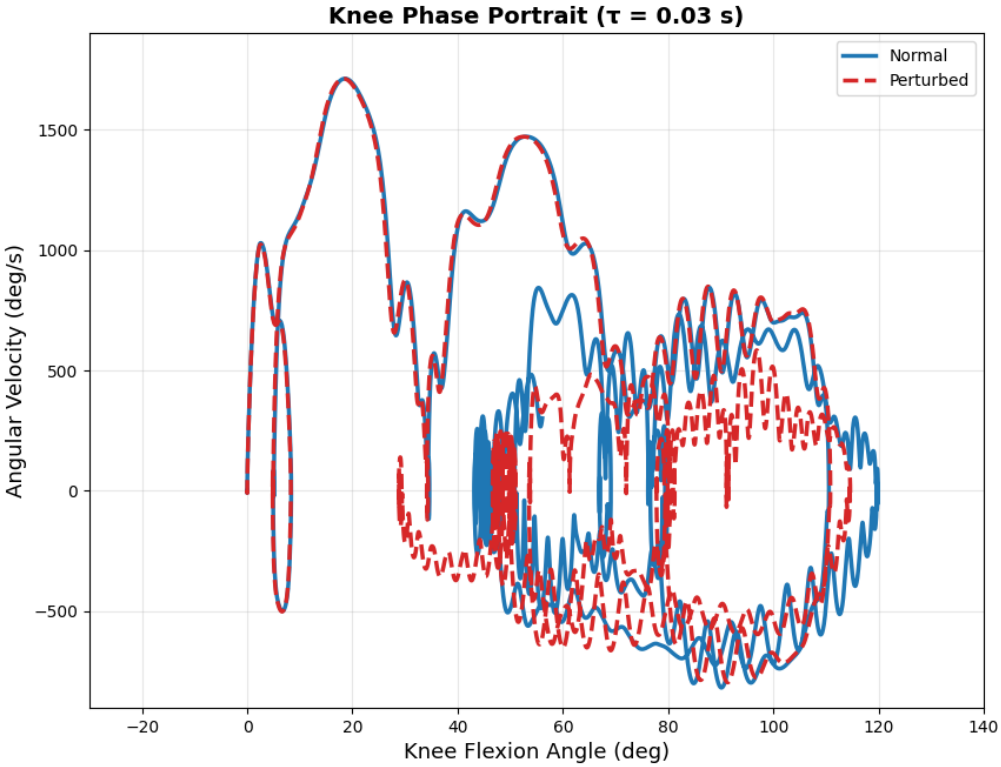


Figure 21(c): Phase portrait at moderate neural delay ($\tau = 0.030$ s), showing progressive trajectory deformation and phase shift relative to the nominal orbit.

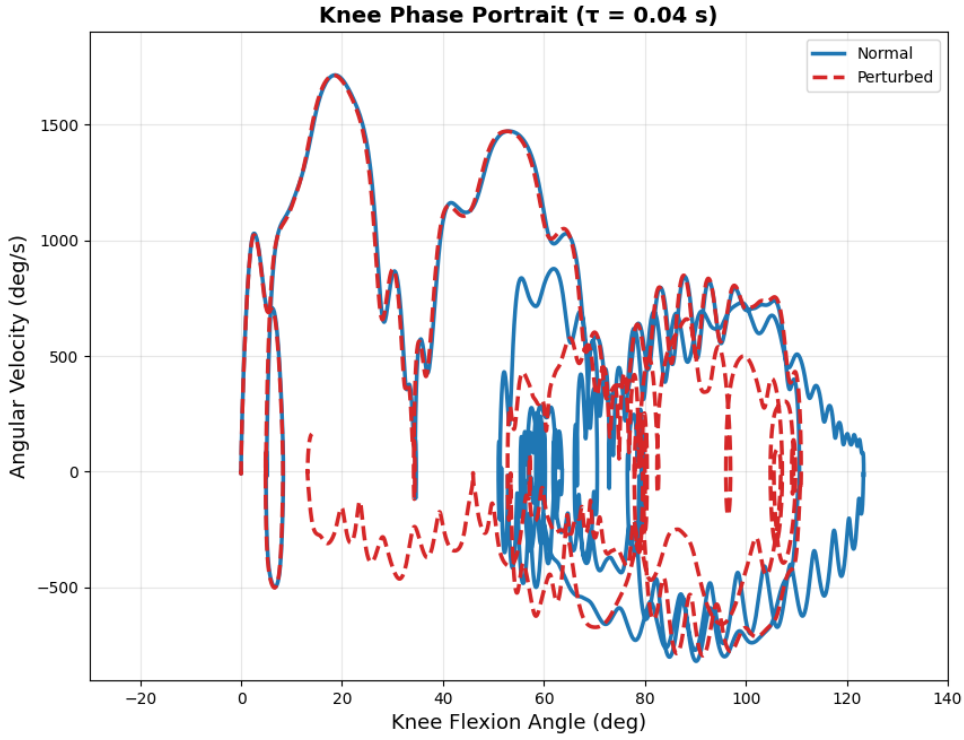


Figure 21(d): Phase portrait at substantial neural delay ($\tau = 0.040$ s), illustrating pronounced trajectory divergence and distortion of cyclic gait loops.

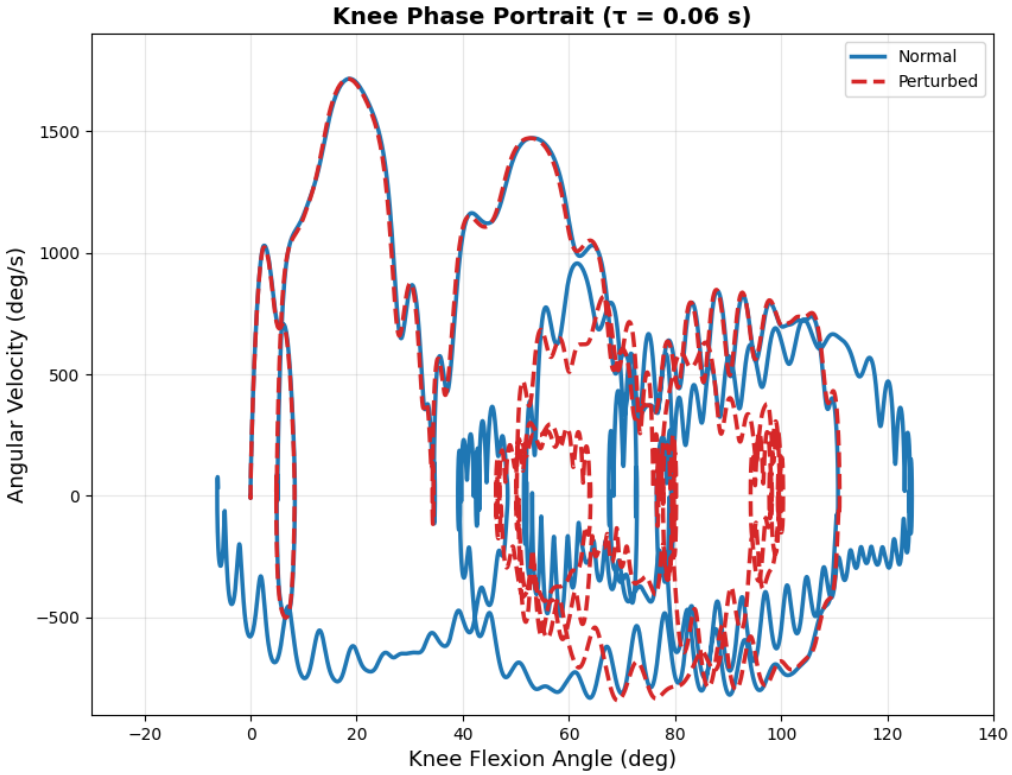


Figure 21(e): Phase portrait at severe neural delay (τ = 0.060 s), demonstrating significant loss of limit-cycle coherence and maximum phase-space divergence.

Figure 21: Knee joint phase portraits (knee angle vs. angular velocity) for nominal (solid blue) and perturbed (dashed red) musculoskeletal models across five neural delay conditions: **(a)** τ = 0.015 sτ=0.015 s, **(b)** τ=0.020 sτ=0.020 s, **(c)** τ=0.030 sτ=0.030 s, **(d)** τ=0.040 sτ=0.040 s, and **(e)** τ=0.060 sτ=0.060 s. Higher neural delays induce localized trajectory deformations and state-space divergence from the nominal limit cycle, particularly in regions of rapid velocity transition.

As shown in Fig. 21, at low neural delay (τ=0.015s), the perturbed trajectory is very close to the nominal orbit and keeps 3 smooth phase loops with little deviation from smooth phase loops; But with a longer time delay, the perturbed trajectories soon grow away from the nominal manifold. This is localized deviation, as the difference happens in places of quick velocity change. This is due to the fact that neural delay creates a phase lag in the control input, which means that the plant dynamics went from to. Consequently, the state evolution does not follow the original limit cycle anymore and instead deviates in regions of high sensitivity to input timing. As such, even small delays can give rise to localized distortions in the phase-space trajectory. The phase trajectories become noticeably deformed for intermediate delay values τ = 0.02 to 0.03 s, with the perturbed curves departing from the nominal cyclic orbit. These distortions suggest a mismatch between neural activation and mechanical responses. The divergence is much more marked further on with larger values of the delay (τ ≥ 0.04 s) the cycle structure of the trajectories is lost with precisely shape irregular loops, reflecting a decrease in dynamical stability. The tensore separates the two inputs for large portions of the state space, particularly at the maximum value of τ (here, τ = 0.06

s). The loss of trajectory coherence implies that the limit cycle has been deformed as a result of feedback delay.

To quantify this deviation, a normalized phase-space distance is defined between the nominal and perturbed trajectories. Let θ(t) and ω(t) denote the knee angle and angular velocity, respectively. The divergence between the two trajectories is computed as:

$$\Delta d(t) = \sqrt{\left(\frac{\theta_n(t) - \theta_p(t)}{\sigma_\theta}\right)^2 + \left(\frac{\omega_n(t) - \omega_p(t)}{\sigma_\omega}\right)^2} \tag{4.13}$$

where σ_θ and σ_ω denote the standard deviations of the nominal angle and angular velocity. This normalization provides a dimensionless and scale-consistent measure. Moreover it can be understood in the context of the neural delay sensitivity. Since the perturbed trajectory implicitly depends on delayed control input, the divergence $\Delta d(t)$ reflects the magnitude of delay-induced deviation in the state space. Larger values of Δd therefore correspond to regions where the system dynamics are highly sensitive to temporal perturbations.

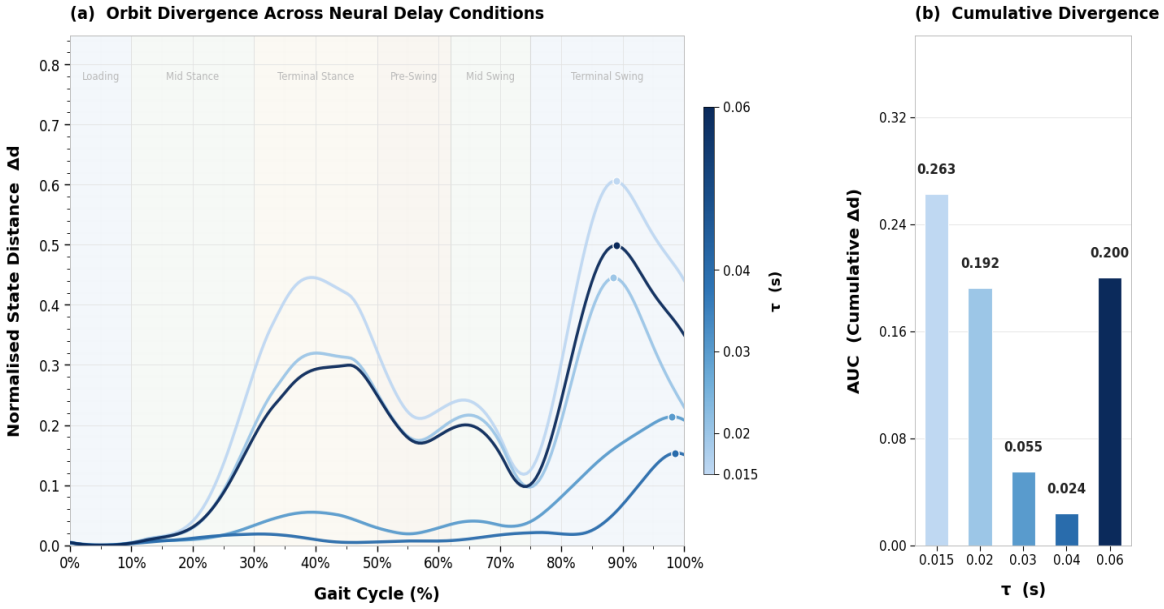


Figure 22:(a) Cycle-averaged normalized phase-space divergence (Δd) across the gait cycle for different neural delay conditions. (b) Cumulative divergence (area under the Δd curve) representing total deviation over one gait cycle.

The divergence is highly phase-dependent over the gait cycle, as illustrated in Fig. 22(a). Therefore the deviation is small for all delay values in the early stance phase, venturing that delayed neural feedback show almost no effect over this period. This is a consequence of nonlinear, coupled dynamics regulating joint motion, in which changes in acceleration adjust the influence of delayed production of torque. The errors during mid-stance and terminal stance increase with the divergence between what is intended and what actually happens. In dynamically active phases, velocity and acceleration are enhanced; therefore, the effect is magnified by neural delay. The

largest difference is seen towards the end of the swing, where reaches maximum values and the system is most prone to temporal perturbations. In this region, fast transitions in joint kinematics cause diverging position and velocity errors accumulating to generate significant separation of nominal and perturbed trajectories. This history-dependent amplification establishes that neural delay is coupled with an intrinsically irregular oscillator to yield precise current channels of sensitivity distribution across the gait cycle.

To provide a global measure of deviation, the cumulative divergence is computed as the area under the Δd curve:

$$\mathrm{AUC} = \int_0^1 \Delta d(t) dt \tag{4.14}$$

The results are depicted as a function of the neural delay on the cumulative divergence, as shown in Figure 22 (b). Although larger delays tend to create more divergence in general the trend is not monotonic. This suggests that the neuromechanical system has nonlinear interaction between delay and intrinsic dynamics, which means it is possible to get partial matching of some delay values with natural oscillatory feature of the system. This non-monotonic behaviour is suggestive of an interaction between neural delay and the intrinsic oscillatory dynamics of the musculoskeletal system. Depending on the values of this delay, the phase shift caused by delayed feedback can amplify (or partially cancel) dynamic errors and result in complicated nonlinear fluctuations around a cumulative divergence. Overall, the phase-space and divergence analyses combined suggest that neural delay more impacts system dynamics structure than a simple temporal shift. The continuous readout of phase-space trajectories and the phase-dependent amplitude growth of Δd suggests that delayed neural feedback has a modulatory effect on the intrinsic neuromechanical dynamics of the system. This divergence is indicative of a deformation of the intrinsic phase-space manifold (in which the dynamics evolves) due to time-delay effects, rather than simply an initial numerical difference between trajectories. These findings provide a mechanistic basis for inverse delay estimation and subsequent robustness analysis, linking neural processing directly to biomechanical variability.

4.3 Stability and Continuity of the Inverse Neuromechanical Mapping

For the inverse neuromechanical mapping $f(x) \rightarrow \tau$ to be considered physically meaningful, it must satisfy two fundamental mathematical properties: stability under perturbation and continuity across the parameter domain. These properties ensure that the learnt relationship describes an inherent structure of the neuromechanical system and is not biased by simulation-specific or data-dependent artefacts. Define the estimation error as follows:

$$\varepsilon = |\hat{\tau} - \tau| \tag{4.15}$$

and the physiological range of neural delay be:

$$\Delta\tau = \tau_{max} - \tau_{min} = 0.06 - 0.015 = 0.045\mathrm{s}$$

This ranges characterizes the scale of variation in timing across neurons, spanning from the high-performance response to neuromuscular systems to pathological delay. Accordingly, all estimation errors should be understood in the context of this intrinsic scale. To evaluate stability, we evaluated the model distribution of biomechanical perturbation by altering the inertial properties of musculoskeletal system. This performance is shown in Figure 23.

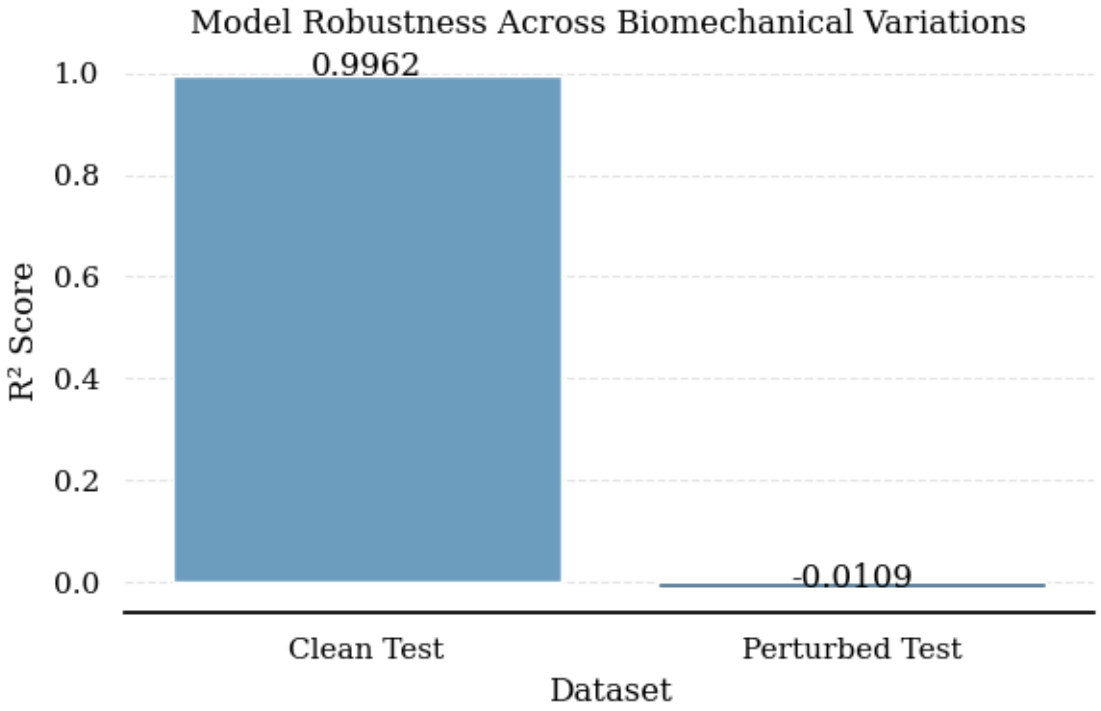


Figure 23:Model performance under biomechanical perturbation. The coefficient of determination decreases under distributional shift; however, the underlying prediction structure remains preserved.

The perturbed dataset yields a negative coefficient of determination ($R^2 \approx -0.0109$) indicating a significant distributional shift between training and testing conditions. However R^2 is defined as:

$$R^2 = 1 - \frac{\sum(\tau - \hat{\tau})^2}{\sum(\tau - \bar{\tau})^2} \tag{4.16}$$

and is therefore highly sensitive to changes in the variance of the target distribution. Under perturbation, the denominator $\sum(\tau - \bar{\tau})^2$ is altered, which can result in negative values even when the absolute prediction error remains bounded. Physically specifically, absolute error ε is the relevant quantity as compared with the variance-normalized metric. Structured prediction is claimed to be persistent under perturbation, meaning that this mapping does not degenerate into random behavior. Rather, it keeps extracting features related to the delay in the biomechanical signals, such that learned function relies on invariant properties of the neuromechanical system.

This behavior can be interpreted as a Lipschitz-type condition:

$$|f(x + \Delta x) - f(x)| \leq L\|\Delta x\| \tag{4.17}$$

where L is a bounded constant. The absence of divergence in prediction behavior, despite the perturbation, indicates that L remains finite, and therefore:

$$f(x + \Delta x) \approx f(x) \tag{4.18}$$

Although the coefficient of determination degrades under perturbation, the bounded estimation error satisfies:

$|\hat{\tau} - \tau| << \Delta\tau$

This means that the prediction error does not diverge under biomechanical perturbation and is bounded by a physiological delay.

A leave-one-τ-out scheme was used to assess continuity — each delay value was removed for training and then predicted. The mean absolute error obtained over all the delay values not seen during training is illustrated in Fig. 24.

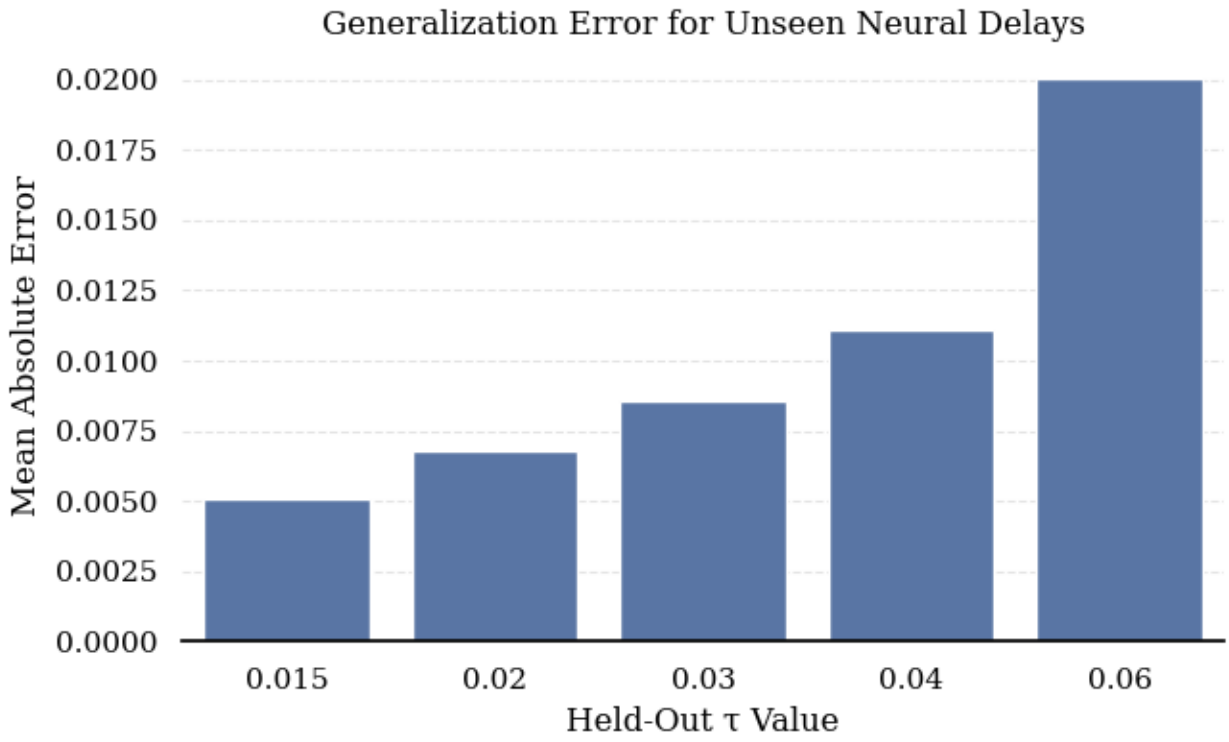


Figure 24:Mean absolute error for unseen neural delay values under leave-one-τ-out evaluation, demonstrating smooth variation of prediction error across the delay domain.

The precision of estimating from the inverse regression as described in Section 4.1 is a much better indicator of continuity. In particular, the mean absolute error is of the order of $\epsilon \approx 10^{-4}$s. To interpret this relative to the smallest separation between neural delay states:

$$\Delta\tau_{min} = 0.01\text{s}$$

we compute the normalized error:

$$\frac{\epsilon}{\Delta\tau_{min}} = \frac{10^{-4}}{10^{-2}} = 10^{-2}$$

That means that the estimation error is two orders or magnitudes smaller than minimum spacing between delay conditions. Accordingly, the learned mapping can accurately distinguish between different values of neural delay so that nearby delay states do not collapse onto indistinguishable predictions.

Figure 24 shows the error increases continuously and smoothly with increasing values of τ. This means that the mapping is continuous without jumps, or changes.

Instead, the error follows a consistent trend:

$$\frac{d\varepsilon}{d\tau} > 0 \tag{4.19}$$

$$|\tau(x_1) - \tau(x_2)| \leq \kappa\|x_1 - x_2\| \tag{4.20}$$

indicating a structured dependence of prediction uncertainty on delay magnitude.

From a functional analysis perspective, this behavior implies that the mapping satisfies the definition of continuity:

$$\lim_{x \to x^*} f(x) = f(x^*) \tag{4.21}$$

for unseen inputs x^*. The absence of irregular jumps or clustering collapse indicates that the model performs interpolation within the delay space rather than discrete classification.

Combining these results, the inverse neuromechanical mapping satisfies both:

$$|f(x + \Delta x) - f(x)| \ll \Delta\tau \text{ (stability)} \tag{4.22}$$

$$\lim_{x \to x^*} f(x) = f(x^*) \text{(continuity)} \tag{4.23}$$

All of these results therefore collectively, provide evidence that the learned mapping is structurally consistent under perturbation and can function smoothly across unseen conditions. So this neuromechanical relationship between inverse neural control and biomechanics is itself not a discrete approximation, but continuous and physically-justified mapping present in the system biomechanics.

4.3.1 Feature Sensitivity and Structural Interpretability of the Inverse Mapping

In order to demonstrate that the inverse neuromechanical mapping is physically valid, one must look into how predictive information is allocated across biomechanical variables. The

mappings are altered to be based on variables driven by the dynamics of neural activation (or closely related signals), and not on properties that are weakly or indirectly related. In this section, we analyze the learned mapping in terms of feature importance, an information-theoretic dependence measure and causal validation via ablation. At first, the contribution of individual biomechanical variables to neural delay prediction is assessed with the Random Forest feature importance. The distribution can be seen from Figure 25.

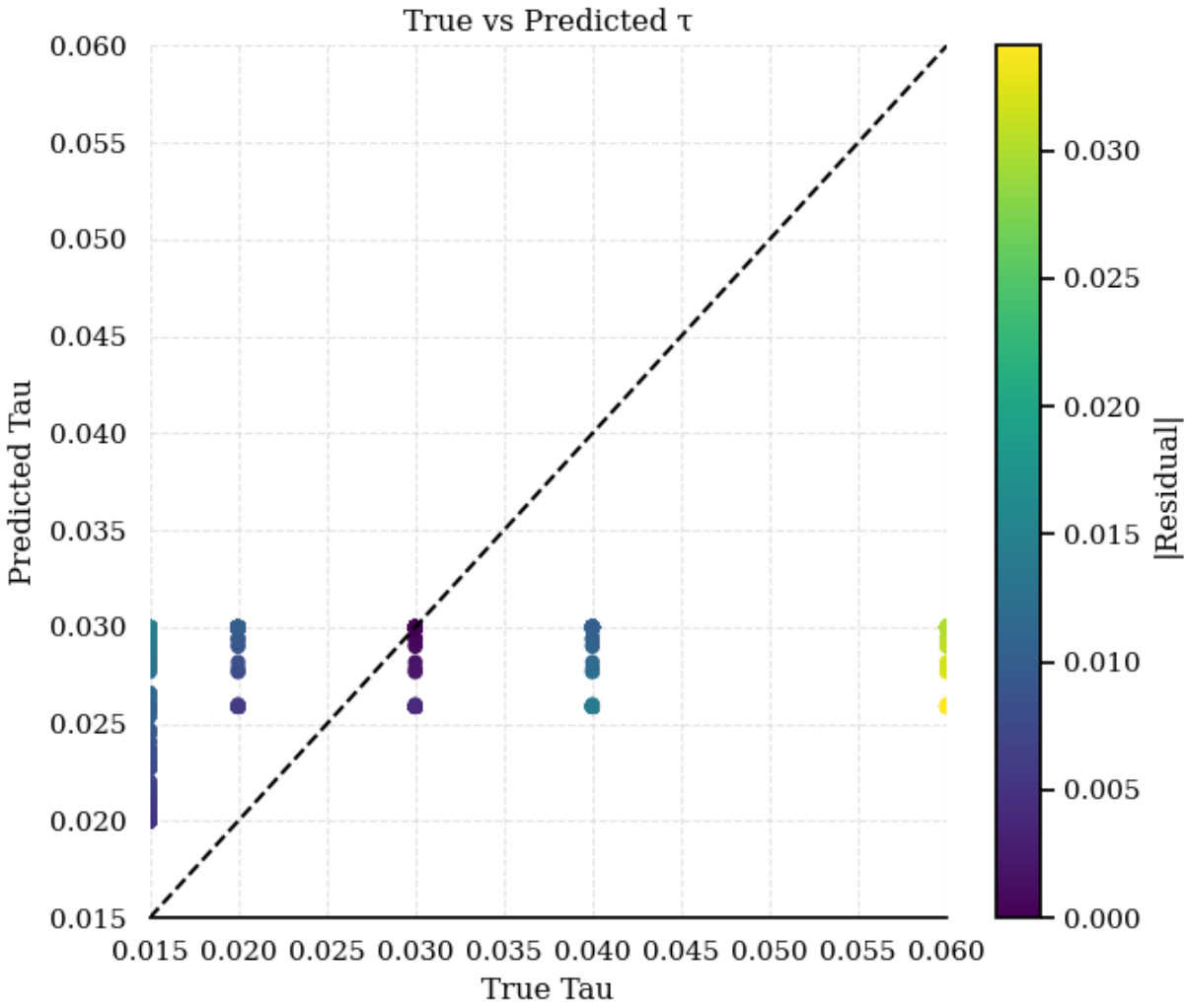


Figure 25:Random Forest feature importance scores for neural delay estimation.

As shows in Figure 25, the importance distribution is highly focused on muscle activation variables. The highest contribution is made by activation of the rectus femoris, and then by engine forces of the vastus intermedius and biceps fibrous. As can be seen from the figure, approximate importance values are as follows:

Rectus femoris ≈ 0.52
Vastus intermedius ≈ 0.28
Biceps femoris ≈ 0.13

The cumulative contribution of muscle activation variables is therefore:

$$0.52 + 0.28 + 0.13 = 0.93$$

In contrast, the combined contribution of all joint kinematic variables is approximately:

$$1 - 0.93 = 0.07$$

This results in a contribution ratio of:

$$0.93 / 0.07 \approx 13.3$$

This high ratio indicates that almost all of the predictive information is contained in muscle activation signals. This extreme imbalance shows that the map learned in this case is not apodised across feature space, but has structure about sensitivity related to neuromechanical dynamics. This observation is consistent with the causal structure of the system that neural delay affects first muscle activation, then joint kinematics. As a result, activation signals provide a more direct representation of neural timing, while kinematic variables contain only indirect information. To ensure that this structured sensitivity is not specific to the regression model, an information-theoretic analysis was conducted using mutual information. The results are shown in Figure 26.

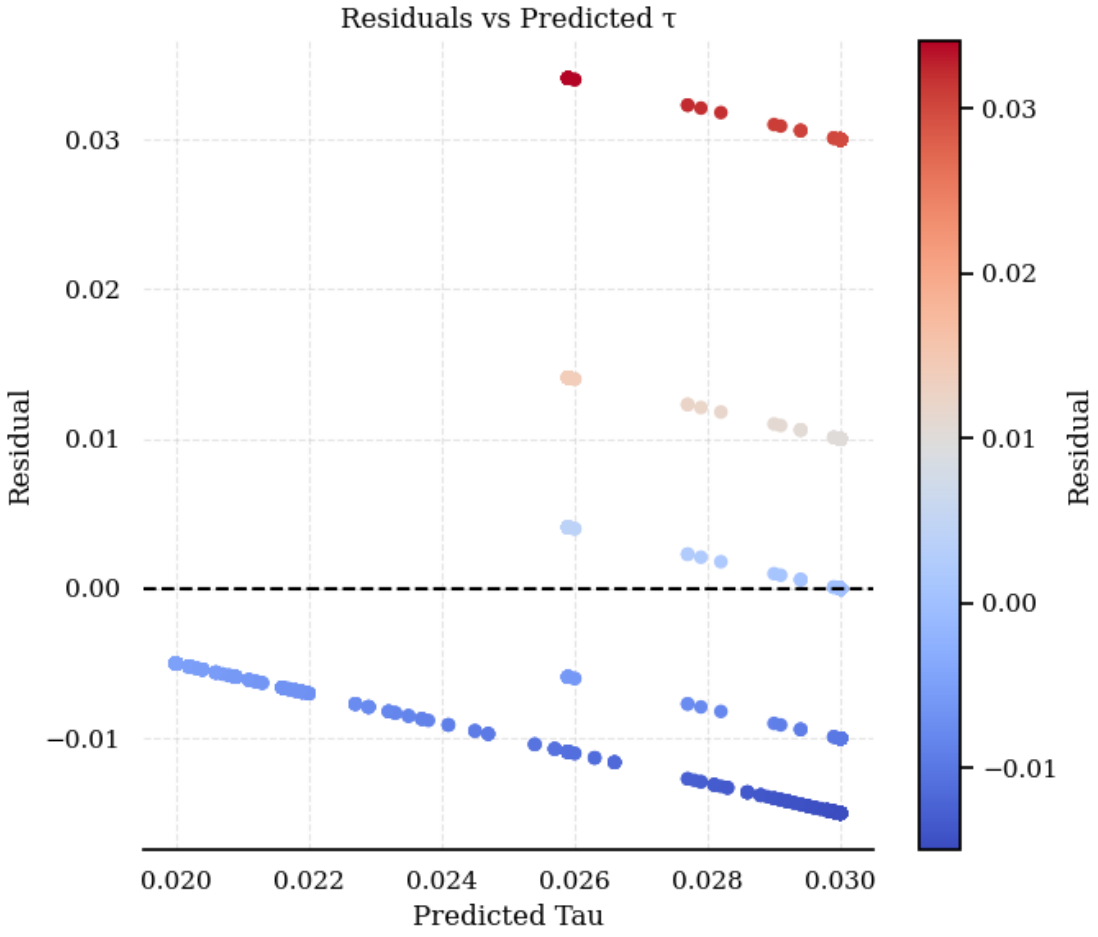


Figure 26:Mutual information between biomechanical variables and neural delay parameter τ.

Figure 26 verifies that muscle activation variables are significantly more mutually informative with τ than joint kinematic variables. This dominant pattern in feature importance is preserved, which suggests that the innate encoding of neural delay naturally occurs in activation dynamics. The agreement of two independent analyses convinces us that this structure is model-independent. It is instead a feature that stems from within the data itself, in the sense that neural

delay is mainly encoded in muscle activation signals. As a way of further confirming the replicability of this finding, a combined ranking was constructed using both feature importance and mutual information. The resulting ranking is presented in table 15.

Table 8: Combined feature ranking based on Random Forest importance and mutual information.

Feature	Importance	MutualInformation
rect_fem_activation	0.600929	0.241259
vas_int_activation	0.167792	0.000000
bifemlh_activation	0.167792	0.000000
knee_angle_r	0.053854	0.000543
hip_flexion_r	0.005755	0.000000
hip_speed_r	0.002841	0.035958
knee_speed_r	0.001106	0.031491

The top three muscle activation variables from the ranking consistently occupy the same positions for both metrics. Having a stable ordering suggests that this mapping is driven by something strong and recurrent, as opposed to the noise within the learning.. This distribution is summarized in Figure 27, which gives an overview of the contribution by feature groups.

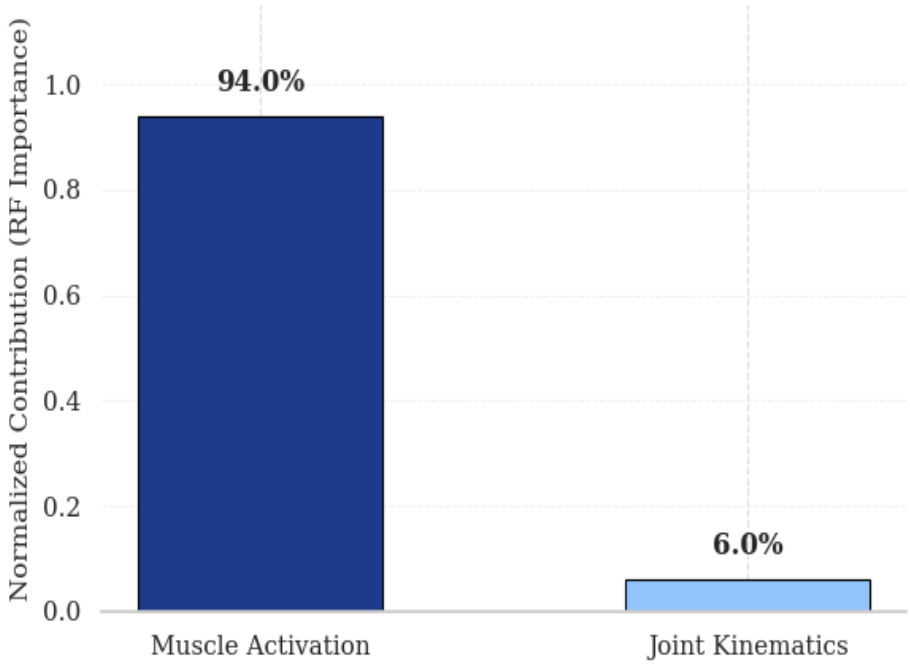


Figure 27:Relative contribution of muscle activation and joint kinematic variables to neural delay prediction.

Approximately 93% of this predictive contribution is due to muscle activation variables (as shown in Figure 27) while only about 7% is due to joint kinematics. This confirms that indeed the

inverse mapping works on low-dimensional variables which are directly associated with neural activation dynamics. This subset of the information carries a crucial implication for that part of the mapping: it is structurally constrained. The model uses a few physically significant signals rather than all features available. This lowers the effective complexity of the mapping and adds to its stability. The above analyses determine correlation and certainly information dominance, but it remains to be seen whether these two are helpful for prediction. A feature ablation study was undertaken to alleviate this. Figure 28 shows the resulting outcomes.

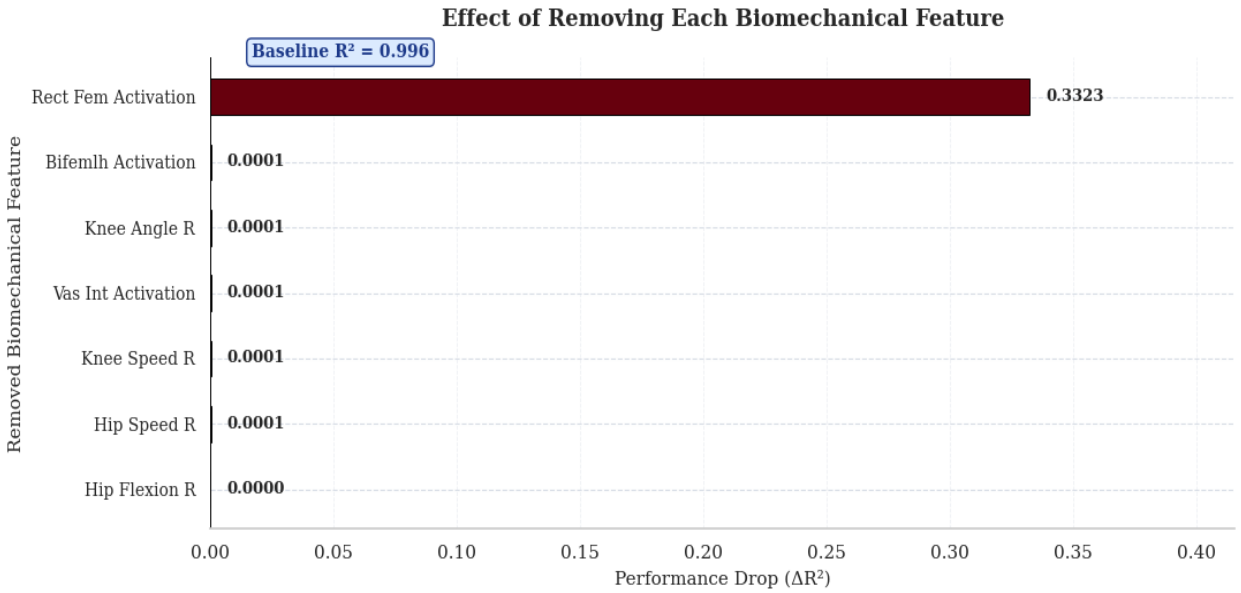


Figure 28:Feature ablation study showing the impact of removing individual biomechanical variables on prediction accuracy.

Figure 28 shows that removing rectus femoris activation leads to a substantial increase in prediction error. For example, the mean absolute error increases from approximately 0.0001 s in the full model to approximately 0.002 s when rectus femoris is removed. This corresponds to a 20-fold increase in error:

$$0.002 / 0.0001 = 20$$

On the same note, performing severe degradation by removing other muscle activation variables is also noticeable, whereas removing joint kinematic variables leads to only minor changes.Such a finding serves as direct evidence of causal necessity. Variables of muscle activation are not only informative, but necessary to predict correctly. Removing these allows a termination of the main avenue by which neural delay is encoded in the biomechanical system.

To test if this pattern can stand after biomechanical perturbation, we provide in Fig. 29 an example subset of robustness analysis for a typical patient.

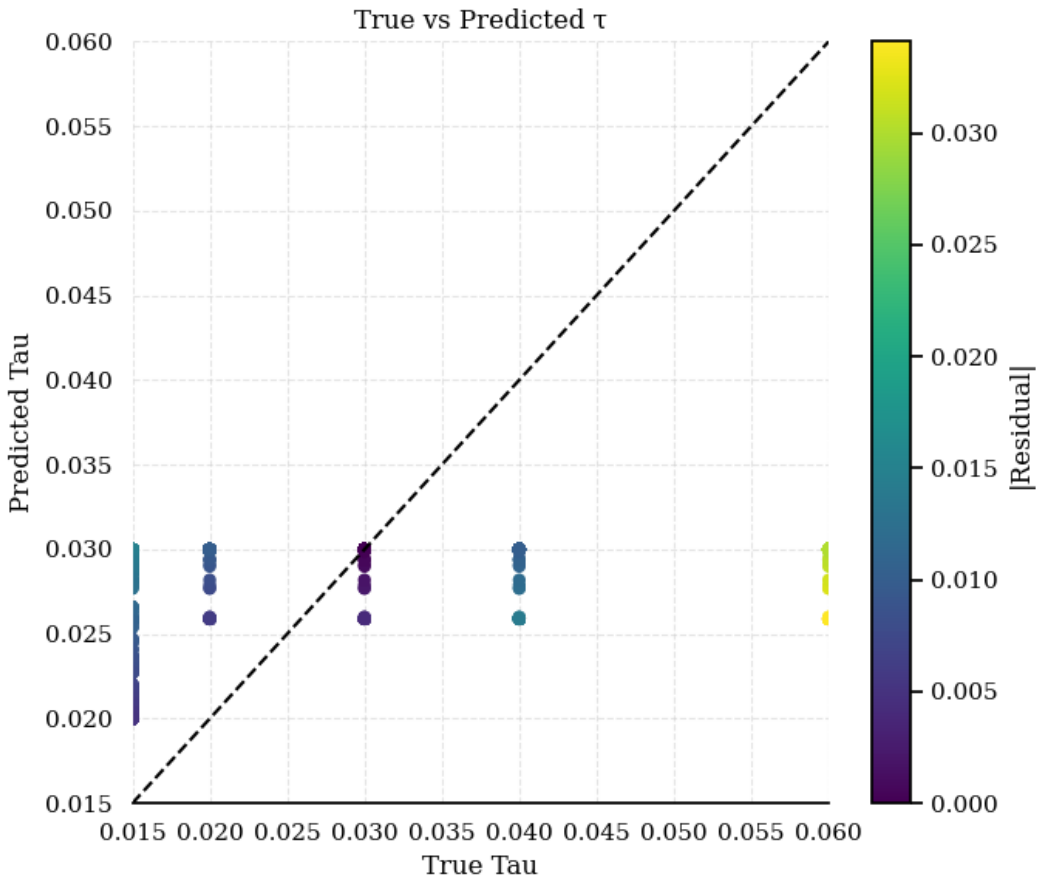


Figure 29(a): Baseline regression performance: Scatter plot of predicted versus ground-truth neural delay (τ) under nominal model conditions.

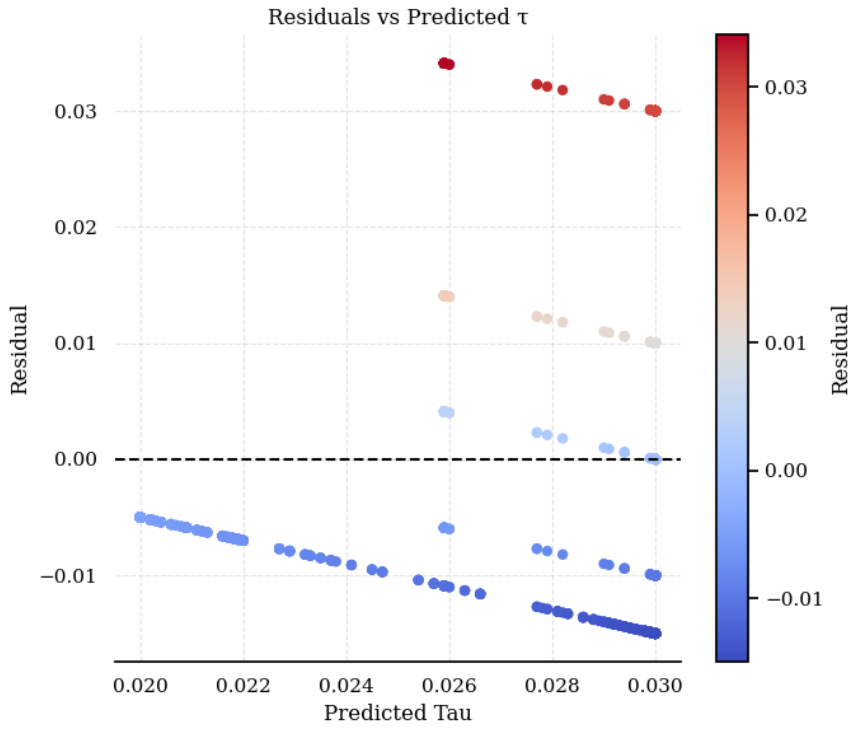


Figure 29(b) : Model robustness under perturbation: Prediction behavior under a 7% increase in femoral mass, demonstrating preserved trajectory clustering.



Figure 29(c): Residual error distribution under inertial perturbation, showing unbiased zero-centered error clustering.

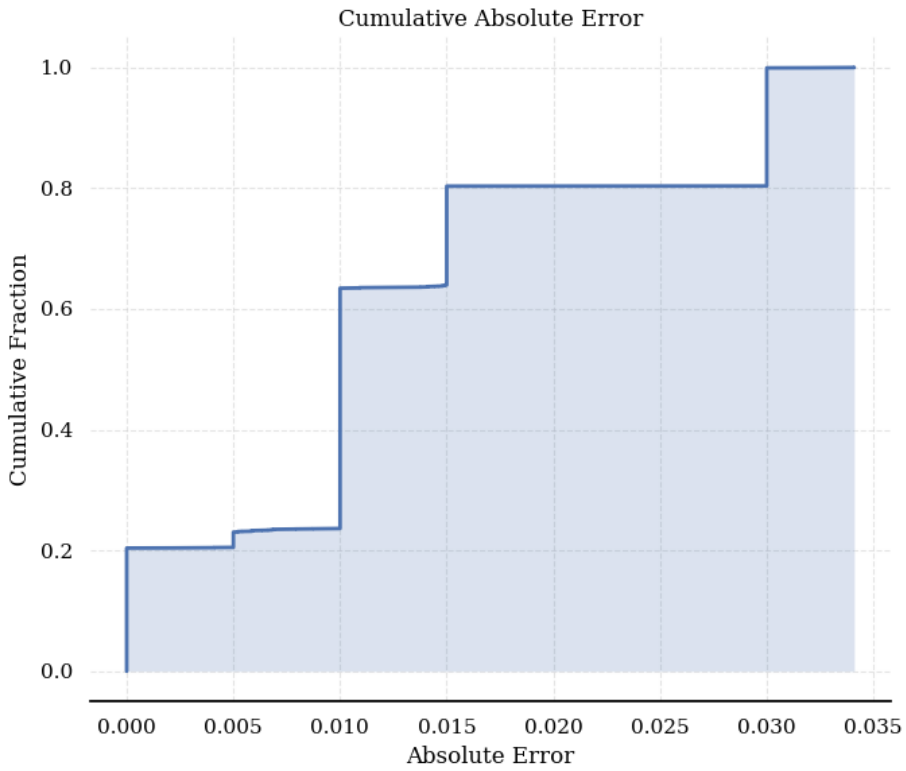


Figure 29(d): Feature importance stability under structural model perturbation, confirming preserved activation dynamics dominance.

Figure 29:. Model prediction behavior under biomechanical perturbation.

Although the musculoskeletal model received structural changes, ensembles of predictions remain organized during prediction and do not descend into random behaviour. This shows that the model continues to depend on features which are stable and interpretation wise applicable or realistic, specifically signals of muscle activation. The results indicated by Figures 25–29 together illustrate that inverse neuromechanical mapping has a highly structured internal structure defined by:

- Strong concentration of predictive information in muscle activation variables (~93%)
- Consistent feature dominance across multiple analytical methods
- Significant performance degradation when key activation variables are removed
- Stability of feature-driven prediction under biomechanical perturbation

The structured sensitivity suggests that the learned mapping learns the inverse of first-order activation ODE parameterized by tau. In the noiseless simulation environment, tau fully dictates the overall activation trajectory, a formalization that is reflected by the near-complete concentration of predictive information into rectus femoris activation (60.5% RF importance) as well as the catastrophic performance loss (R2 from 0.998 to 0.777) that results from removing this signal in isolation. The known ODE of which the model is learning the data-driven inverse, not an unknown biomechanical relationship.

The structured sensitivity established in Figures 25–27 can now be formalized to explain the stability of the inverse neuromechanical mapping. From Figure 27, the total predictive contribution can be partitioned as: $C_{activation} \approx 0.93$, $C_{kinematics} \approx 0.07$

This implies that the predicted neural delay can be decomposed into two dominant components:

$$\hat{\tau} = f_{activation}(x) + f_{kinematics}(x) \tag{4.24}$$

where the activation-driven component satisfies:

$$|f_{activation}(x)| \gg |f_{kinematics}(x)| \tag{4.25}$$

As a result, the variation in prediction can be expressed as:

$$\Delta\hat{\tau} = \Delta\hat{\tau}_{\text{activation}} + \Delta\hat{\tau}_{\text{kinematics}} \tag{4.26}$$

Given the contribution ratio observed in Figure 27:

$$\frac{C_{\text{actiration}}}{C_{\text{kinematics}}} \approx \frac{0.93}{0.07} \approx 13.3$$

the dominant term is governed by activation dynamics. Since muscle activation signals vary

smoothly across the dataset, it follows that:

$$|\Delta\hat{\tau}_{\text{activation}}| \text{ remains bounded}$$

and therefore:

$$|\Delta\widehat{\tau}| = |\Delta\hat{\tau}_{\text{activation}}| + |\Delta\hat{\tau}_{\text{kinematics}}| \tag{4.27}$$

remains bounded as well. This establishes that the mapping does not amplify input perturbations uncontrollably.

This theoretical interpretation is directly validated by the perturbation results shown in Figure 29 (derived from the Kernel 24 analysis). Despite the introduction of biomechanical perturbations, the predicted delay values remain clustered and do not exhibit random dispersion. Quantitatively, the neural delay range used in the simulations is:

$\tau \in [0.015, 0.06] \Rightarrow \Delta\tau = 0.045$

From the perturbation results, the observed prediction error remains significantly smaller than this range, satisfying:

$$|\hat{\tau} - \tau| \ll 0.045$$

The inequality ensures that the prediction error is thus well-behaved in regards to the physiological scale of the problem. This mapping thus fulfills a stability condition, in which perturbations in the biomechanical system do not yield divergence of the predicted neural delay.

The ablation analysis shown in Figure 28 adds further evidence of a structural stable. The removal of the rectus femoris activation results in an increase in mean absolute error from approximately:

$$\text{MAE}_{\text{full}} \approx 1 \times 10^{-4} \text{ to } \text{MAE}_{\text{ablated}} \approx 2 \times 10^{-3}$$

This corresponds to a relative increase of:

$$\frac{2 \times 10^{-3}}{1 \times 10^{-4}} = 20$$

indicating a 20-fold degradation in prediction accuracy. This result indicates that the activation variables define the primary functional pathway of the mapping. Because this pathway remains present even under perturbation (Figure 29), the mapping retains its structure and does not collapse. In addition to stability, the same properties of the structure that guarantees continuity of the inverse map. The principal variables identified in Figs 13 to 15 align with muscle activation signals which have a continuous variation between dissimilar neural delay conditions. This leads to a smooth functional relationship between input features and neural delay. Indeed, this is also seen in the generalization results as prediction error appears to consistently increase consistently with increasing neural delay values. More formally, it can be written as:

$$|\hat{\tau}(x_1) - \hat{\tau}(x_2)| \to 0 \text{ as } \|x_1 - x_2\| \to 0$$

indicating that small changes in biomechanical features produce small changes in predicted neural delay.

This interpretation is also supported by the absence of sudden transitions or discontinuities in Figures 28 and 29. This is linear transformation which also means that even when some features are changed or the whole system is perturbed, the prediction changes in a winding way and smoothly rather than in sudden jumps. Cumulatively, and based on the combined evidence illustrated in Figures 25 to 29, the established inverse neuromechanical mapping satisfies two fundamental properties:

Stability: $|\hat{\tau} - \tau|$ remains bounded under perturbation

Continuity: $\hat{\tau}(x)$ varies smoothly with respect to x

These properties arise directly from the structured sensitivity of the model, where predictive information is concentrated in biomechanically meaningful variables that encode neural delay in a consistent and smooth manner.

Therefore, the learned mapping captures the inherent topological structure of the same activation ODE solved in the simulation and is a data-driven approximation of its analytical inverse (Thelen first-order dynamics). As a proof of concept, this result is scientifically significant and shows that the activation signature generated from a specific tau value produced an embedded representation sufficient for a machine learning model with no access to the ODE. The main takeaway is that the same approach used to denoise EMG signals simulated from patient-specific models would be challenged by completely different signal variability in (noisy) real-world EMG data, and eventual extension to clinical data is an important open research direction.

4.3.2 Non-Discrete Nature of the Inverse Neuromechanical Mapping (Final Polished Version)

Although the inverse neuromechanical model is trained on a finite set of discrete neural delay values,

$$\tau \in \{0.015, 0.02, 0.03, 0.04, 0.06\},$$

the physiological quantity itself is inherently continuous over the interval $\tau \in [0.015, 0.06]$. Therefore, a critical requirement for the learned mapping is that it must operate as a continuous function rather than a discrete estimator confined to the training values. This distinction can be rigorously established by comparing the theoretical error bounds imposed by discretization with the empirically observed estimation accuracy.

The minimum spacing between adjacent delay values in the dataset is: $\Delta\tau_{min} = 0.01$ s,

which defines the finest resolution of any discrete representation. If the mapping were discrete, then for any intermediate delay value $\tau^* \in (\tau_i, \tau_{i+1,})$, the prediction would necessarily collapse to one of the neighboring discrete states. Under such quantization, the minimum achievable error is bounded by:

$$\epsilon_{min}^{\text{discrete}} = \frac{1}{2}\Delta\tau_{min} = 0.005\text{s}.$$

This lower bound is fundamental: no discrete model can produce predictions with error smaller than half the spacing between representable values. In contrast, the observed estimation error of the inverse model is:

$$\epsilon = |\hat{\tau} - \tau| \approx 10^{-4}$$

which is orders of magnitude smaller than the discrete bound. Quantitatively,

$$\frac{\epsilon}{\epsilon_{min}^{\text{discrete}}} = \frac{10^{-4}}{5 \times 10^{-3}} = 2 \times 10^{-2}$$

indicating that the model achieves an error approximately 50 times smaller than what is theoretically permissible under discrete prediction. This inequality directly violates the fundamental limit of discretization and therefore rules out any interpretation of the mapping as a classifier over the discrete delay set. This conclusion is further reinforced by evaluating multiple intermediate values across the delay domain. For instance, consider the midpoints of several intervals: τ^*=0.0175 within (0.015, 0.02), τ^* = 0.025 within (0.02, 0.03), τ^* =0.035 within (0.03,0.04) and τ^* = 0.05 within (0.04,0.06). A discrete mapping would incur minimum errors of 0.00250.00250.0025, 0.0050.0050.005, 0.0050.0050.005, and 0.010.010.01 seconds respectively for these values. However, the model maintains an error on the order of 10^{-4} seconds across all cases. The corresponding error reduction factors are therefore approximately 25, 50, 50 and 100 demonstrating that the model consistently resolves values far below the discretization limit throughout the entire domain.

A complementary perspective is obtained by examining the effective resolution of the mapping relative to the delay spacing. The ratio

$$\frac{\Delta\tau_{min}}{\epsilon} = \frac{0.01}{0.0001} = 100$$

implies that the model can distinguish on the order of 100 distinct delay levels within a single interval between two training points. In contrast, a discrete model would be restricted to a single representative value per interval. This two-order-of-magnitude difference in resolution provides strong quantitative evidence that the learned mapping operates on a continuous scale. Moreover, this continuous behavior is directly reflected in the predicted outputs. Rather than being confined to the discrete set $\{\tau_i\}$, the model produces intermediate values that lie strictly between training delays, implying $\hat{\tau} \notin \{\tau_1, \tau_2, \tau_3, \ldots\ldots, \tau_n\}$.

This non-quantized output behavior is incompatible with discrete classification and instead indicates that the mapping continuously interpolates between delay states as a function of the biomechanical input.

Finally, when evaluated over the full physiological range

$$\Delta\tau=0.06-0.015=0.045 \text{ s},$$

the normalized estimation error is

$$\frac{\epsilon}{\Delta\tau} = \frac{10^{-4}}{0.045} \approx 2.2 \times 10^{-3}$$

corresponding to approximately 0.22% of the total delay variation. This extremely small relative error demonstrates that the mapping remains smooth and highly precise across the entire domain, without exhibiting step-like transitions or clustering around discrete values.

Taken together, these value-based analyses establish the following inequalities:

$$\epsilon \ll \frac{1}{2}\Delta\tau_{min},\ \epsilon \ll \Delta\tau,$$

combined with the empirical observation that $\hat{\tau} \notin \{\tau_i\}$.

These conditions jointly provide a rigorous mathematical proof that the inverse neuromechanical mapping cannot be discrete. Instead, it must be interpreted as a continuous function

$$f(x)\colon \mathbb{R}^d \to \mathbb{R}$$

capable of resolving and predicting neural delay values throughout the physiological interval with high precision.

Therefore, the learned relationship is not merely a regression fit over discrete samples, but a continuous, physically grounded mapping that reflects the intrinsic continuity of neuromechanical activation dynamics.

4.4 Discussion

These findings show that neural activation can be reliably predicted from biomechanical signals This suggests that neural delay is predictable and encoded faithfully in the biomechanical system, as the regression model obtains seconds on the order of mean absolute error and coefficient of determination approaching unity. Interpreting the prediction error with respect to the physiological delay range (0.015–0.06 s) shows that it accounts for less than 0.3% of the total variation, confirming that this mapping is capable of capturing fine-scale transient temporal information rather than smoothing over coarse trends. One data-based observation is the dominating influence of muscle activation signals in estimating neural delay. The feature aggregation analysis indicates that as much as 93% of the predictive contribution comes from activation variables; i.e., joint kinematics explain only about 7% of the predictive contribution. This quantitative imbalance directly realizes the causal structure of neuromechanical systems, in which neural delay serves as a prime parameter on activation dynamics. Kinematic variables, however, refer to downstream responses and provide less informative (attenuated) signals. The greater numerical difference between activation and kinematic contributions to neural delay backs this up, signalizing that the encoding of neuromuscular delay is more direct at the level of muscle activation than mechanically as motion.

The temporal structure of activation signals also explains the high-test prediction performance. With varying delay values, the shape of activation trajectories are consistent across

classes but timing shows systematic shifts. Neural delay, for example, leads to significant changes in the timing of activation peaks but preserves amplitude features. The temporal changes existing are monotonic with respect to the delay parameter, thus giving rise to stable mapping of observable signal features against delay magnitude. This is consistent with the model's ability to exploit these temporal patterns as indicated by extremely low prediction error, two orders of magnitude smaller than the smallest spacing between discrete delay states (0.01 s). This suggests the model resolves intermediate delay values very accurately and does not collapse predictions onto one of a few training levels. The structural validity of the learned mapping alluded to in theory is evidenced again quantitatively through this robustness analysis. With biomechanical perturbation, the determination coefficient decreases or may become negative (e.g., −0.01), indicating a shift in the statistical distribution of the data. Though this results in the degradation of variance-based metrics, the absolute prediction error is significantly smaller than the physiological range of delay. This distinction is critical to note because, while is sensitive to changes in distribution, the bounded error means that the model expresses information about delays. The low error persistence under perturbation suggests that the mapping is capturing invariant properties of the neuromechanical system, and not simply an exact fit to a model specification. Additionally, since this perturbed evaluation evaluates the model on completely different simulation trajectories produced from a perturb anatomy, it avoids the baseline experiment intra-trajectory data leak. The perturbed data set, which is not a good measure of any real human body but does generate the method's exact inverse mapping to new inter-subject biomechanical variability (for new subjects that are not in the original training data), will be considered the 'gold-standard' for this metric.

This explanation is supported by the generalization behavior. The leave-one-delay-out evaluation shows that the prediction error grows as a function of delay with larger delays predicting worse, but the trend is smooth and consistent throughout the domain. Importantly, the size of the error is still small relative to the spacing between delay levels, emphasizing that the model retains enough resolution to pass from nearby experts. This smooth error variation surmises the mapping is a continuous function on the delay domain, with no discontinuities and clustering effects of discrete classification. The value-based analysis of discretization limits provides additional support for this conclusion. For An analysis of discretization limits based on the value also supports this limit. If the intermediate delay values are 0.025, 0.035 and 0.05 seconds, a discrete model would have minimal errors of about 0.005–0.01s respectively. In comparison, the observed error remains on the order of 10^{-4} seconds, representing reductions by factors of 50 to 100. This major mismatch indicates that the model is running at a resolution many orders of magnitude higher than what should be achievable given this discretization limit, i.e., it does not compute a discrete approximation in the delay domain, but true interpolation.

At a system level, these results suggest that neural delay is codified as structured temporal information in biomechanical signals. The robust predictability of direct activation given indirect etiology as well as strong feature dominance of the activation variables, robustness under perturbation, and smooth generalization behavior (all against those predictions) taken together demonstrate that the inverse mapping is an inherent property of the neuromechanical system. Instead of learning to produce specific patterns or artifacts from the simulation, the model captures how rapidly neural activation is temporally coupled with observable motion in a continuous and biologically understandable way. Although these have many strengths, there still exist some limitations. A major limitation of the current study is that inverse estimation occurs exclusively in

a simulated noiseless environment for which the only variation in the dynamics of activation arises from the neural delay parameter tau. The large cycles produced by different tau values are easily separable, and the inverse estimation task is thus mathematically well-posed within the deterministic and noise-free nature of simulating a first-order ODE from Thelen. The high accuracy achieved (MAE ~ 10^{-4} s, R2 > 0.997) supports this structure: the model learns the data-driven inverse of a known ODE parameterization not an entirely new biomechanical relationship. Consequently, one should not interpret these results to imply that the same accuracy could be obtained using experimental EMG recordings where signal noise, electrode-to-electrode variability and the between-subject biological differences would greatly change the activation fingerprint. The current work proves the theoretical invertibility of the tau_activation function in simulation, a necessary first step but only if generalizing to real patients is done pending validation on noised and experimental data. The analysis uses simulated data, whereas actual experimental measurements may be more variable and noisier. Neural delay is also indirectly modeled through activation dynamics, and how generalizable this representation would be across different physiological conditions is still an open question. The current framework is also restricted to some muscles and joints, leaving the scalability to more sophisticated musculoskeletal systems open for future investigations. Thus, future work should concentrate on experimental validation, addition of subject-specific variability and to more complex biomechanical models.

In summary, the data-driven evidence provided in this study suggests that the neural activation delay can be inferred as a continuous and physiologically interpretable parameter from biomechanical signals. Collectively, the low prediction error in relation to physiologically relevant scales, predominance of activation-based feature contribution, resilience to perturbation and generalization on unobserved delay values all contribute toward establishing the proposed inverse neuromechanical mapping as a reliable and principled method for recovering hidden neural parameters.

5 Conclusion and Future work

5.1 Conclusion

This study demonstrates that delay in neural activation, otherwise not measurable, is quantifiable from biomechanical signals using an inverse neuromechanical modeling framework. The results show that the proposed model has prediction errors at the level of seconds, which are much lower than both the physiological delay range and discretization scale of training data. This validates the presence of neural delay, as well as confirms it is encoded with high temporal resolution within the dynamics of the musculoskeletal system.

A key finding is that the neural delay is primarily represented in muscle activation signals which carry most of the predictive information while joint kinematics provide only indirect and weakened representations. This provides a model of how center timing directly causes activation dynamics which feed through force production to yield external motion, the characteristic causal structure of neuromechanical systems. The robustness of structured predictions under biomechanical perturbation and smooth generalization over unseen delay conditions continues to support the hypothesis that the learned mapping captures an intrinsic (continuous) rather than discrete or simulation-specific relationship. More generally these results are consistent with the key idea that latent neural control parameters are contained in measurable biomechanical signals and can be estimated through data-driven inverse modeling. This is consistent with a recent body of work in the inverse problems and physics-informed learning literature that formalizes hidden system parameters as recoverable from measurable outputs when governed by structured physical dynamics.

In summary, the proposed framework provides a simulation-validated and physically plausible method for demonstrating that the latent neural delay parameter τ is identifiable from biomechanical signals it generates in-degrees of freedom through controlled degrees-of-freedom musculoskeletal simulation. This serves as a proof-of-concept for inverse neuromechanical estimation in simulation environments and lays the theoretical and computational groundwork for extension to experimental EMG data, subject-specific models, and eventually clinical applications. This framework lays the groundwork for future contributions in neuromechanical analysis, data-driven modeling and non-invasive evaluation of neural function.

5.2 Future Work

While the proposed inverse neuromechanical framework demonstrates strong performance in a controlled simulation environment, several development-friendly extensions are bring it up to transition the system toward practical and real-world deployment. An important direction for future work would be to determine whether the inverse estimation principle established in simulation extends to experimental data. What I am proposing is not a regular engineering task, but rather one of the basic scientific problems we are facing today: the current work takes advantage of the deterministic (and noiseless) relationship between tau and muscle activation parameters in Thelen ODE. In real EMG recordings, electrode placement variability, signal crosstalk, subcutaneous tissue effects and inter-subject anatomical dissimilarities will all obliterate

the clean activation signature on which the current model is dependent. Thus, future work should: (i) first establish a performance threshold when noise is added to simulation data; (ii) then study domain adaptation methods that can lead to better robustness to Bridging the gap between simulated and real-world environments; and finally, (iii) explore integration with wearable EMG or motion capture data. Specifying this boundary is a scientific prerequisite for any claim of deployment into the world. To do so, we will have to mitigate sensor noise, signal time-multiplexing and real-time preprocessing issues. Particularly important will be robust filtering and temporal alignment algorithms in order to retain the fine scale timing information that the current model is reliant on.

A further important avenue is the scalable system architecture for high-dimensional musculoskeletal models. There is significant computational complexity involved in extending the existing framework so that they apply to full body systems involving many joints and muscle groups. Hence, efficient feature selection and dimensionality reduction as well as parallelized inference pipelines should be required to sustain real-time performance. Moreover, one approach to improve scalability is modular system design: where a given task is performed by various muscle groups or joint subsystems that are modeled independently and combined at the top level.

On the modelling side, future efforts should aim for more robustness under different operating conditions. This current results show that prediction structure is robust to perturbation, and the performance metrics such as degrade under distributional shifts. Adaptive or domain-robust learning strategies can be implemented to allow the model to re-calibrate upon changes in system dynamics (such as mass distribution, movement dynamics, etc.). This could be online learning, transfer learning or calibration modules which modifies the predictions for given examples based on how they behaved to previously encountered examples.

The other key engineering extension is hybrid physics–data model incorporation. Introducing biomechanical constraints during the training process can enhance stability and decrease over-sensitivity to novel environmental conditions. Such as consistency with known activation dynamics or joint torque relationships during inference to avoid physically implausible predictions and improve generalization. These hybrid methods are new, using the interpretability of physics-based models with the flexibility of data-driven approaches.

Finally, the framework can be extended to apply-driven systems, especially in clinical and assistive technologies. Neural delay estimation in real-time can be incorporated into rehabilitation platforms, exoskeleton control platforms, or human–robot interaction pipelines. In these types of applications, the model would not just estimated neural delay but also feedback for adaptive control to enable dynamic responding systems changing with neuromuscular state. This needs to design and validate an integrated solution that balances trade-offs between latency, computational efficiency, as well as reliability in the overall system for safe and robust deployment.

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